Principles of Genetics

GPB 201 — Lecture-wise Theory Study Material

B.Sc. (Hons.) Agriculture · Second Year, Third Semester · 2025 Syllabus

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About this study material

This handbook follows the official GPB 201 lecture list one lecture at a time. Each of the 34 numbered lectures is treated as a chapter. Inside a chapter, every sub-topic named in the syllabus line is expanded in the same order in which it appears in the syllabus, so the material can be read alongside classroom teaching and used directly for the semester examination and for later competitive examinations.

The style is deliberately simple. Definitions are given in plain English, comparisons are placed in tables, ratios and crosses are worked out step by step, and structures that are easier to see than to describe are drawn as text diagrams. Every chapter closes with an **Exam focus** box containing key points, short-answer questions with model answers, and long-answer questions with model outlines.

Agricultural examples are used wherever they arise naturally. Classical experiments are reported with their standard textbook values; where a value or a variety name is not certain, the material says so rather than inventing one.

How to use each lecture chapter

- **Read the definition box first.** It gives the one-line meaning of the topic before the detail.
- **Study the tables.** Most exam questions in genetics ask for comparisons, ratios or classifications, and these are exactly what the tables hold.
- **Redo the worked examples on paper.** Genetics numericals are learned by writing the cross, not by reading it.
- **Answer the Exam focus questions aloud.** If the model answer can be produced from memory, the chapter is finished.

Course map — the 34 lectures at a glance

Lecture	Topic
1	Introduction and definition of cytology, genetics, heredity, inheritance and cytogenetics; brief history of genetics
2	Cell and cell organelles; differences between prokaryotes and eukaryotes
3	Cell cycle; cell division — mitosis and meiosis and their significance
4	Sporogenesis, gametogenesis and syngamy in plants; identical and fraternal twins
5	Architecture of chromosomes — chemical composition, chromonemata, matrix, nucleosome, centromere, telomere, euchromatin, heterochromatin, NOR, secondary constriction, satellite chromosome, karyotype, ideogram
6	Types of chromosomes by position and number of centromeres; special chromosomes — polytene, lampbrush, B, ring, isochromosomes
7	Chromosomal aberration — structural: deletion, duplication, inversion, translocation
8	Chromosomal aberration — numerical: aneuploidy and its origin; Klinefelter and Turner syndromes
9	Euploidy — haploids, dihaploids, double haploids; polyploidy — auto- and allopolyploids; evolution of wheat, Triticale, cotton, tobacco, Brassica
10	Pre-Mendelian concepts of heredity — vapour and fluid theory, magnetic power theory, preformation, Lamarckism, Darwinism, germplasm theory, mutation theory
11	Model organisms — Drosophila, Arabidopsis, garden pea, *E. coli*, mice
12	Probability and chi-square
13	Mendel's experiments and laws of inheritance; rediscovery of Mendel's work
14	Terminologies — gene, allele, locus, homozygous, heterozygous, hemizygous, genotype, phenotype, monohybrid, dominance, recessive, test cross, back cross, dihybrid, trihybrid, polyhybrid
15	Chromosomal theory of inheritance; allelic interactions — dominance vs recessive, complete dominance, codominance, incomplete dominance, threshold characters
16	Non-allelic interaction without modification of Mendelian ratio — Bateson and Punnett's fowl comb experiment
17	Non-allelic interaction with modification — dominant epistasis (12:3:1), recessive epistasis (9:3:4), duplicate/additive (9:6:1), duplicate dominant (15:1)
18	Mid Semester Examination
19	Duplicate recessive epistasis (9:7); dominant and recessive epistasis (13:3); summary of epistatic ratios
20	Qualitative and quantitative traits; polygenes and continuous variation; transgressive segregation; multiple factor hypothesis — Nilsson-Ehle's wheat kernel colour; corolla length in *Nicotiana longiflora*
21	Lethal genes, pleiotropism, penetrance and expressivity, multiple alleles, blood groups, coat colour in rabbits, self-incompatibility, pseudoalleles, isoalleles
22	Crossing over — mechanism, significance, cytological proof (Stern's experiment), factors controlling crossing over
23	Linkage and its estimation — coupling and repulsion, Bateson and Punnett's experiment; Morgan's chromosomal theory; complete and incomplete linkage; linkage group

24	Strength of linkage and recombination; double crossover; interference and coincidence
25	Two-point and three-point test cross; genetic map; physical map
26	Sex determination — autosomes and sex chromosomes; human, fowl, butterfly, grasshopper, honey bee, *Fumea*; plants — *Melandrium*, papaya, maize; Bridges' genic balance theory; gynandromorphs
27	Sex-linked inheritance — criss-cross inheritance, reciprocal difference; holandric genes; sex-influenced and sex-limited traits
28	Mutation — classification, methods of inducing mutations, mutagenic agents, induction of mutation
29	Maternal effects — snail shell coiling; cytoplasmic inheritance; chloroplast and mitochondrial inheritance; plastid colour in *Mirabilis jalapa*; CMS in maize; kappa particles in *Paramecium*
30	DNA as genetic material — Griffith, Avery-MacLeod-McCarty, Hershey-Chase; RNA as genetic material — Fraenkel-Conrat and Singer
31	Types of DNA and RNA (mRNA, tRNA, rRNA); Watson-Crick model; proof for semi-conservative replication
32	Models of DNA replication; steps involved in DNA replication
33	Genetic code, cistron, muton, recon; gene concept — gene structure, molecular concept, molecular definition, number of genes per chromosome, size of a gene, fine structure, types of genes
34	Central dogma; transcription and protein synthesis; regulation of gene expression; Operon model of Jacob and Monod — lac and trp operons

Note: Lecture 18 of the syllabus is the Mid Semester Examination. It is retained in this document in its correct position as a chapter that lists the exam pattern and revision strategy, so that no syllabus number is skipped.

Lecture 1. Introduction, Definitions and Brief History of Genetics and Cytogenetics

> **In one line:** Genetics is the science of heredity and variation. Cytology is the study of the cell. Cytogenetics joins the two by explaining heredity in terms of chromosomes and their behaviour.

1.1 Definitions

Cytology (Cell Biology)

Cytology is the branch of biology that deals with the **structure, composition and functions of the cell** and its organelles. The word comes from the Greek *kytos* (cell) and *logos* (study). Modern cytology overlaps heavily with biochemistry and molecular biology, and is now often called cell biology.

Genetics

Genetics is the branch of biology that deals with **heredity and variation** — how characters are transmitted from parents to offspring, and why offspring differ from parents and from one another. The term was coined by **William Bateson** in 1905 (from the Greek *gen*, meaning to become or to grow). Genetics studies the gene: its structure, function, transmission and variation.

Heredity

Heredity (also called inheritance) is the **transmission of characters from parents to offspring**. The resemblance between parents and offspring is due to heredity. The physical basis of heredity is the gene, carried on chromosomes, which are passed through gametes.

Inheritance

Inheritance is the **process by which the hereditary material (genes) is passed** from one generation to the next. The pattern in which a character appears in successive generations is called its mode of inheritance (for example, dominant, recessive, sex-linked, cytoplasmic).

Variation

Variation is the **difference among individuals** of the same species. It may be continuous (quantitative, for example plant height) or discontinuous (qualitative, for example flower colour). Variation is of two kinds:

- **Heritable (genetic) variation** — due to differences in genes; caused by mutation, recombination, hybridisation and polyploidy. This is the raw material of evolution and of plant breeding.
- **Non-heritable (environmental) variation** — due to nutrition, climate, soil, management; not passed to the next generation.

Cytogenetics

Cytogenetics is the branch that **combines cytology and genetics** to study chromosomes and their relation to heredity. It deals with chromosome number, structure, behaviour during cell division, and how changes in chromosomes produce changes in characters. Techniques

include karyotyping, banding, fluorescence in situ hybridisation (FISH) and genome in situ hybridisation (GISH).

1.2 Interaction among cytology, genetics and cytogenetics

The three subjects are nested, not separate. Cytology supplies the physical picture of the cell; genetics supplies the rules of transmission; cytogenetics explains the rules using the picture. When Mendel's laws were rediscovered in 1900, cytologists had already described the behaviour of chromosomes in meiosis. In 1902–03 Sutton and Boveri saw that chromosomes behave exactly as Mendel's "factors" were predicted to behave, and the **chromosomal theory of inheritance** was born — the first great product of cytogenetics.

Table 1.1 — Scope of the three disciplines and how they connect

Discipline	Chief question	Main tools	Typical output
Cytology	What is the cell and how does it work?	Microscopy, staining, cell fractionation	Cell structure and organelle function
Genetics	How are characters transmitted and varied?	Crosses, pedigrees, statistical tests, molecular methods	Laws of inheritance, gene maps, breeding strategy
Cytogenetics	How do chromosomes explain heredity?	Karyotyping, banding, FISH, meiotic analysis	Chromosome maps, aberration diagnosis, ploidy manipulation

1.3 Brief history of developments in genetics and cytogenetics

The history is best remembered in phases: (i) early cell and fertilisation studies, (ii) the birth of Mendelism, (iii) the union of genetics with cytology, (iv) the molecular era, and (v) the genomics era.

Phase 1 — Early observations (1665–1865)

- **1665 — Robert Hooke** observed cork cells under a compound microscope and coined the term **cell**.
- **1670s — Antonie van Leeuwenhoek** first saw living cells, bacteria and spermatozoa.
- **1831 — Robert Brown** discovered the **nucleus**.
- **1838–39 — Schleiden and Schwann** proposed the **cell theory** (all organisms are made of cells).
- **1855 — Rudolf Virchow** added *Omnis cellula e cellula* — every cell arises from a pre-existing cell.
- **1865 — Gregor Johann Mendel** presented his paper *Experiments on Plant Hybrids* to the Natural History Society of Brünn. It contained the laws of inheritance but was ignored until 1900.
- **1869 — Friedrich Miescher** isolated "nuclein" (DNA) from pus cells.

Phase 2 — Birth of genetics (1880–1910)

- **1882 — Walther Flemming** described mitosis and coined the term **chromatin**.
- **1888 — Waldeyer** coined the term **chromosome**.
- **1890s — van Beneden, Boveri and Weismann** worked out meiosis and the continuity of chromosomes; Weismann proposed the **germplasm theory**.

- **1900 — Rediscovery of Mendel's laws** independently by **Correns, de Vries and Tschermak**.
- **1902–03 — Sutton and Boveri** proposed the **chromosomal theory of inheritance**.
- **1905 — Bateson** coined the word **genetics** and, with Punnett, discovered non-allelic interaction in fowl comb shape.
- **1909 — Wilhelm Johannsen** coined the terms **gene**, **genotype** and **phenotype**.
- **1910–11 — T. H. Morgan** and his students (Sturtevant, Bridges, Muller) established linkage, sex linkage, crossing over and the first genetic maps in *Drosophila*.

Phase 3 — Cytogenetics matures (1920–1950)

- **1927 — H. J. Muller** induced mutations in *Drosophila* with X-rays — the beginning of induced mutation.
- **1930s — Barbara McClintock and Harriet Creighton** gave **cytological proof of crossing over** in maize; Sturtevant had already made genetic maps.
- **1936–41 — Blakeslee, Avery** and others used **colchicine** to induce polyploidy.
- **1937 — C. D. Darlington** published *Recent Advances in Cytology*, defining cytogenetics as a subject.
- **1941 — Beadle and Tatum** gave the **one gene–one enzyme** hypothesis using *Neurospora*.
- **1944 — Avery, MacLeod and McCarty** proved DNA is the transforming principle.

Phase 4 — Molecular genetics (1950–1980)

- **1952 — Hershey and Chase** confirmed DNA as genetic material using bacteriophage T2.
- **1953 — Watson and Crick** proposed the **double helix** model of DNA.
- **1958 — Meselson and Stahl** proved **semi-conservative replication**.
- **1961 — Jacob and Monod** proposed the **operon model** of gene regulation; Khorana, Nirenberg and Ochoa cracked the **genetic code**.
- **1970s — Restriction enzymes, DNA sequencing (Sanger) and gene cloning** laid the base of genetic engineering.

Phase 5 — Genomics and modern breeding (1980–present)

- **1983 — Kary Mullis** invented **PCR**.
- **1990–2003 — Human Genome Project**; the **rice genome** was sequenced in the early 2000s.
- **1996 onwards — Marker-assisted selection and GM crops** (Bt cotton in India from 2002).
- **2010s — Next-generation sequencing, genomic selection, CRISPR-Cas genome editing** (2020 Nobel to Charpentier and Doudna).

Table 1.2 — Milestones worth remembering for the examination

Year	Scientist	Contribution
1865	G. J. Mendel	Laws of inheritance
1869	F. Miescher	Isolation of nuclein (DNA)
1900	Correns, de Vries, Tschermak	Rediscovery of Mendelism
1902–03	Sutton and Boveri	Chromosomal theory of inheritance
1909	Johannsen	Gene, genotype, phenotype
1910–11	Morgan, Sturtevant	Linkage, sex linkage, gene mapping
1927	Muller	Induced mutation by X-rays
1931	Creighton and McClintock	Cytological proof of crossing over
1944	Avery, MacLeod, McCarty	DNA is the transforming principle
1953	Watson and Crick	Double-helix model of DNA
1961	Jacob and Monod	Operon model
2003	International consortium	Human genome sequence completed

1.4 Why genetics matters in agriculture

- Plant breeding is applied genetics: selection and hybridisation manipulate genes to raise yield, quality and resistance. Hybrid rice, hybrid maize and hybrid cotton all rest on genetic principles.
- Disease and pest resistance (for example, bacterial blight resistance in rice) is inherited and can be transferred by crossing or by gene transfer.
- Mutation breeding has produced improved varieties in rice, groundnut, pulses and sugarcane; induced polyploidy is used in fodder crops and ornamentals.
- Marker-assisted selection and genomic selection now speed up breeding of crops such as rice, maize and chickpea by selecting at the DNA level before the plant is grown.

Exam focus — Lecture 1

Key points

1. Genetics = study of heredity and variation; the word was coined by Bateson in 1905.
2. Heredity is the transmission of characters; inheritance is the process; the gene is the unit.
3. Cytology studies the cell; cytogenetics links chromosome behaviour with heredity.
4. Sutton and Boveri (1902–03) gave the chromosomal theory of inheritance.
5. 1900 rediscovery by Correns, de Vries and Tschermak marks the birth of modern genetics.
6. Johannsen (1909) gave gene, genotype and phenotype; Morgan's group gave linkage and mapping.
7. Avery (1944) and Hershey-Chase (1952) established DNA as the genetic material; Watson-Crick (1953) gave its structure.
8. Variation may be heritable (genetic) or non-heritable (environmental); only heritable variation responds to selection.

Short-answer questions with model answers

Q1. Define cytogenetics and state its two principal contributions to genetics.
Model answer: Cytogenetics is the branch that combines cytology and genetics to study the relationship between chromosomes and heredity. Two contributions: (i) it provided the chromosomal theory of inheritance by showing that chromosomes behave as Mendel's factors do, and (ii) it enabled gene mapping and the diagnosis of chromosomal aberrations such as aneuploidy and translocation.

Q2. Differentiate heredity from variation.
Model answer: Heredity is the transmission of characters from parents to offspring and explains the resemblance between them. Variation is the difference among individuals of the same species and explains why offspring are not identical to parents or to each other. Heredity conserves the type; variation provides the material for selection and evolution.

Q3. Name the three scientists who rediscovered Mendel's laws in 1900.
Model answer: Hugo de Vries (Holland), Carl Correns (Germany) and Erich von Tschermak (Austria).

Long-answer questions with model outlines

Q1. Trace the development of genetics from Mendel to the genomic era.
Outline: (i) Mendel 1865 — laws from pea experiments. (ii) 1900 rediscovery. (iii) 1902–03 chromosomal theory (Sutton, Boveri). (iv) 1909–11 gene concept, linkage and mapping (Johannsen, Morgan). (v) 1927–41 induced mutation, cytological proof of crossing over, one gene–one enzyme. (vi) 1944–53 DNA as genetic material and its structure. (vii) 1958–61 semi-conservative replication, operon, genetic code. (viii) 1980s–present PCR, sequencing, genomics, GM crops, genome editing.

Q2. "Genetics and cytology are two sides of the same coin." Discuss with examples.
Outline: Define both; explain that chromosomes carry genes; give the parallel between Mendelian segregation and meiotic separation of homologues, and between independent assortment and random orientation of bivalents; cite Sutton-Boveri, Creighton-McClintock

and Stern as evidence; conclude with agricultural applications such as ploidy breeding and marker-assisted selection.

Lecture 2. The Cell and Cell Organelles; Prokaryotes versus Eukaryotes

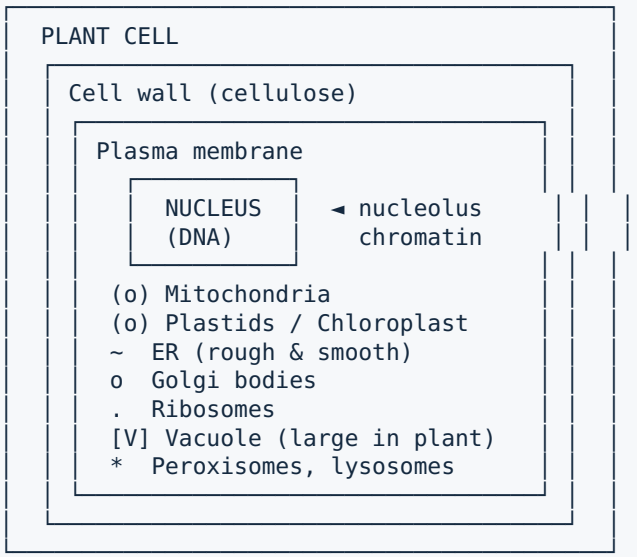
In one line: The cell is the smallest structural and functional unit of life; prokaryotic cells lack a true nucleus and membrane-bound organelles, whereas eukaryotic cells have both.

2.1 Cell theory

- All living organisms are composed of one or more cells (Schleiden and Schwann, 1838–39).
- The cell is the basic structural and functional unit of life.
- All cells arise from pre-existing cells by division (Virchow, 1855).
- The cell contains the hereditary material (DNA) which is passed to daughter cells.

2.2 The plant cell and its organelles

A typical plant cell has three main regions: the **cell wall** (outer), the **protoplasm** (protoplast) which includes cytoplasm and nucleus. The cytoplasm holds the cytosol and the organelles.



Major organelles and their functions

Table 2.1 — Organelles of the plant cell and their functions

Organelle	Structure	Function
Cell wall	Cellulose, hemicellulose, pectin; rigid, non-living	Gives shape and rigidity, prevents bursting, allows turgor
Plasma membrane	Phospholipid bilayer with proteins (fluid-mosaic)	Selective permeability, transport, cell recognition
Nucleus	Nuclear envelope, nucleoplasm, chromatin, nucleolus	Stores DNA; controls cell activity; site of transcription
Nucleolus	Dense region inside nucleus	rRNA synthesis and ribosome assembly
Mitochondrion	Double membrane, cristae; own DNA	Powerhouse; aerobic respiration, ATP synthesis
Plastids	Chloroplast, chromoplast, leucoplast; own DNA	Photosynthesis, storage, pigmentation
Endoplasmic reticulum	Rough (with ribosomes) and smooth	Protein synthesis and transport; lipid synthesis
Golgi apparatus	Stacked cisternae	Packaging, secretion, cell-wall material
Ribosome	rRNA + protein, 80S in eukaryotes	Site of protein synthesis (translation)
Vacuole	Large fluid-filled sac with tonoplast	Stores sap, maintains turgor, waste storage
Peroxisome / glyoxysome	Single membrane	Photorespiration, lipid metabolism
Cytoskeleton	Microtubules, microfilaments	Shape, movement, spindle formation

2.3 Prokaryotic and eukaryotic cells

Prokaryotes (bacteria, blue-green algae) are the oldest and simplest cells. Eukaryotes (fungi, plants, animals, protists) are larger and compartmentalised. The chief difference is the absence or presence of a **true nucleus bounded by a nuclear membrane**, and of membrane-bound organelles.

Table 2.2 — Differences between prokaryotic and eukaryotic cells

Feature	Prokaryotic cell	Eukaryotic cell
Examples	Bacteria, cyanobacteria (blue-green algae)	Fungi, algae, plants, animals, protists
Size	Small, 1–10 µm	Larger, 10–100 µm
Nucleus	Absent; nucleoid, no nuclear membrane	True nucleus with nuclear membrane
DNA	Single circular naked DNA; no histones (mostly)	Linear DNA complexed with histones into chromosomes
Membrane-bound organelles	Absent (no mitochondria, ER, Golgi, plastids)	Present
Ribosomes	70S	80S (organellar ribosomes 70S)
Cell wall	Peptidoglycan (murein)	Cellulose (plants), chitin (fungi); absent in animals
Cell division	Binary fission (amitosis)	Mitosis and meiosis
Genetic recombination	Transformation, conjugation, transduction	Meiosis and syngamy (sexual reproduction)
Respiratory site	Plasma membrane / mesosomes	Mitochondria
Photosynthetic site	Thylakoids in cytoplasm	Chloroplast
Flagella	Simple, single fibril	Complex, 9+2 microtubule arrangement
Introns	Rare	Common

Worked comparison task

Task: A student must classify three organisms — *E. coli*, *Saccharomyces cerevisiae* (yeast) and a rice plant cell — as prokaryotic or eukaryotic, and give one reason each.
Solution:

- *E. coli* — prokaryotic. Reason: no nuclear membrane; circular DNA; 70S ribosomes.
- *Saccharomyces cerevisiae* — eukaryotic. Reason: true nucleus present, mitochondria and 80S ribosomes, though it is unicellular and has a chitin wall.
- Rice plant cell — eukaryotic. Reason: true nucleus, cellulose wall, chloroplasts and mitochondria.

Exam focus — Lecture 2

Key points

1. Cell theory: Schleiden and Schwann (1838–39); Virchow added "every cell from a pre-existing cell".
2. Prokaryotes have no true nucleus and no membrane-bound organelles; eukaryotes have both.
3. Ribosomes are 70S in prokaryotes and 80S in eukaryotes.
4. Mitochondria and plastids are semi-autonomous: they have their own DNA and 70S ribosomes.
5. Cell wall is peptidoglycan in bacteria, cellulose in plants, chitin in fungi, absent in animals.
6. Prokaryotes divide by binary fission; eukaryotes divide by mitosis and meiosis.

Short-answer questions with model answers

Q1. Why are mitochondria and chloroplasts called semi-autonomous organelles?
Model answer: Because they contain their own circular DNA and 70S ribosomes and can synthesise some of their own proteins, yet most of their proteins are coded by nuclear genes, so they are only partly self-governing. This supports the endosymbiotic theory of their origin.

Q2. Give any four differences between prokaryotic and eukaryotic cells.
Model answer: (i) nucleus absent vs present; (ii) circular naked DNA vs linear histone-bound DNA; (iii) no membrane-bound organelles vs organelles present; (iv) 70S vs 80S ribosomes.

Q3. What is the function of the nucleolus?
Model answer: The nucleolus is the site of ribosomal RNA synthesis and of ribosome assembly.

Long-answer questions with model outlines

Q1. Describe the structure of a plant cell with the functions of any six organelles.
Outline: Draw the cell; describe cell wall, plasma membrane, nucleus, mitochondrion, chloroplast, ER, Golgi, ribosome, vacuole; give function of each; mention that plant cells additionally have a rigid cellulose wall and a large central vacuole.

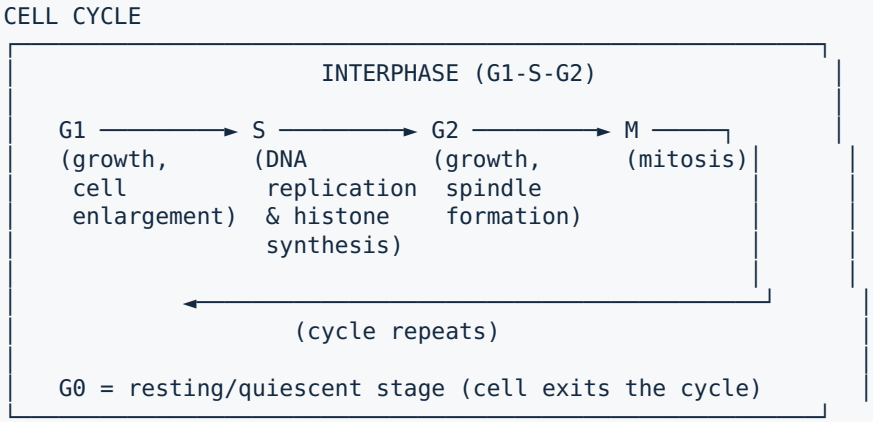
Q2. Compare prokaryotic and eukaryotic cells under at least eight headings.
Outline: Use Table 2.2 headings — size, nucleus, DNA, organelles, ribosomes, wall, division, recombination, respiratory site, flagella — and end with the significance of compartmentalisation in eukaryotes.

Lecture 3. Cell Cycle; Cell Division — Mitosis and Meiosis and Their Significance

> **In one line:** The cell cycle is the ordered sequence of events by which a cell grows and divides. Mitosis produces two genetically identical cells for growth; meiosis produces four genetically different haploid cells for reproduction.

3.1 The cell cycle

The cell cycle is the period from the end of one division to the end of the next. In eukaryotes it has two main phases: **interphase** (the preparatory, non-dividing phase) and the **M phase** (the dividing phase). Interphase occupies about 90–95% of the cycle.



Stages of interphase

Table 3.1 — Events of the cell cycle

Stage	Main events	DNA content
G_1 (first gap)	Cell grows, synthesises RNA and proteins, prepares for DNA synthesis	2C (diploid amount)
S (synthesis)	DNA replication; each chromosome becomes two chromatids; histones synthesised	2C → 4C
G_2 (second gap)	Further growth, spindle proteins made, repair checked	4C
M (mitosis)	Nuclear and cell division	4C → 2C per daughter cell
G_0	Quiescent stage; cell leaves the cycle (common in mature tissues)	2C

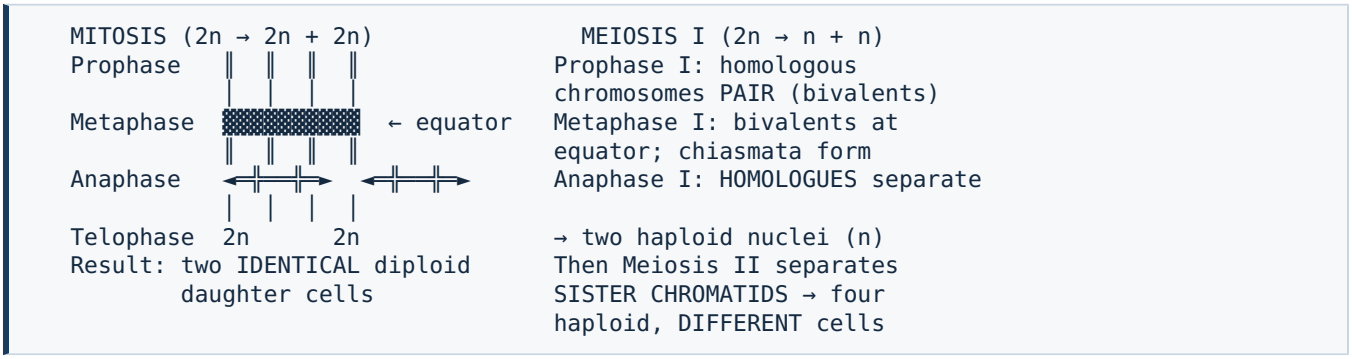
3.2 Mitosis

Mitosis is the division of a nucleus in which the daughter nuclei receive the **same number and kind of chromosomes** as the parent. It is also called **equational division**. It occurs in somatic cells and in meristems of plants.

Stages of mitosis (Karyokinesis)

Table 3.2 — The four stages of mitosis

Stage	What happens
Prophase	Chromosomes condense and become visible; nucleolus and nuclear membrane disappear; spindle fibres begin to form
Metaphase	Chromosomes line up at the equator; spindle fibres attach to centromeres at the kinetochore
Anaphase	Centromeres split; sister chromatids move to opposite poles (shortest stage)
Telophase	Chromosomes decondense; nuclear membrane and nucleolus reappear; spindle disappears



Cytokinesis

Cytokinesis is the division of the cytoplasm. In plants it occurs by formation of a **cell plate** (phragmoplast) that grows outward and becomes the middle lamella; in animals it occurs by a cleavage furrow.

3.3 Meiosis

Meiosis is the special division that occurs in germ cells to form gametes. It has **two successive divisions (Meiosis I and Meiosis II) with only one DNA replication**, so a diploid (2n) cell gives four haploid (n) cells. Meiosis I is **reductional** (halves the chromosome number); Meiosis II is **equational**.

Prophase I — the most important stage

Table 3.3 — Sub-stages of prophase I

Sub-stage	Event
Leptotene	Chromosomes become visible as thin threads; homologous chromosomes start to approach
Zygotene	Synapsis: homologous chromosomes pair to form bivalents; synaptonemal complex forms
Pachytene	Chromosomes thicken; crossing over occurs between non-sister chromatids; tetrads visible
Diplotene	Synaptonemal complex dissolves; homologues separate but remain joined at chiasmata
Diakinesis	Chiasmata terminalise; nuclear membrane breaks; spindle forms

Meiosis I and Meiosis II

- **Metaphase I:** bivalents arrange at the equator; the two homologues face opposite poles.

- **Anaphase I:** homologous chromosomes separate (disjunction) to opposite poles; sister chromatids stay together.
- **Telophase I:** two haploid nuclei form; may be followed by interkinesis (no DNA replication).
- **Meiosis II:** prophase II, metaphase II (chromosomes at equator), anaphase II (centromeres split, chromatids separate), telophase II. Four haploid cells result.

3.4 Comparison of mitosis and meiosis

Table 3.4 — Mitosis versus meiosis

Feature	Mitosis	Meiosis
Occurs in	Somatic cells, meristems	Germ cells / sporogenous tissue
Number of divisions	One	Two (I and II)
DNA replication	Once per division	Once for two divisions
Products	2 cells	4 cells
Chromosome number	Same as parent (2n → 2n)	Halved (2n → n)
Genetic result	Daughter cells identical to parent	Daughter cells genetically different
Homologous pairing	Absent	Present (synapsis)
Crossing over	Absent	Present in pachytene
Chiasmata	Absent	Present
Anaphase separation	Sister chromatids separate	Anaphase I homologues; anaphase II chromatids
Significance	Growth, repair, maintains chromosome number	Gamete formation, variation, evolution

3.5 Significance of mitosis and meiosis

Significance of mitosis

- Maintains the chromosome number constant in all somatic cells of a species.
- Responsible for growth of the plant body (meristematic division).
- Repairs and replaces worn-out cells; heals wounds in woody tissue.
- Keeps genetic constitution identical — basis of vegetative propagation in sugarcane, banana, potato, and of clonal multiplication of elite planting material.
- Basis of asexual reproduction and of tissue culture (micropropagation).

Significance of meiosis

- Produces haploid gametes/spores so that the diploid number is restored at fertilisation (alternation of generations).
- Crossing over and independent assortment generate **new gene combinations** — the chief source of variation.
- Provides the genetic variation on which natural selection and plant breeding depend.
- Maintains the constancy of chromosome number from generation to generation.

- Abnormal meiosis leads to aneuploidy and polyploidy, which are used in breeding (for example, triploid banana and triploid sugar-beet).

Worked example — chromosome and DNA content bookkeeping

Problem: A diploid plant has 2n = 24. State the chromosome number and DNA content (in C units) at G_1, metaphase of mitosis, and at the end of meiosis II in each product.

Stage	Chromosomes	Chromatids	DNA (C)
G_1	24	24	2C
Metaphase (mitosis)	24	48	4C
Each daughter cell after mitosis	24	24	2C
Each product of meiosis II	12	12	1C

Answer: G_1 = 24 chromosomes, 2C. Metaphase = 24 chromosomes, 48 chromatids, 4C. Mitosis daughters = 24 chromosomes, 2C each. Meiosis products = 12 chromosomes, 1C each.

Exam focus — Lecture 3

Key points

1. Cell cycle = interphase (G_1, S, G_2) + M phase; interphase is ~90% of the cycle.
2. DNA replicates only in the S phase; chromosome number does not change in S phase.
3. Mitosis is equational (2n → 2n); meiosis is reductional in division I (2n → n).
4. Crossing over occurs in pachytene of prophase I; synapsis in zygotene.
5. Chiasmata become visible in diplotene and terminalise in diakinesis.
6. Mitosis gives genetic constancy; meiosis gives genetic variation.
7. Plant cytokinesis occurs by cell-plate formation; animal cytokinesis by cleavage furrow.

Short-answer questions with model answers

Q1. Why is meiosis called reduction division?
Model answer: Because during meiosis I the homologous chromosomes separate into two daughter nuclei, reducing the chromosome number from diploid (2n) to haploid (n), so the products carry half the parental chromosome number.

Q2. Name the stage in which crossing over occurs and state its cytological marker.
Model answer: Crossing over occurs in pachytene of prophase I. Its cytological markers are the synaptonemal complex of paired homologues and the later appearance of chiasmata in diplotene.

Q3. State two differences between mitosis and meiosis regarding chromosome number and number of products.
Model answer: Mitosis keeps the chromosome number unchanged (2n → 2n) and produces two cells; meiosis halves the chromosome number (2n → n) and produces four cells.

Long-answer questions with model outlines

Q1. Describe meiosis with the help of a diagram and explain its significance.
Outline: Define meiosis; describe prophase I sub-stages, metaphase I, anaphase I, telophase I and meiosis II; draw the two-division scheme; then give significance (gamete formation, variation via crossing over and independent assortment, constancy of number, origin of aneuploidy/polyploidy).

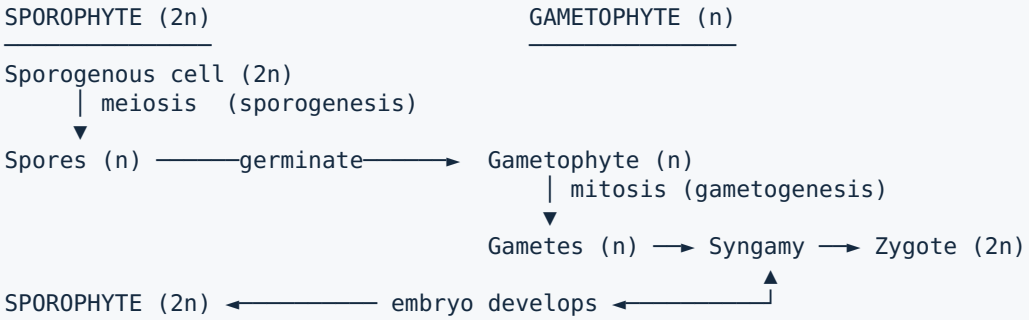
Q2. Compare mitosis and meiosis under at least eight headings.
Outline: Use Table 3.4 headings — site, number of divisions, DNA replication, products, chromosome number, genetic result, pairing, crossing over, chiasmata, anaphase behaviour, significance.

Lecture 4. Sporogenesis, Gametogenesis and Syngamy in Plants; Identical and Fraternal Twins

In one line: Sporogenesis is the formation of spores (in the sporophyte) and gametogenesis the formation of gametes (in the gametophyte); syngamy is the fusion of gametes to form a zygote. Twins are of two kinds: identical (from one zygote) and fraternal (from two zygotes).

4.1 The alternation of generations

In plants there is an alternation between a diploid **sporophyte** generation that produces spores by meiosis, and a haploid **gametophyte** generation that produces gametes by mitosis. Sporogenesis and gametogenesis occur in different generations.

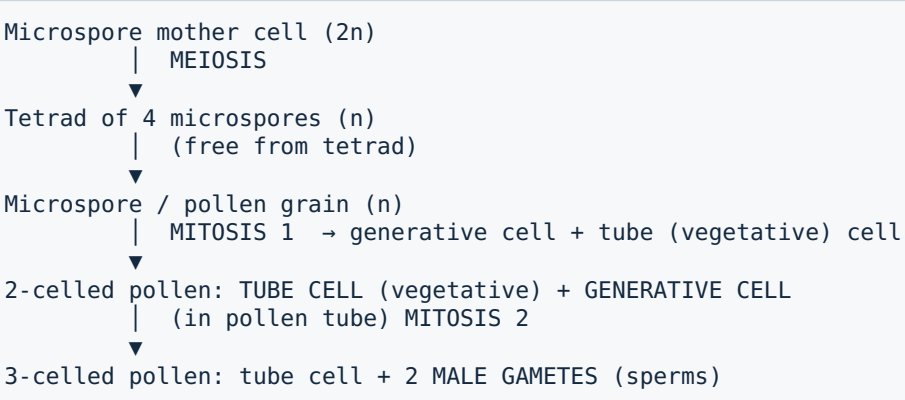


4.2 Sporogenesis

Sporogenesis is the process of formation of spores from the sporogenous tissue by meiosis. Two important types in higher plants:

- **Microsporogenesis** — formation of microspores (pollen grains) inside the **anther**. The diploid microspore mother cell (pollen mother cell) undergoes meiosis to give a tetrad of four haploid microspores.
- **Megasporogenesis** — formation of megaspores inside the **ovule**. The diploid megaspore mother cell undergoes meiosis to give a tetrad of four haploid megaspores; usually only one survives and the other three degenerate.

Microsporogenesis and development of the male gametophyte



Megasporogenesis and development of the female gametophyte (embryo sac)

```
Megaspore mother cell (2n)
         |  MEIOSIS
         ▼
Tetrad of 4 megaspores (n)  →  3 degenerate, 1 survives
         |
         ▼
Functional megaspore (n)
         |  3 successive FREE-NUCLEAR MITOSES (8 nuclei)
         ▼
Embryo sac (Polygonum type, 8 nuclei, 7 cells):
   ┌──────────────────────────────────────┐
   | 3 ANTIPODALS (n)          [chalazal end]  |
   | 2 POLAR NUCLEI → CENTRAL CELL (2n)        |
   | EGG CELL (n)  +  2 SYNERGIDS (n)          |
   |                            [micropylar]   |
   └──────────────────────────────────────┘
```

4.3 Gametogenesis

Gametogenesis is the formation of gametes from the gametophyte by mitosis.

- **Male gametogenesis (microgametogenesis):** the pollen grain germinates on the stigma, the generative cell divides to form **two male gametes (sperms)**, which travel through the pollen tube.
- **Female gametogenesis (megagametogenesis):** the embryo sac develops; the **egg cell** is the female gamete and the two polar nuclei fuse to form the diploid secondary nucleus.

4.4 Syngamy, double fertilisation and triple fusion

Syngamy is the fusion of one male gamete with the egg cell to form the diploid **zygote**, which develops into the embryo. In flowering plants a second male gamete fuses with the two polar nuclei (secondary nucleus) — this is **triple fusion**, forming the triploid (3n) **primary endosperm nucleus**. Together the two fusions are called **double fertilisation**, a unique feature of angiosperms.

```
Pollen tube enters embryo sac
     |
     ├── Male gamete 1  +  Egg (n)  ──► ZYGOTE (2n) ──► EMBRYO
     |         (syngamy)
     └── Male gamete 2  +  2 polar nuclei (n+n) ──► PRIMARY ENDOSPERM
               (triple fusion)                       NUCLEUS (3n) ──► ENDOSPERM
```

Fertilisation in plants — sequence of events

1. Pollination — transfer of pollen to the stigma (self or cross).
2. Pollen germination — pollen tube grows through the style.
3. Entry of pollen tube into the embryo sac (usually through a synergid).
4. Discharge of two male gametes.
5. Syngamy (gamete + egg) and triple fusion (gamete + polar nuclei) = double fertilisation.
6. Zygote develops into the embryo; primary endosperm nucleus develops into endosperm.

4.5 Identical and fraternal twins

Twins are two offspring produced in the same pregnancy. In humans they are of two genetic types.

Table 4.1 — Identical versus fraternal twins

Feature	Identical (monozygotic) twins	Fraternal (dizygotic) twins
Origin	One fertilised egg (one zygote) splits into two embryos	Two eggs fertilised by two sperms (two zygotes)
Genetic constitution	Genetically identical	No more alike than ordinary siblings
Sex	Always the same sex	May be same or different sex
Placenta / membranes	May share one placenta and membranes	Always separate placentas and membranes
Resemblance	Very close (same blood group, fingerprints nearly alike)	Ordinary resemblance
Cause	Random splitting of the embryo; not hereditary	Release of two eggs (double ovulation); tends to run in families and increases with maternal age
Frequency	About one-third of all twins	About two-thirds of all twins

In plants, "identical twins" arise by vegetative propagation or apomixis — for example, sugarcane setts and banana suckers give genetically identical plants (clones), which are the plant equivalent of monozygotic twins. Polyembryony (more than one embryo in a seed, as in citrus and mango) also gives identical seedlings when the embryos are of nucellar origin.

Worked example — chromosome numbers through the plant life cycle

Problem: A crop has 2n = 20. Give the ploidy and chromosome number of: (a) pollen grain, (b) egg cell, (c) zygote, (d) endosperm, (e) nucellus, (f) antipodal cell.

Structure	Ploidy	Chromosome number	Formed by
Pollen grain (microspore)	n	10	Meiosis (microsporogenesis)
Egg cell	n	10	Mitosis in embryo sac
Zygote	2n	20	Syngamy
Endosperm	3n	30	Triple fusion
Nucellus	2n	20	Parental somatic tissue
Antipodal cell	n	10	Mitosis in embryo sac

Exam focus — Lecture 4

Key points

1. Sporogenesis (meiosis) gives spores; gametogenesis (mitosis) gives gametes.
2. Microspore mother cell → tetrad of microspores → pollen grain (2-celled → 3-celled).
3. Megaspore mother cell → tetrad → one functional megaspore → 8-nucleate embryo sac (Polygonum type).
4. Double fertilisation = syngamy (2n zygote) + triple fusion (3n endosperm).
5. Endosperm in angiosperms is triploid (3n).
6. Identical twins are monozygotic, same sex, genetically identical; fraternal twins are dizygotic.
7. In plants, clones (vegetative propagation) and nucellar polyembryony are the equivalent of identical twins.

Short-answer questions with model answers

Q1. What is double fertilisation? Why is it unique to angiosperms?
Model answer: Double fertilisation is the occurrence of two fusions in the same embryo sac — syngamy (male gamete + egg → 2n zygote) and triple fusion (male gamete + two polar nuclei → 3n primary endosperm nucleus). It is unique to angiosperms and results in a nutritive triploid endosperm alongside the embryo.

Q2. Distinguish identical from fraternal twins.
Model answer: Identical twins develop from one zygote that splits, are genetically identical and always of the same sex. Fraternal twins develop from two separate zygotes (two eggs, two sperms), are no more alike than siblings, and may be of different sexes.

Q3. Give the ploidy of the zygote and endosperm of a crop with 2n = 24.
Model answer: Zygote is 2n = 24; endosperm is 3n = 36.

Long-answer questions with model outlines

Q1. Describe microsporogenesis and the development of the male gametophyte.
Outline: Start from the microspore mother cell in the anther; meiosis to a tetrad; release of microspores; pollen mitosis 1 (tube + generative cell); mitosis 2 in the tube (two sperms); draw the sequence and state ploidy at each step.

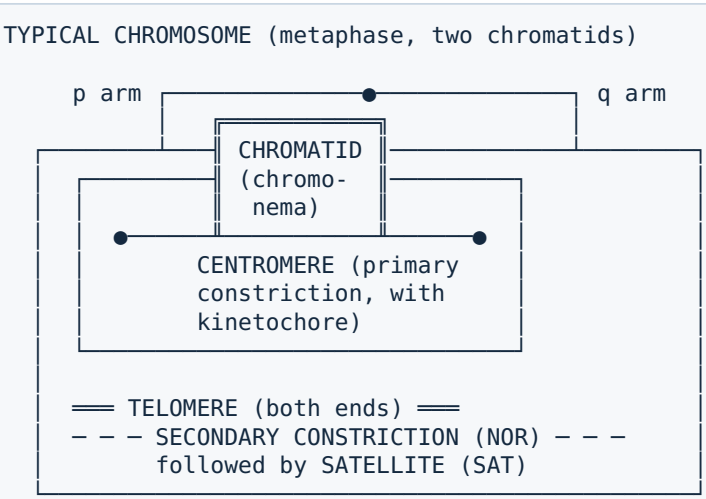
Q2. Explain sporogenesis, gametogenesis and syngamy in plants with a flow diagram.
Outline: Define each term; show alternation of generations; describe micro- and megasporogenesis; describe male and female gametogenesis; describe syngamy and double fertilisation; end with the ploidy of zygote (2n) and endosperm (3n).

Lecture 5. Architecture of Chromosomes

In one line: A chromosome is a thread-like structure of DNA and protein that carries genes. Its essential parts are chromatids (chromonemata), centromere, telomeres and, in some cases, secondary constrictions and satellites.

5.1 Chromosome — definition and morphology

A chromosome is a self-replicating, thread-like structure present in the nucleus, composed mainly of DNA and proteins (histones and non-histones), and carrying the genes in a linear order. The word was coined by Waldeyer (1888). The number of chromosomes is constant for a species: garden pea 2n = 14, rice 2n = 24, maize 2n = 20, tomato 2n = 24, onion 2n = 16, wheat 2n = 42, human 2n = 46.



5.2 Chemical composition of chromosomes

Table 5.1 — Chemical composition of a chromosome

Component	Approximate share	Role
DNA	~30–40%	Carries the genetic information
Histones (basic proteins)	~30–40%	Package DNA into nucleosomes; regulate transcription
Non-histone proteins	~5–15%	Scaffold, enzymes of replication and transcription
RNA	~5–10%	Transcripts in the process of synthesis
Metals (Ca, Mg)	Trace	Structural stability

The five types of histone are H1, H2A, H2B, H3 and H4. Two each of H2A, H2B, H3 and H4 form the core of the nucleosome; H1 binds the linker DNA and packs nucleosomes into higher-order structure.

5.3 Chromonemata

Each chromatid contains one or more fine threads called **chromonemata** (singular chromonema). The chromonema is the actual gene-bearing thread; the surrounding material is the matrix. Chromomeres are the beaded granules on the chromonema; they are characteristic of the pachytene stage and help in the identification of chromosomes.

5.4 Chromosome matrix

The matrix (chromosome matrix or calymma) is the **non-genetic, achromatic material** surrounding the chromonema. It is made mainly of proteins and is not stained by basic dyes that stain the chromatin. It gives mechanical support and helps maintain chromosome shape.

5.5 Nucleosome — the structural unit of chromatin

A nucleosome is the basic packing unit of chromatin. It consists of about **146 base pairs of DNA wound around an octamer of eight histone proteins** (two each of H2A, H2B, H3, H4), forming a "beads-on-a-string" structure. The linker DNA between beads (about 20–60 bp) is bound by histone H1.

```
NUCLEOSOME ("beads on a string")

linker DNA      core particle (H2A,H2B,H3,H4 ×2)
~~~●~~~~~~~~~~~~[═══════════]~~~~~~~~●~~~~~~~~~~[═══════════]~~~
   │            146 bp DNA        │            146 bp DNA
H1 histone      wrapped 1.75     H1
                turns around
                histone octamer
```

5.6 Centromere

The centromere is the **primary constriction**, a specialised region at which the two sister chromatids are held together and to which the spindle fibres attach. The disc-shaped protein structure at the centromere that binds the spindle is the **kinetochore**. The centromere has no genes for ordinary characters but controls chromosome movement in division.

- It is the point of attachment of spindle fibres.
- It determines the shape of the chromosome (metacentric, sub-metacentric, acrocentric, telocentric).
- Each chromosome has one centromere (monocentric); some have two (dicentric) or none (acentric).

5.7 Telomere

The telomere is the **end region of a chromosome**. It has two functions: it **seals the ends** so that chromosomes do not fuse with one another or with broken ends, and it allows the end to be replicated fully (prevents shortening). Telomeres consist of short, repeated DNA sequences (in humans, TTAGGG repeats) with associated proteins.

5.8 Euchromatin and heterochromatin

Table 5.2 — Euchromatin versus heterochromatin

Feature	Euchromatin	Heterochromatin
Staining	Lightly stained (less condensed)	Deeply stained (highly condensed)
Gene content	Rich in genes	Few genes; mostly repetitive DNA
Activity	Genetically active (transcription occurs)	Genetically inactive
Replication	Replicates early in S phase	Replicates late in S phase
Occurrence	Mostly in the arms	Near centromere and telomeres
Types	—	Constitutive (permanent) and facultative (e.g. Barr body)

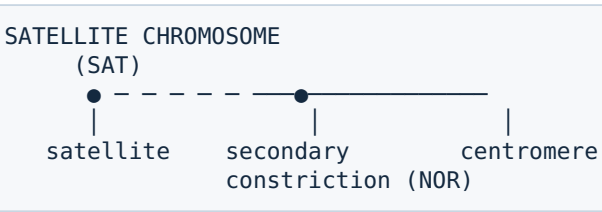
Constitutive heterochromatin is permanently condensed in all cells (for example, the region near the centromere). **Facultative heterochromatin** is condensed in only some cells or at some times; the inactive X chromosome (Barr body) in mammalian females is the classic example.

5.9 Nucleolar Organiser Region (NOR)

The nucleolar organiser region is a chromosomal segment carrying the genes for **ribosomal RNA**. At this region the nucleolus is formed during interphase. It usually appears as a **secondary constriction**. In a diploid cell there are normally two NORs (one per homologue), each organising a nucleolus.

5.10 Secondary constriction and satellite chromosome

- **Secondary constriction:** a constriction other than the centromere, usually associated with the NOR. It is a gap in the chromosome where the nucleolus forms.
- **Satellite (SAT):** a small, rounded terminal segment of a chromosome separated from the main body by a secondary constriction.
- **Satellite chromosome (SAT chromosome):** a chromosome that bears a satellite. SAT chromosomes carry the NOR and take part in nucleolus formation.



5.11 Karyotype and ideogram

- **Karyotype:** the complete set of chromosomes of a cell or an organism, arranged in a standard order (pairs, in decreasing order of length) showing their number, size, shape and centromere position. Karyotyping is done on a metaphase spread.
- **Ideogram (idiogram):** a diagrammatic, standardised representation of a karyotype — the chromosomes are drawn to scale with their centromere positions, relative lengths and banding patterns marked. It is the "map picture" of the chromosome complement.
- **Karyotyping is used** to detect chromosome abnormalities (aneuploidy, translocation, deletion), to identify species and to study evolutionary relationships.

Table 5.3 — Chromosome numbers of some common crops and organisms

Organism	2n	Organism	2n
Garden pea (*Pisum sativum*)	14	Rice (*Oryza sativa*)	24
Maize (*Zea mays*)	20	Tomato (*Solanum lycopersicum*)	24
Onion (*Allium cepa*)	16	Wheat (*Triticum aestivum*)	42
Groundnut (*Arachis hypogaea*)	40	Cotton (*Gossypium hirsutum*)	52
Drosophila melanogaster	8	Human (*Homo sapiens*)	46
Arabidopsis thaliana	10	*E. coli* (single chromosome)	1

Worked example — reading a karyotype

Problem: A human karyotype shows 45 chromosomes with a single X (44 + X). Identify the condition and state the chromosome number expected normally.
Solution: Normal human diploid number is 46 (44 autosomes + XX or XY). A complement of 44 autosomes + a single X = 45 chromosomes indicates **Turner syndrome** (monosomy of the sex chromosome, 2n = 45, X). It is an aneuploid condition (details in Lecture 8).

Exam focus — Lecture 5

Key points

1. A chromosome is made of DNA + histones + non-histone proteins; DNA ~35%, protein ~65%.
2. Nucleosome = 146 bp DNA around an octamer of 8 histones (H2A, H2B, H3, H4 ×2); H1 binds linker DNA.
3. Centromere = primary constriction, site of spindle attachment via the kinetochore.
4. Telomere seals chromosome ends and prevents end-to-end fusion and shortening.
5. Euchromatin is loosely packed and active; heterochromatin is condensed and inactive.
6. NOR carries rRNA genes and forms the nucleolus at a secondary constriction.
7. A chromosome with a satellite is a SAT (satellite) chromosome.
8. Karyotype = ordered set of chromosomes; ideogram = its diagrammatic map.

Short-answer questions with model answers

Q1. What is a nucleosome? Give its composition.
Model answer: A nucleosome is the basic packing unit of chromatin, made of about 146 base pairs of DNA wound about 1.75 turns around a core octamer of eight histone molecules (two each of H2A, H2B, H3 and H4). Histone H1 binds the linker DNA between nucleosomes.

Q2. Differentiate euchromatin from heterochromatin.
Model answer: Euchromatin is loosely packed, lightly stained, gene-rich, transcriptionally active and replicates early. Heterochromatin is tightly packed, deeply stained, gene-poor, transcriptionally inactive and replicates late; it may be constitutive or facultative (e.g. Barr body).

Q3. What is a satellite chromosome?
Model answer: A satellite chromosome is one that carries a small terminal segment (satellite) separated from the rest of the chromosome by a secondary constriction, which is usually the nucleolar organiser region.

Long-answer questions with model outlines

Q1. Describe the structure of a chromosome with a labelled diagram.
Outline: Define chromosome; describe chromatids and chromonemata, chromomeres, matrix, centromere and kinetochore, telomeres, secondary constriction, NOR and satellite; state chemical composition; draw a labelled metaphase chromosome.

Q2. Explain euchromatin, heterochromatin and the nucleosome model of chromatin packaging.
Outline: Define both chromatin types and give examples; describe the nucleosome and "beads on a string"; describe higher-order coiling (solenoid, loops) briefly; relate packaging to gene activity.

Lecture 6. Types of Chromosomes; Special Types of Chromosomes

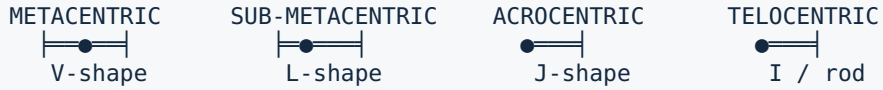
In one line: Chromosomes are classified by the position of the centromere and by the number of centromeres. Special chromosomes — polytene, lampbrush, B, ring and isochromosomes — have unusual structure and behaviour.

6.1 Classification by position of the centromere

The position of the centromere fixes the ratio of the two arms (arm ratio) and hence the shape seen at anaphase.

Table 6.1 — Types of chromosomes by centromere position

Type	Centromere position	Arm ratio (long:short)	Shape at anaphase
Metacentric	At the middle	1 : 1	V-shaped
Sub-metacentric	Slightly off-centre	~2 : 1	L-shaped
Acrocentric	Near one end	~4 : 1 (very unequal)	J-shaped
Telocentric	At the terminal end	No short arm	I-shaped (rod)



6.2 Classification by number of centromeres

Table 6.2 — Types of chromosomes by number of centromeres

Type	Number of centromeres	Behaviour
Monocentric	One	Normal; moves as one unit to one pole
Dicentric	Two	Forms a bridge at anaphase (both centromeres pulled to opposite poles); usually breaks
Polycentric	Many	Seen in some plants; behaves as a diffuse centromere
Acentric	None	Cannot attach to the spindle; lost during division

6.3 Structure and functions of chromosomes (summary)

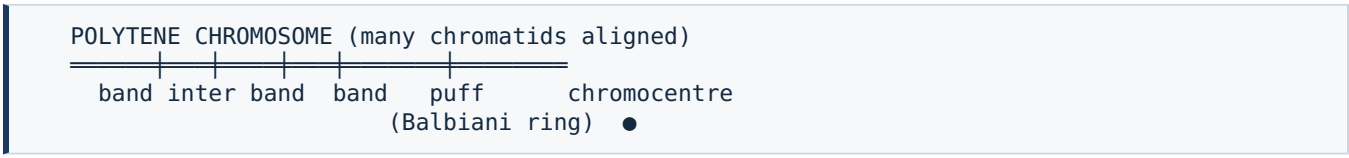
- **Structure:** chromatid/chromonema threads, centromere, telomeres, NOR, satellite; DNA packaged with histones into nucleosomes and higher-order coils.
- **Functions:** (i) carry genes and transmit hereditary information; (ii) maintain the species chromosome number through division; (iii) control cell metabolism and development through gene expression; (iv) generate variation through crossing over, recombination and mutation; (v) take part in sex determination in many organisms.

6.4 Special types of chromosomes

(a) Polytene (salivary gland) chromosomes

Polytene chromosomes are giant, multi-stranded chromosomes formed when **DNA replicates repeatedly without cell or nuclear division** (endomitosis). The many chromatids stay side by side, giving a cable-like structure. They occur in the salivary glands of *Drosophila* larvae and in some plant cells.

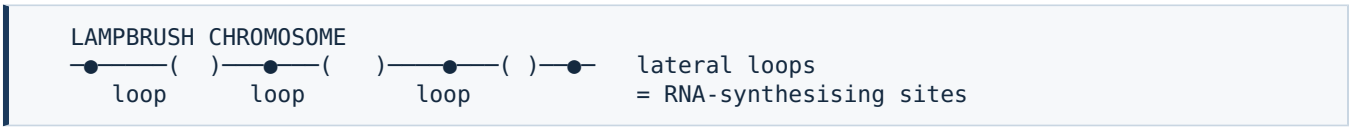
- Show characteristic **bands and interbands**; bands correspond to genes and are used in gene mapping.
- Show **Balbiani rings** and **puffs** — sites of intense RNA synthesis (gene activity).
- The **chromocentre** is the point at which all polytene chromosomes are joined (their centromeres fuse).
- They have no normal pairing and no crossing over.



(b) Lampbrush chromosomes

Lampbrush chromosomes are large, paired chromosomes found in the **oocytes (diplotene stage) of amphibians and some birds**, and also in some plant meiotic cells. They have a central axis with numerous lateral loops that give a brush-like appearance.

- The loops are sites of intense **RNA synthesis**; they provide the large amount of mRNA needed for the growing egg.
- They are a modified form of the bivalent (paired homologues) of prophase I.



(c) B chromosomes (supernumerary)

B chromosomes are **extra chromosomes over and above the normal (A) complement**. They are not essential for the organism, are usually smaller, and are often heterochromatic.

- They do not pair with A chromosomes; they pair among themselves.
- They are found in maize, rye, and several grasses; they affect vigour and fertility when present in many copies.
- Their number varies between individuals and even between cells (they are unstable).

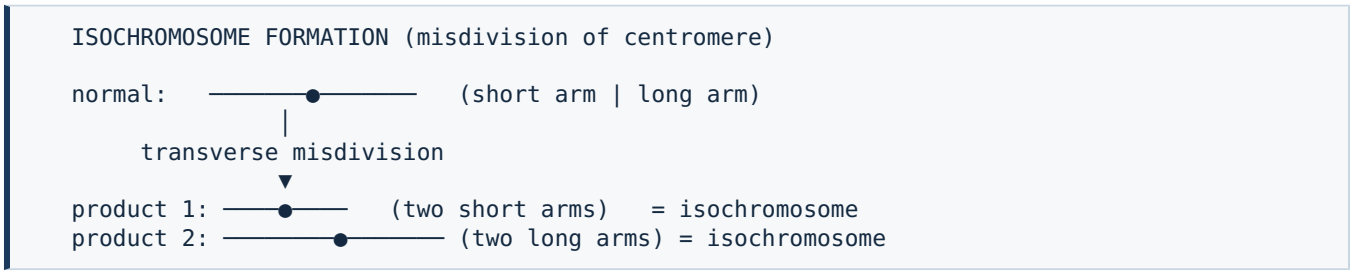
(d) Ring chromosomes

A ring chromosome forms when a chromosome **loses both telomeres** and its two broken ends fuse, making a closed ring. Ring chromosomes occur after deletion and during some translocations. They are unstable at anaphase and are often lost, so cells carrying them show mosaicism.

(e) Isochromosomes

An isochromosome is an abnormal chromosome in which the two arms are **mirror images of each other**. It arises by **misdivision of the centromere** in a transverse (instead of longitudinal) plane, so that one daughter chromosome gets two identical arms (either two

long or two short arms) and the other is lost. An isochromosome therefore carries a duplication of one arm and a deficiency of the other.



Exam focus — Lecture 6

Key points

1. By centromere position: metacentric (middle, V), sub-metacentric (L), acrocentric (J), telocentric (rod).
2. By number of centromeres: monocentric, dicentric (forms bridge), polycentric, acentric (lost).
3. Polytene chromosomes arise by endomitosis; occur in *Drosophila* salivary glands; show bands and Balbiani rings.
4. Lampbrush chromosomes occur in oocytes at diplotene; loops synthesise RNA.
5. B chromosomes are supernumerary, non-essential and unstable.
6. Ring chromosomes form by fusion of two broken ends after telomere loss.
7. Isochromosomes arise by transverse misdivision of the centromere, giving two identical arms.

Short-answer questions with model answers

Q1. How are chromosomes classified by centromere position?
Model answer: Metacentric (centromere in the middle, V-shaped), sub-metacentric (slightly off-centre, L-shaped), acrocentric (near one end, J-shaped) and telocentric (at the end, rod-shaped).

Q2. What are polytene chromosomes and where do they occur?
Model answer: Polytene chromosomes are giant chromosomes formed by repeated DNA replication without nuclear division (endomitosis), in which the daughter chromatids remain aligned. They occur in the salivary glands of *Drosophila* larvae and in some plant cells, and show bands, puffs and Balbiani rings.

Q3. How does an isochromosome arise?
Model answer: By misdivision of the centromere in a transverse plane instead of a longitudinal one, so that a chromosome receives two identical arms (both long or both short), producing a duplication of one arm and a deficiency of the other.

Long-answer questions with model outlines

Q1. Describe the special types of chromosomes with diagrams.
Outline: Define and describe polytene, lampbrush, B, ring and isochromosomes; for each give origin, occurrence, structure and significance; draw simple diagrams.

Q2. Classify chromosomes by centromere position and number, and give their functions.
Outline: Use Tables 6.1 and 6.2; draw the four shapes; then list the general functions of chromosomes (carrying genes, transmission, variation, sex determination).

Lecture 7. Chromosomal Aberration — Structural Changes (Deletion, Duplication, Inversion, Translocation)

> **In one line:** Structural aberrations change the arrangement or the amount of chromosomal material without changing the chromosome number; the four main types are deletion, duplication, inversion and translocation.

7.1 What are chromosomal aberrations?

Chromosomal aberrations (chromosomal mutations) are changes in the **number or structure** of chromosomes. They are of two broad classes:

- **Structural aberrations** — change in the arrangement of genes within chromosomes (deletion, duplication, inversion, translocation).
- **Numerical aberrations** — change in the number of chromosomes (aneuploidy, euploidy) — covered in Lectures 8 and 9.

Structural changes arise when a chromosome breaks and the broken ends reunite abnormally. Breaks may be spontaneous or induced by X-rays, gamma rays and chemicals. A single break can cause deletion; two breaks on one chromosome can cause inversion; breaks on two chromosomes can cause translocation.

7.2 Deletion (deficiency)

A **deletion** is the loss of a segment of a chromosome, so that the genes in that segment are missing. It requires one break (terminal deletion) or two breaks (interstitial deletion).

- **Terminal deletion:** a single break near the end; the segment without a centromere (acentric) is lost.
- **Interstitial deletion:** two breaks; the middle segment is lost and the two outer ends rejoin.

Effects: a deletion makes the chromosome shorter; it produces a **deficiency loop** at pachytene when paired with a normal homologue; it removes the genes in the deleted segment, so it is usually **harmful** and often lethal in the homozygous condition. A well-known example is the deletion of the short arm of chromosome 5 in humans causing **cri-du-chat syndrome**.

```
DELETION
Normal:  A B C D E F G H
Terminal deletion (loss of G H):     A B C D E F       ← shorter chromosome
Interstitial deletion (loss of D E): A B C   F G H     ← loop at pachytene
```

7.3 Duplication

A **duplication** is the presence of a chromosomal segment **more than once** in the same chromosome or in its homologue. It is the opposite of deletion and usually arises from unequal crossing over or from an aberration that adds a segment.

- **Tandem duplication:** the repeated segment lies immediately next to the original (A B C B C D).
- **Reverse tandem duplication:** the repeat is in reverse order.

- **Displaced (non-tandem) duplication:** the repeat lies at a distance from the original.
- **Transposed duplication:** the repeat is moved to another chromosome.

Effects: the chromosome becomes longer; a **duplication loop** forms at pachytene. Duplications are usually less harmful than deletions and are an important source of **new genes and evolution** — the extra copy can mutate freely while the original continues to work. The classic example is the Bar-eye duplication in *Drosophila*, which reduces eye size.

7.4 Inversion

An **inversion** is a rearrangement in which a segment of a chromosome is rotated by 180° and reinserted, so the order of genes in that segment is reversed. It requires two breaks. Inversions do not change the amount of genetic material, so they are usually not harmful to the individual, but they change gene order and suppress crossing over inside the inverted segment.

- **Paracentric inversion:** both breaks are in the **same arm**, so the inversion does not include the centromere.
- **Pericentric inversion:** the breaks are in **opposite arms**, so the inversion includes the centromere and may change the arm ratio and hence the chromosome shape.

Effects: pairing at pachytene produces an **inversion loop**. A single crossover inside a paracentric inversion gives a **dicentric bridge** and an acentric fragment (the fragment is lost, gametes are inviable); a single crossover inside a pericentric inversion gives **duplication-deficiency** chromatids. Inversions therefore act as **cross-over suppressors** and hold favourable gene blocks together.

```
INVERSION
Normal:  A B C D E F G H
Paracentric (same arm): A B E D C F G H     (C D E reversed)
Pericentric (both arms): A B E D C F G H with centromere inside the loop
At pachytene an INVERSION LOOP is formed:
     ——A B C D E F G H——
     ——A B E D C F G H——
            └——loop——┘
```

7.5 Translocation

A **translocation** is the transfer of a segment of one chromosome to a **non-homologous chromosome**. It requires breaks in two chromosomes.

- **Simple (non-reciprocal) translocation:** a segment moves from one chromosome to another without exchange.
- **Reciprocal translocation:** two non-homologous chromosomes exchange segments (the common type).
- **Intercalary (shift) translocation:** a segment is inserted into the middle of another chromosome.

Effects: a reciprocal translocation does not change gene number, so the individual is normal but forms a **cross-shaped (quadrivalent) configuration** at pachytene. Segregation from this configuration gives duplication-deficiency gametes, so fertility is reduced. Translocation homozygotes are normal and are used to link characters of different linkage groups; translocations also help in locating genes and in the **evolution of chromosome numbers** (for example, the reduction from 2n = 14 in *Pisum* to lower numbers in related species is explained by translocations).

7.6 Comparison of the four structural aberrations

Table 7.1 — Structural aberrations at a glance

Type	Change	Breaks needed	Pachytene figure	Effect
Deletion	Loss of a segment	1 (terminal) or 2 (interstitial)	Deficiency loop	Harmful, often lethal; loss of genes
Duplication	Segment repeated	2 (or unequal crossing over)	Duplication loop	Mildly harmful; source of new genes
Inversion	Segment reversed	2	Inversion loop	No gene loss; suppresses crossing over
Translocation	Segment moved to non-homologue	2	Cross (quadrivalent)	Reduced fertility; used in mapping

7.7 Genetic and cytological implications

- **Genetic implications:** deletions remove genes (pseudo-dominance of the recessive allele on the normal homologue); duplications add genes and can change phenotype dosage; inversions suppress recombination and keep gene complexes intact; translocations link genes of different chromosomes and reduce fertility.
- **Cytological implications:** all four produce characteristic pairing figures at pachytene (loop or cross), and inversions and translocations produce bridges, fragments and abnormal segregation at anaphase.
- **Agricultural implications:** translocations are used to transfer a desired segment (for example, the rye segment carrying disease resistance into wheat, as in Triticale and some wheat varieties); induced inversions help in building "balanced lethal" systems used in hybrid seed production.

Worked example — gene order after inversion

Problem: A chromosome has gene order A B C D E F G. An inversion reverses the segment C D E. Write the new gene order and state the type if the centromere lies between C and D.

Solution: Reversing C D E gives E D C, so the new order is **A B E D C F G**. Since the centromere (between C and D) lies inside the reversed segment, the inversion is **pericentric**.

Exam focus — Lecture 7

Key points

1. Structural aberrations change gene arrangement, not chromosome number.
2. Deletion = loss of a segment; deficiency loop at pachytene; usually harmful.
3. Duplication = extra segment; duplication loop; source of new genes in evolution.
4. Inversion = segment reversed; paracentric (same arm) or pericentric (both arms).
5. Paracentric inversion with a crossover gives a dicentric bridge and an acentric fragment.
6. Translocation = segment moved to a non-homologous chromosome; cross-shaped quadrivalent at pachytene.
7. Inversions suppress crossing over; translocations reduce fertility and help in mapping.

Short-answer questions with model answers

Q1. Differentiate deletion from duplication.
Model answer: A deletion is the loss of a chromosomal segment (genes are missing), producing a deficiency loop at pachytene and usually being harmful. A duplication is the repetition of a segment (genes are extra), producing a duplication loop, being less harmful and serving as a source of new genes.

Q2. What is the difference between paracentric and pericentric inversion?
Model answer: In paracentric inversion both breaks lie in the same arm and the centromere is outside the inverted segment; in pericentric inversion the breaks are in opposite arms and the centromere lies inside the inverted segment, which may change the arm ratio of the chromosome.

Q3. Why do inversions suppress crossing over?
Model answer: A crossover inside an inverted segment produces abnormal chromatids — a dicentric bridge and an acentric fragment in paracentric inversions, and duplication-deficiency chromatids in pericentric inversions — which are usually inviable, so the recombinant products are not recovered and the inversion appears to suppress recombination.

Long-answer questions with model outlines

Q1. Describe the four types of structural chromosomal aberrations with their genetic and cytological effects.
Outline: Define each of deletion, duplication, inversion and translocation; give breaks required; draw pairing figures; give effects on the individual and on fertility; end with their significance in plant breeding.

Q2. Explain how structural aberrations are detected at meiosis and why they matter in agriculture.
Outline: Pairing figures (loops, cross) and anaphase figures (bridges, fragments); sterility and duplication-deficiency gametes; uses in transferring genes, linking characters and producing hybrid seed.

Lecture 8. Chromosomal Aberration — Numerical Changes: Aneuploidy, Klinefelter and Turner Syndromes

> **In one line:** Aneuploidy is the gain or loss of one or a few chromosomes; it arises mainly by non-disjunction at meiosis. Klinefelter (47, XXY) and Turner (45, X) syndromes are human sex-chromosome aneuploids.

8.1 Numerical aberrations — an overview

Numerical aberrations change the number of chromosomes. They are of two classes:

- **Aneuploidy:** the chromosome number is not an exact multiple of the haploid number — one or a few chromosomes are added or lost (for example, 2n + 1, 2n − 1).
- **Euploidy:** the chromosome number is an exact multiple of the haploid number (haploid, diploid, triploid, tetraploid) — treated in Lecture 9.

8.2 Aneuploidy

An aneuploid is an organism whose chromosome number is not an exact multiple of the basic (haploid) number. The main types are given below.

Table 8.1 — Types of aneuploids

Type	Formula	Meaning	Remarks
Nullisomic	2n − 2	A pair of homologous chromosomes missing	Usually lethal in diploids; used in wheat analysis
Monosomic	2n − 1	One chromosome missing	Viable in polyploids; used to locate genes to chromosomes
Trisomic	2n + 1	One extra chromosome	Example: Down syndrome in humans; used in mapping
Tetrasomic	2n + 2	One pair duplicated (four copies of one chromosome)	Viable in polyploids
Double trisomic	2n + 1 + 1	Two different extra chromosomes	Two independent trisomics

8.3 Origin of aneuploids

Aneuploids arise mainly by **non-disjunction** — the failure of homologous chromosomes (anaphase I) or sister chromatids (anaphase II) to separate and go to opposite poles. Both members then pass into one gamete (n + 1) while the other gamete gets none (n − 1). Other causes are:

- **Non-disjunction** at meiosis I or meiosis II — the principal cause.
- **Anaphase lag** — a chromosome or chromatid fails to attach to the spindle and is left behind.
- **Failure of pairing (asynapsis/desynapsis)** — unpaired chromosomes are distributed at random.
- **Structural changes** such as translocation or inversion causing abnormal segregation.

- **Polyploidy** — aneuploids are more easily tolerated in polyploids (for example, in wheat and sugarcane) because of gene duplication.

```
NON-DISJUNCTION AT MEIOSIS I
Normal (disjunction)          Non-disjunction
(n)  ←— A   a —→              (n+1) ←— A a —→  (n-1) none
gametes: n and n              gametes: n+1 and n-1
     |                              |
     ▼                              ▼
zygote 2n                    zygote 2n+1 (trisomic)
                             zygote 2n-1 (monosomic)
```

8.4 Effects of aneuploidy

- Aneuploidy disturbs the gene balance, so it usually reduces vigour and fertility; most aneuploids in diploid crops are weak or sterile.
- Aneuploids are used in **gene mapping**: a gene is located on a particular chromosome by studying how its phenotype changes in trisomics, monosomics and nullisomics. This is how the chromosomes of wheat were mapped (Sears' monosomic analysis).
- Aneuploidy in crops causes the production of aneuploid gametes and hence partial sterility, which is a problem in seed crops but is exploited in sugar-beet (triploids) and some ornamentals.

8.5 Human sex-chromosome aneuploids

Klinefelter syndrome (47, XXY)

Table 8.2 — Klinefelter syndrome

Feature	Description
Karyotype	47, XXY (44 autosomes + XXY) — an extra X chromosome
Sex	Male (phenotypically male because of the Y chromosome)
Cause	Non-disjunction of sex chromosomes during meiosis in either parent
Physical features	Tall, long limbs, sparse body hair, small testes (hypogonadism), gynaecomastia (breast development)
Fertility	Usually sterile (azoospermia)
Other	Mild intellectual difficulty in some cases; Barr body present (one extra X)

Turner syndrome (45, X)

Table 8.3 — Turner syndrome

Feature	Description
Karyotype	45, X (44 autosomes + a single X) — one sex chromosome missing
Sex	Female (phenotypically female)
Cause	Non-disjunction of sex chromosomes, usually in the father
Physical features	Short stature, webbed neck (pterygium colli), broad chest, low hairline, undeveloped ovaries (streak gonads)
Fertility	Sterile (no oogenesis)
Other	Normal intelligence usually; no Barr body (only one X)

Comparison: Both are sex-chromosome aneuploids arising by non-disjunction. Klinefelter is a male with an extra X (47, XXY); Turner is a female with a missing sex chromosome (45, X). Klinefelter has a Barr body; Turner has none.

8.6 Implications

- These syndromes show that the **Y chromosome determines maleness** in humans (XXY is male), while the number of X chromosomes affects other characters.
- They demonstrate the **importance of correct chromosome number** for normal development and fertility.
- In agriculture, the same principle explains why aneuploids are weak and sterile and why **balanced chromosome number** is essential in seed crops; it also explains the use of aneuploids as tools in gene mapping.

Worked example — predicting gametes from non-disjunction

Problem: In a plant with 2n = 14, non-disjunction of one chromosome pair occurs at meiosis I. What gametes are formed, and what zygotes result after fertilisation with a normal n = 7 gamete?

Solution: Non-disjunction at meiosis I gives gametes with n + 1 = 8 and n − 1 = 6 chromosomes (the other five pairs segregate normally). Fertilisation with a normal gamete (n = 7) gives:

Gamete	+ normal n=7	Zygote	Type
n+1 = 8	8 + 7	15 = 2n + 1	Trisomic
n−1 = 6	6 + 7	13 = 2n − 1	Monosomic

Exam focus — Lecture 8

Key points

1. Aneuploidy = chromosome number not an exact multiple of n; euploidy = exact multiple.
2. Nullisomic 2n−2, monosomic 2n−1, trisomic 2n+1, tetrasomic 2n+2.
3. Chief cause of aneuploidy is non-disjunction at meiosis I or II.
4. Aneuploids are usually weak and sterile; better tolerated in polyploids.
5. Klinefelter = 47, XXY, male, sterile, Barr body present.
6. Turner = 45, X, female, sterile, no Barr body.
7. Aneuploids (monosomics, trisomics) are powerful tools for locating genes on chromosomes.

Short-answer questions with model answers

Q1. Define aneuploidy and name four of its types.
Model answer: Aneuploidy is a condition in which the chromosome number is not an exact multiple of the haploid number, one or a few chromosomes being added or lost. Types: nullisomic (2n−2), monosomic (2n−1), trisomic (2n+1) and tetrasomic (2n+2).

Q2. What is non-disjunction and when does it occur?
Model answer: Non-disjunction is the failure of homologous chromosomes (anaphase I) or sister chromatids (anaphase II) to separate and move to opposite poles. It produces gametes with n+1 and n−1 chromosomes, which on fertilisation give trisomics and monosomics.

Q3. Give the karyotype of Klinefelter and Turner syndromes.
Model answer: Klinefelter syndrome: 47, XXY (male). Turner syndrome: 45, X (female).

Long-answer questions with model outlines

Q1. Explain the origin of aneuploidy and describe its types with examples.
Outline: Define aneuploidy; explain non-disjunction, anaphase lag, asynapsis and structural causes; tabulate nullisomic, monosomic, trisomic, tetrasomic and double trisomic; give examples and uses in mapping.

Q2. Describe Klinefelter and Turner syndromes and their implications.
Outline: Karyotype, cause, features, fertility of each; compare in a table; implications — Y determines maleness, importance of chromosome number, use of aneuploids in genetics.

Lecture 9. Euploidy — Haploids, Dihaploids and Double Haploids; Polyploidy — Auto- and Allopolyploids

> **In one line:** Euploidy is a change in chromosome number by whole sets. A haploid carries one set (n), an autopolyploid has multiple copies of one genome, and an allopolyploid has combined sets from two or more different species.

9.1 Euploidy — the concept of the genome

A **genome** is the complete set of chromosomes of a species, denoted by the basic (monoploid) number x. Euploidy is the condition in which the chromosome number is an exact multiple of x. The chief types:

- **Haploid / monoploid (n = x):** one set.
- **Diploid (2n = 2x):** two sets (normal).
- **Triploid (3n = 3x):** three sets.
- **Tetraploid (4n = 4x):** four sets; higher polyploids similarly.

Table 9.1 — Euploid series and terminology

Type	Sets	Examples / remarks
Haploid (monoploid)	x	Gametophyte; haploids of diploid crops (weak, sterile)
Diploid	2x	Normal somatic cells of most crops
Triploid	3x	Banana (3n=33), triploid sugar-beet, seedless watermelon
Tetraploid	4x	Autotetraploid rye, potato (4x), durum wheat (4x=28)
Pentaploid	5x	Some *Triticum* and *Brassica* forms
Hexaploid	6x	Bread wheat (6x = 42)
Octoploid	8x	Strawberry (8x), some sugarcane forms

9.2 Haploids

A **haploid** is an organism with the gametic chromosome number (n). It is produced by the development of an unfertilised egg (parthenogenesis), by anther/pollen culture, by ovule culture, or rarely by chromosome elimination.

- **Characters:** haploids are weak and small, have low vigour, are usually sterile (no pairing of homologues at meiosis) and show no dominance effects (all genes are present in a single dose).
- **Uses:** haploids are used to study gene action without dominance, to detect mutations easily (a recessive mutation shows at once), and to produce **homozygous diploids** by chromosome doubling.
- **Examples:** haploids have been produced in rice, wheat, maize, barley, tobacco, tomato and several other crops mainly by anther culture.

9.3 Dihaploids and double haploids

- **Dihaploid:** a diploid obtained by doubling the chromosome number of a haploid (or by parthenogenetic development of a diploid egg). It is **completely homozygous** at every locus.
- **Double haploid (DH):** a fertile, homozygous diploid produced by doubling the chromosome set of a haploid, usually with **colchicine**. In a DH line every gene pair is homozygous, so the line is a pure line obtained in a single generation.

Steps in producing double haploids (anther culture method):

1. Collect anthers with microspores at the uninucleate stage.
2. Culture them on a suitable medium so that microspores develop into haploid callus or embryos.
3. Regenerate haploid plantlets.
4. Treat with colchicine to double the chromosome number, giving a homozygous diploid (double haploid).
5. Grow and self the DH line; it breeds true and can be used directly in breeding or as a parent.

Importance of double haploids in breeding: they shorten the breeding cycle (a homozygous line is obtained in one step instead of six or more generations of selfing), they are useful in hybrid breeding and in marker studies, and they are produced in rice, wheat, barley, maize, rapeseed-mustard and several vegetables. Anther culture of rice and wheat is a standard TNAU-relevant technique.

9.4 Polyploidy

Polyploidy is the presence of **more than two complete sets of chromosomes** in a cell or organism. It is common in plants and rare in animals. It is of two main kinds: autopolyploidy and allopolyploidy.

Autopolyploidy

An **autopolyploid** has multiple sets of chromosomes **from the same species** (for example, autotriploid AAA, autotetraploid AAAA). It arises by:

- Failure of chromosome separation in mitosis or meiosis, giving an unreduced (2n) gamete.
- Fusion of an unreduced gamete with a normal gamete (triploid) or with another unreduced gamete (tetraploid).
- Induced by **colchicine**, which blocks spindle formation and doubles the chromosome number.

Characters of autopolyploids: gigas effect (larger cells, thicker stems, bigger flowers and fruits), increased water content, later maturity, poor seed set and partial sterility because multivalents form at meiosis and segregate irregularly. Autotetraploids are more fertile than autotriploids.

Allopolyploidy

An **allopolyploid** (amphiploid) has chromosome sets from **two or more different species**, combined by hybridisation followed by chromosome doubling. It arises when an interspecific

hybrid (which is sterile because the chromosomes do not pair) undergoes chromosome doubling, so that each set now has a partner and pairing is regular.

- Steps: two species cross → sterile F_1 hybrid → chromosome doubling → fertile allopolyploid.
- **Characters:** usually more vigorous and more fertile than autopolyploids, often combines useful characters of both parents, and is a new species (for example, bread wheat, Triticale, tobacco, cotton).

Table 9.2 — Autopolyploidy versus allopolyploidy

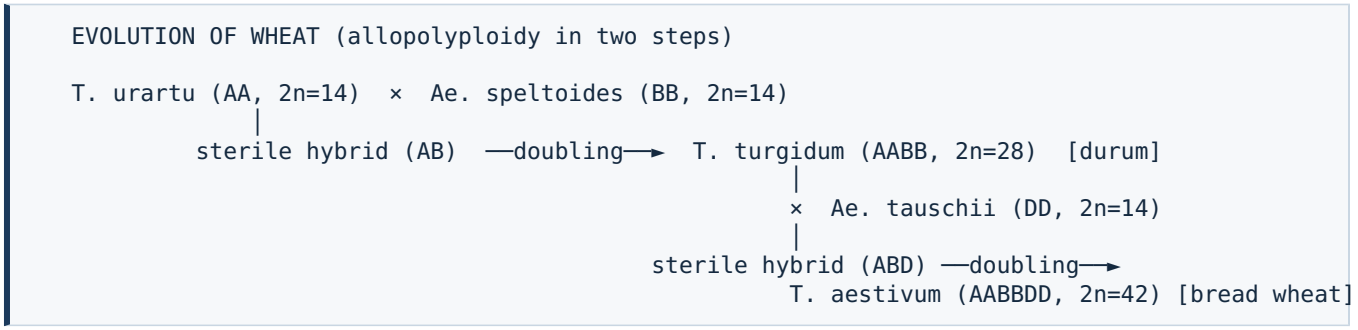
Feature	Autopolyploidy	Allopolyploidy
Source of sets	Same species (homologous genomes)	Different species (homoeologous genomes)
Origin	Chromosome doubling within a species	Interspecific hybridisation + chromosome doubling
Pairing at meiosis	Multivalents (irregular)	Bivalents (regular, mostly)
Fertility	Reduced	Usually good
Vigour	Increased (gigas effect)	Increased; may combine parental traits
Examples	Autotetraploid rye, autotriploid sugar-beet, triploid banana	Bread wheat, Triticale, tobacco, upland cotton, rapeseed

9.5 Evolution of wheat

Wheat is the classic example of evolution by allopolyploidy. Two diploid wild species of the genus *Triticum* and *Aegilops* combined in two steps:

1. **Einkorn to Emmer:** a diploid einkorn-type wheat (*Triticum urartu*, genome AA, 2n = 14) crossed with a wild goat grass (*Aegilops speltoides*, genome BB, 2n = 14). The hybrid was sterile; chromosome doubling gave a fertile **tetraploid** (AABB, 2n = 28) — the emmer/durum group, for example *Triticum turgidum* (durum wheat).
2. **Emmer to Bread wheat:** the tetraploid AABB wheat crossed with another wild goat grass (*Aegilops tauschii*, genome DD, 2n = 14). Doubling gave the fertile **hexaploid** bread wheat *Triticum aestivum* (AABBDD, 2n = 42).

So bread wheat is a hexaploid allopolyploid with three genomes (A, B, D). Durum wheat is a tetraploid (AABB, 2n = 28). This natural allopolyploidy gave wheat its adaptability, its large grain and its broad cultivation.



9.6 Triticale

Triticale is a **man-made allopolyploid** from a cross between wheat (*Triticum*) and rye (*Secale cereale*). It combines the high yield and grain quality of wheat with the hardiness and disease resistance of rye.

- **Octoploid Triticale:** bread wheat (AABBDD, 2n = 42) × rye (RR, 2n = 14) → doubling → AABBDDRR, 2n = 56.
- **Hexaploid Triticale:** durum wheat (AABB, 2n = 28) × rye (RR, 2n = 14) → doubling → AABBRR, 2n = 42 (this is the type most widely grown).
- Triticale is grown as a fodder and grain crop on poor soils and in cool climates; its main defects are shrivelled grain and poor flour quality.

9.7 Polyploidy in other crops

Table 9.3 — Allopolyploid origin of some important crops

Crop	Genomic constitution	2n	Nature
Bread wheat (*T. aestivum*)	AABBDD	42	Hexaploid allopolyploid
Durum wheat (*T. turgidum*)	AABB	28	Tetraploid allopolyploid
Triticale	AABBRR (hexaploid)	42	Man-made allopolyploid (wheat × rye)
Upland cotton (*Gossypium hirsutum*)	AADD	52	New World allopolyploid
Asiatic cotton (*G. arboreum*)	AA	26	Old World diploid
Tobacco (*Nicotiana tabacum*)	TTSS	48	Allopolyploid from two *Nicotiana* species
Oilseed rape (*Brassica napus*)	AACC	38	Amphidiploid of *B. oleracea* × *B. rapa*

Cotton

The cultivated cottons fall into two groups. The Old World cottons (*Gossypium arboreum* and *G. herbaceum*) are diploids with 2n = 26 (genome AA). The New World cottons (*G. hirsutum*, upland cotton, and *G. barbadense*) are **allotetraploids with 2n = 52** (AADD), formed by hybridisation of an Old World A-genome species with an American D-genome species followed by doubling. Upland cotton (*G. hirsutum*, 2n = 52) is the most widely grown cotton in India.

Tobacco

Cultivated tobacco *Nicotiana tabacum* (2n = 48) is a natural allopolyploid (amphidiploid) formed from *N. sylvestris* (2n = 24, genome SS) and *N. tomentosiformis* (2n = 24, genome TT). The hybrid was sterile and chromosome doubling gave the fertile 2n = 48 tobacco.

Brassica

The **triangle of U** (Nagaharu U, 1935) explains the relationships in *Brassica*. Three diploid species give three amphidiploids:

- *B. rapa* (AA, 2n = 20) × *B. oleracea* (CC, 2n = 18) → *B. napus* (AACC, 2n = 38, oilseed rape/rutabaga).

- *B. rapa* (AA, 2n = 20) × *B. nigra* (BB, 2n = 16) → *B. juncea* (AABB, 2n = 36, Indian mustard).
- *B. oleracea* (CC, 2n = 18) × *B. nigra* (BB, 2n = 16) → *B. carinata* (BBCC, 2n = 34, Ethiopian mustard).

```
TRIANGLE OF U (Brassica)
         B. nigra (BB, 16)
          /           \
B. carinata (BBCC,34)   B. juncea (AABB,36)
        |                   |
B. oleracea (CC,18) ── B. napus (AACC,38) ── B. rapa (AA,20)
```

9.8 Significance of polyploidy in agriculture

- Polyploidy has produced many of our major crops (wheat, cotton, tobacco, Brassica, potato, strawberry).
- Induced polyploidy increases size (gigas effect) — useful in fodder crops, ornamentals and some fruits.
- Triploids give seedlessness (banana, watermelon) and high sugar (triploid sugar-beet).
- Allopolyploidy allows the combination of characters of different species (Triticale) and is a route to transferring disease resistance.
- Polyploids buffer against mutation and aneuploidy because of duplicated genes.

Worked example — chromosome number of an allopolyploid

Problem: Species X has 2n = 14 (genome AA) and species Y has 2n = 18 (genome CC). A hybrid is made and its chromosome number doubled. Give the genome formula and chromosome number of the resulting allopolyploid.

Solution: The F_1 hybrid has one set from each species: 7 + 9 = 16 chromosomes (AC). Doubling gives the amphidiploid with genome formula **AACC** and chromosome number 2 × 16 = **32** (2n = 32). This resembles the *B. napus* type (AACC, 2n = 38) in principle.

Exam focus — Lecture 9

Key points

1. Euploidy = exact multiples of x; aneuploidy = not an exact multiple.
2. Haploids are weak and sterile; they reveal recessive genes and give homozygotes after doubling.
3. Double haploids are fully homozygous diploids obtained by doubling haploids, shortening the breeding cycle.
4. Autopolyploids have sets from the same species, form multivalents and are less fertile.
5. Allopolyploids combine sets from different species, form bivalents and are fertile.
6. Bread wheat (AABBDD, 2n=42) evolved by two rounds of allopolyploidy; durum wheat is AABB (2n=28).
7. Triticale is wheat × rye; upland cotton is AADD (2n=52); tobacco is TTSS (2n=48); the triangle of U explains *Brassica*.
8. Colchicine is used to induce chromosome doubling.

Short-answer questions with model answers

Q1. What is a double haploid and why is it useful in plant breeding?
Model answer: A double haploid is a fully homozygous diploid produced by doubling the chromosome number of a haploid (usually with colchicine). It is useful because a completely homozygous line is obtained in one generation, shortening the breeding cycle and helping in hybrid and marker-assisted breeding.

Q2. Distinguish auto- from allopolyploidy.
Model answer: An autopolyploid has multiple chromosome sets from the same species, forms multivalents at meiosis and is partly sterile (e.g. autotetraploid rye). An allopolyploid has sets from two or more species combined by hybridisation and doubling, forms bivalents, and is fertile (e.g. bread wheat, Triticale).

Q3. Give the genomic constitution of bread wheat and durum wheat.
Model answer: Bread wheat (*T. aestivum*) = AABBDD, 2n = 42 (hexaploid). Durum wheat (*T. turgidum*) = AABB, 2n = 28 (tetraploid).

Long-answer questions with model outlines

Q1. Describe the role of polyploidy in the evolution of wheat, cotton and *Brassica*.
Outline: Define allopolyploidy; explain wheat's two-step evolution (AA→AABB→AABBDD); cotton's AADD origin (2n = 52); the triangle of U in *Brassica*; conclude with the agricultural value of these polyploids.

Q2. Explain haploids, dihaploids and double haploids and their importance.
Outline: Define each; describe how haploids arise (anther culture, parthenogenesis); describe DH production by colchicine; give uses in breeding and genetics; give crop examples.

Lecture 10. Pre-Mendelian Concepts of Heredity

> **In one line:** Before Mendel, several theories tried to explain heredity; they were mostly speculative and wrong, but they set the questions that Mendel finally answered by experiment.

10.1 Vapour and fluid theories

- **Vapour theory (Pythagoras, ~500 BC):** proposed that "vapours" from various parts of the body, especially the reproductive organs, are collected and transmitted to the offspring, so the offspring resembles the parent.
- **Fluid theory (Hippocrates and Democritus):** suggested that "humours" or fluids from all parts of the body pass to the semen and carry the characters of those parts to the offspring. This explained resemblance but could not explain why a character lost by a parent (for example, a lost limb) is not passed on.

10.2 Magnetic power theory

This theory held that a **magnetic or vital force** attracts the male and female elements together and controls the development of the embryo. It was a vague, untestable idea with no experimental basis.

10.3 Preformation theory

Proposed in the 17th–18th centuries, the **preformation theory** stated that the **fully formed organism (a miniature "homunculus") is present in the sperm or the egg**, and development merely enlarges it. Two schools arose: the **animalculists** (spermists), who claimed the homunculus was in the sperm, and the **ovists**, who claimed it was in the egg. It was disproved by the theory of **epigenesis** (Wolff, 1759), which showed that the embryo develops gradually from an undifferentiated mass.

10.4 Lamarck's theory (Theory of inheritance of acquired characters, 1809)

Jean Baptiste Lamarck proposed two ideas:

1. **Use and disuse:** organs used more become stronger and better developed, while unused organs weaken and disappear (for example, the giraffe's long neck from stretching to reach high leaves).
2. **Inheritance of acquired characters:** these changes acquired during the lifetime of an individual are passed to the offspring.

Criticism: acquired characters are changes in the **somatic** cells, and they do not alter the **germ cells (gametes)**, so they are not inherited. Weismann's experiment of cutting off the tails of mice for many generations, without any shortening of the tail in the offspring, disproved the theory.

10.5 Darwin's theory (Theory of natural selection, 1859)

Charles Darwin, in *The Origin of Species*, proposed that:

- Organisms produce far more offspring than can survive (overproduction).
- Individuals vary, and some variations are better suited to the environment.
- In the **struggle for existence**, the fittest survive (**survival of the fittest**) and reproduce (**natural selection**).
- Favourable variations accumulate over generations, leading to evolution and the origin of new species.

Contribution: Darwin gave the mechanism of evolution but did not know the material basis of heredity. He proposed **pangenesis** (gemmules from all body parts collect in the gametes), which was later disproved. Darwin's weakness was supplied by Mendel's laws and by the mutation theory.

10.6 Germplasm theory (Weismann, 1892)

August Weismann distinguished two kinds of protoplasm:

- **Germplasm:** the substance of the germ cells (gametes), which is set aside early and is continuous from generation to generation.
- **Somatoplasm:** the substance of the body (somatic) cells, which is derived from the germplasm but does not influence it.

Key point: the germplasm is **independent of and uninfluenced by the somatoplasm**, so changes in the body are not transmitted. This is the **barrier between soma and germ cells** and it **disproved the inheritance of acquired characters**. Weismann also proposed that chromosomes are the carriers of heredity.

10.7 Mutation theory (de Vries, 1901)

Hugo de Vries, working on evening primrose (*Oenothera lamarckiana*), proposed that:

- New species arise suddenly by **discontinuous, large changes called mutations**, not gradually.
- Mutations are the source of heritable variation and hence of evolution.
- Evolution proceeds by the appearance of mutations followed by natural selection.

Contribution and criticism: de Vries' work, together with the rediscovery of Mendel's laws, restored the importance of heredity in evolution and gave the modern theory (neo-Darwinism) its material. However, many of the "mutations" he saw in *Oenothera* were actually due to chromosomal rearrangements, and not all evolution is sudden — small continuous variation (polygenic) also matters.

10.8 Summary comparison

Table 10.1 — Pre-Mendelian and related theories of heredity

Theory	Author	Main idea	Status
Vapour theory	Pythagoras	Vapours from body parts form the seed	Speculative; rejected
Fluid theory	Hippocrates	Humours from all body parts reach the semen	Rejected
Magnetic power theory	—	A magnetic force directs development	Vague; rejected
Preformation	17th–18th century	Miniature organism already present in gamete	Replaced by epigenesis
Lamarckism	Lamarck (1809)	Inheritance of acquired characters	Disproved (germplasm barrier)
Darwinism	Darwin (1859)	Natural selection, survival of the fittest	Accepted, but lacked heredity basis
Germplasm theory	Weismann (1892)	Germplasm is separate and continuous	Accepted; explains non-inheritance of somatic changes
Mutation theory	de Vries (1901)	New species by sudden mutation	Partly accepted; basis of neo-Darwinism

Exam focus — Lecture 10

Key points

1. Early theories (vapour, fluid, magnetic, preformation) were speculative and could not explain heredity.
2. Preformation (homunculus) was replaced by epigenesis (Wolff).
3. Lamarckism: use and disuse + inheritance of acquired characters; disproved.
4. Darwinism: overproduction, variation, struggle for existence, natural selection.
5. Weismann's germplasm theory separates germplasm from somatoplasm — the barrier against acquired characters.
6. de Vries' mutation theory: new species arise by sudden, heritable mutations.
7. Darwinism + mutation theory + Mendelism = neo-Darwinism (modern synthetic theory).

Short-answer questions with model answers

Q1. Why is the inheritance of acquired characters not accepted?
Model answer: Because acquired characters are changes in the somatic (body) cells, while heredity is transmitted only through the germ cells (gametes). Weismann's germplasm theory shows that the germplasm is separate and uninfluenced by the somatoplasm, so somatic changes are not passed to the offspring.

Q2. State the main features of Darwin's theory of natural selection.
Model answer: Overproduction of offspring, occurrence of variation, struggle for existence, survival of the fittest (natural selection) and gradual accumulation of favourable variations leading to evolution.

Q3. What is preformation theory?
Model answer: The preformation theory held that a fully formed miniature organism (homunculus) is already present in the sperm or the egg and that development is merely an enlargement of it; it was disproved by epigenesis.

Long-answer questions with model outlines

Q1. Describe the pre-Mendelian theories of heredity and explain why they failed.
Outline: Describe vapour, fluid, magnetic power and preformation theories; explain their weaknesses (no experimental basis, cannot explain resemblance and variation); mention epigenesis as the corrective idea.

Q2. Discuss Lamarckism, Darwinism and the germplasm theory, and show how they lead to neo-Darwinism.
Outline: Explain each theory; give the criticism of Lamarckism; give the strength and weakness of Darwinism; show how Weismann's germplasm theory and de Vries' mutation theory, together with Mendelism, supplied the missing hereditary basis and produced the modern synthetic theory.

Lecture 11. Study of Model Organisms — Drosophila, Arabidopsis, Garden Pea, E. coli and Mice

In one line: A model organism is a species that is easy to grow, has a short life cycle, produces many offspring and is well studied genetically, so that results obtained in it can be applied to other organisms.

11.1 What makes a good model organism?

- Short life cycle and rapid reproduction.
- Large number of offspring per generation (so that ratios can be tested statistically).
- Small size and easy, inexpensive culture.
- Small genome with few chromosomes, and a known karyotype.
- Availability of mutants and of genetic tools (crosses, transformation, mapping).
- Ability to be kept in the laboratory under controlled conditions.

Table 11.1 — Chief model organisms used in genetics

Organism	2n	Life cycle	Chief contribution
Drosophila melanogaster	8	~10–12 days	Linkage, sex linkage, mapping, mutation, gene regulation (polytene)
Arabidopsis thaliana	10	~6–8 weeks	Plant molecular genetics, flowering, genome sequenced (first plant)
Garden pea (*Pisum sativum*)	14	~3–4 months	Mendel's laws of inheritance
Escherichia coli	1 (single circular)	~20 minutes	Gene regulation (operon), molecular genetics, transformation
House mouse (*Mus musculus*)	40	~2–3 months	Mammalian genetics, immunology, disease models

11.2 Drosophila melanogaster (fruit fly)

- **Description:** a small fly about 2–3 mm long; has 2n = 8 (four pairs: two large autosome pairs, one small pair and the sex chromosomes).
- **Life cycle:** egg → larva (three instars) → pupa → adult, in about 10–12 days at 25 °C.
- **Why useful:** short generation time, many offspring, easy culture on simple medium, clear sexual dimorphism and, above all, a wealth of visible mutants.
- **Contributions:** T. H. Morgan used it to prove the chromosomal theory of inheritance; Sturtevant made the first genetic map; Bridges studied non-disjunction and sex determination; Muller induced mutations with X-rays; its salivary-gland polytene chromosomes helped in gene mapping.
- **Key traits used:** eye colour (red/white), wing shape (normal/vestigial), body colour, bristles, and sex-linked characters.

11.3 Arabidopsis thaliana (thale cress)

- **Description:** a small flowering plant of the mustard family (Brassicaceae); 2n = 10; genome of about 135 megabase pairs with about 25,000 genes.
- **Why useful:** small size, short life cycle (6–8 weeks), self-pollinating, large seed set, small genome and easy transformation by *Agrobacterium*.
- **Contributions:** the **first plant genome to be fully sequenced** (2000); used to study flowering time, hormone action, disease resistance, gene function and genome editing. It is the plant equivalent of *Drosophila*.

11.4 Garden pea (Pisum sativum)

- **Description:** an annual legume; 2n = 14; normally self-pollinated but easily cross-pollinated by hand.
- **Why chosen by Mendel:** it has clear-cut, contrasting characters (round/wrinkled seed, yellow/green cotyledon, tall/dwarf plant, axial/terminal flower); it is self-pollinating so pure lines are easy to maintain; it is easy to cross artificially; and it has a short life cycle.
- **Contributions:** Mendel's seven pairs of contrasting characters gave the laws of dominance, segregation and independent assortment — the foundation of genetics.

11.5 Escherichia coli

- **Description:** a rod-shaped bacterium; single circular chromosome (about 4.6 million base pairs, about 4,400 genes); divides by binary fission every 20 minutes under good conditions.
- **Why useful:** extremely fast growth, simple nutrition, easy to grow in large numbers, and simple genetics with a single chromosome and no nucleus.
- **Contributions:** the **operon model** of gene regulation (Jacob and Monod, lac operon); the mechanism of DNA replication and the genetic code; transformation, conjugation and transduction; the chief host for gene cloning and recombinant DNA work.

11.6 House mouse (Mus musculus)

- **Description:** a small mammal; 2n = 40; short generation time of about 2–3 months; produces many young.
- **Why useful:** as a mammal it shares many genes and pathways with humans, so it is the leading model for mammalian genetics, immunology and disease (cancer, diabetes, obesity).
- **Contributions:** coat-colour genetics (agouti, albino series), the H2 (histocompatibility) complex, transgenic and knockout mice, and the study of sex-linked and lethal genes.

Table 11.2 — Comparison of the model organisms

Feature	Drosophila	Arabidopsis	Garden pea	E. coli	Mouse
Kingdom	Animal (insect)	Plant	Plant	Bacterium	Animal (mammal)
2n	8	10	14	1	40
Genome size	~180 Mb	~135 Mb	~4,300 Mb	~4.6 Mb	~2,700 Mb
Life cycle	10–12 days	6–8 weeks	3–4 months	20 minutes	2–3 months
Chief use	Linkage, mapping, mutation	Plant molecular genetics	Mendel's laws	Gene regulation, cloning	Mammalian genetics, disease

Exam focus — Lecture 11

Key points

1. A model organism has a short life cycle, many offspring, small size and a well-known genome.
2. *Drosophila* (2n = 8) gave linkage, sex linkage, mapping and induced mutation.
3. *Arabidopsis* (2n = 10) was the first plant genome sequenced.
4. Garden pea (2n = 14) was Mendel's material; self-pollinated with clear contrasting characters.
5. *E. coli* (single circular chromosome) gave the operon model and the genetic code.
6. The mouse (2n = 40) is the chief mammalian model for disease genetics.

Short-answer questions with model answers

Q1. Why was garden pea a good choice for Mendel's experiments?
Model answer: Because it has clear-cut contrasting characters, is naturally self-pollinated (so pure lines are easily maintained), can be easily cross-pollinated by hand, has a short life cycle, and produces many seeds in a small space.

Q2. State two contributions of *Drosophila* to genetics.
Model answer: (i) It was used by Morgan to prove the chromosomal theory of inheritance and to discover sex-linked inheritance; (ii) its salivary-gland polytene chromosomes and its mutants were used by Sturtevant and Bridges for genetic mapping and the study of non-disjunction.

Q3. Why is *E. coli* important in molecular genetics?
Model answer: Because of its rapid growth, simple genome and easy culture, it was used to establish the operon model of gene regulation, the genetic code and the mechanisms of replication, and it is the chief host for gene cloning and recombinant DNA technology.

Long-answer questions with model outlines

Q1. Describe any four model organisms and their contributions to genetics.
Outline: For each of *Drosophila*, *Arabidopsis*, garden pea and *E. coli* (or mouse), give description, chromosome number, life cycle, why it is suitable, and its chief contributions; finish with a comparison table.

Lecture 12. Probability and Chi-Square

In one line: Probability is the chance that an event will happen; the chi-square test compares observed results with those expected from a genetic hypothesis and tells us whether the difference is due to chance alone.

12.1 Probability — basic ideas

Probability is the ratio of the number of favourable outcomes to the total number of equally likely outcomes. For an event A:

P(A) = number of favourable outcomes / total number of outcomes

Probability lies between 0 and 1. A probability of 1 means the event is certain; 0 means it is impossible. In genetics, probability is used to predict the ratios expected from a cross and to test whether observed results agree with a hypothesis.

Rules of probability

Table 12.1 — Rules of probability used in genetics

Rule	Statement	When used	Formula
Product (multiplication) rule	The probability of two or more independent events occurring together is the product of their separate probabilities	Independent events (e.g. two separate fertilisations)	P(A and B) = P(A) × P(B)
Sum (addition) rule	The probability of either of two mutually exclusive events is the sum of their probabilities	Alternative ways of getting the same result	P(A or B) = P(A) + P(B)

Applications in genetics

- In a monohybrid cross (Tt × Tt) the probability of a tall offspring is 3/4 and of a dwarf offspring 1/4.
- The probability that two independent events occur together (for example, a tall plant with yellow seeds in a dihybrid cross) is found by the product rule: 3/4 × 3/4 = 9/16.
- The sum rule is used when a result can be obtained in more than one way (for example, in a cross of Aa × Aa, a dominant phenotype arises as AA or Aa: 1/4 + 2/4 = 3/4).

Worked example 12.1 — simple probability

Problem: In a family of four children, what is the probability that all four are boys? (Assume equal chance of a boy or a girl.)

Solution: Probability of a boy in one birth = 1/2. The four births are independent, so by the product rule:

$$P(4 \text{ boys}) = 1/2 \times 1/2 \times 1/2 \times 1/2 = \mathbf{1/16}.$$

Problem: What is the probability of exactly two boys and two girls in four births?

Solution: The number of ways of choosing which two of the four children are boys is 6 (use the binomial coefficient C(4,2) = 6). Each arrangement has probability $(1/2)^4 = 1/16$. By the sum rule:

P = 6 × 1/16 = **6/16 = 3/8**.

12.2 The chi-square (χ^2) test

The chi-square test is a statistical test of **goodness of fit**. It measures how well the observed numbers agree with the numbers expected from a genetic hypothesis. It was applied to genetics by R. A. Fisher.

$$\chi^2 = \Sigma\,(O - E)^2 / E$$

where O = observed number in a class, E = expected number in that class, and the sum is taken over all classes.

Steps in the chi-square test

1. State the hypothesis (for example, a 3:1 ratio is expected).
2. Calculate the expected numbers (E) for each class from the hypothesis and the total.
3. Find the deviation (O − E) in each class and square it.
4. Divide each squared deviation by the corresponding E and add all values to get χ^2.
5. Find the **degrees of freedom (df)** = number of classes − 1.
6. Compare the calculated χ^2 with the table value at 5% (0.05) probability.
7. If calculated χ^2 is less than the table value, the hypothesis is **accepted** (difference is due to chance); if it is greater, the hypothesis is **rejected**.

Table 12.2 — Chi-square table values at the 5% level of significance

Degrees of freedom (df)	1	2	3	4	5
χ^2 value (P = 0.05)	3.84	5.99	7.82	9.49	11.07

Worked example 12.2 — chi-square for a monohybrid ratio

Problem: In a cross of tall × tall pea plants, 787 tall and 277 dwarf offspring were obtained. Test whether the results agree with a 3:1 ratio.

Solution: Total = 787 + 277 = 1064. Expected 3:1 → tall = 3/4 × 1064 = 798, dwarf = 1/4 × 1064 = 266.

Class	Observed (O)	Expected (E)	O − E	$(O - E)^2$	$(O - E)^2/E$
Tall	787	798	−11	121	0.152
Dwarf	277	266	+11	121	0.455
Total	1064	1064	0	—	**χ^2 = 0.607**

df = 2 − 1 = 1. Table value of χ^2 at df 1 and P = 0.05 is **3.84**. Since calculated χ^2 (0.607) is **less** than 3.84, the deviation is due to chance alone and the 3:1 ratio is **accepted**.

Worked example 12.3 — chi-square for a dihybrid ratio

Problem: In a dihybrid cross, the following offspring were obtained: round-yellow 315, round-green 108, wrinkled-yellow 101, wrinkled-green 32 (Mendel's actual data). Test the 9:3:3:1 ratio.

Solution: Total = 556. Expected: 9/16 × 556 = 312.75, 3/16 × 556 = 104.25, 3/16 × 556 = 104.25, 1/16 × 556 = 34.75.

Class	O	E	O − E	$(O - E)^2$	$(O - E)^2/E$
Round-yellow	315	312.75	+2.25	5.06	0.016
Round-green	108	104.25	+3.75	14.06	0.135
Wrinkled-yellow	101	104.25	−3.25	10.56	0.101
Wrinkled-green	32	34.75	−2.75	7.56	0.218
Total	556	556	0	—	**$\chi^2 = 0.470$**

df = 4 − 1 = 3. Table value at df 3, P = 0.05 is **7.82**. Calculated $\chi^2 = 0.470 < 7.82$, so the 9:3:3:1 ratio is **accepted** — the observed results fit independent assortment.

Worked example 12.4 — a case where the hypothesis is rejected

Problem: A 9:3:3:1 ratio is expected but observed numbers are 300, 100, 100, 60 (total 560). Test the ratio.

Solution: Expected: 315, 105, 105, 35.

Class	O	E	O−E	$(O-E)^2/E$
1	300	315	−15	0.714
2	100	105	−5	0.238
3	100	105	−5	0.238
4	60	35	+25	17.857
Total				**$\chi^2 = 19.05$**

df = 3; table value = 7.82. Calculated $\chi^2 = 19.05 > 7.82$, so the hypothesis is **rejected** — the observed results do not fit a 9:3:3:1 ratio (the excess in the last class suggests another interaction, such as a modified ratio).

12.3 Uses and limitations of chi-square

- **Uses:** to test goodness of fit of observed ratios to expected Mendelian ratios; to test linkage (independence of two genes); to test the homogeneity of data; to compare genetic models.
- **Limitations:** it needs adequate numbers (each expected class should generally be 5 or more); it only tells whether a difference is significant, not why; and it must be applied only to counts, not to percentages or measurements.

Exam focus — Lecture 12

Key points

1. Probability = favourable outcomes / total outcomes; ranges from 0 to 1.
2. Product rule for independent events; sum rule for mutually exclusive events.
3. $\chi^2 = \Sigma(O-E)^2/E$; used to test goodness of fit.
4. df = number of classes − 1.
5. Accept the hypothesis if calculated χ^2 < table value; reject if greater.
6. Table value at 5% for df 1 is 3.84, for df 3 is 7.82.
7. Each expected class should generally be 5 or more.

Short-answer questions with model answers

Q1. State the product and sum rules of probability with one genetic example each.

Model answer: Product rule — the probability of two independent events together is the product of their probabilities; e.g. in a dihybrid cross the chance of a tall-yellow plant is 3/4 × 3/4 = 9/16. Sum rule — the probability of either of two mutually exclusive events is their sum; e.g. a dominant phenotype in Aa × Aa arises as AA or Aa, so 1/4 + 2/4 = 3/4.

Q2. Write the chi-square formula and explain each term.

Model answer: $\chi^2 = \Sigma(O-E)^2/E$, where O is the observed number, E the expected number, and Σ means summing over all classes. It measures the deviation of observation from expectation.

Q3. How is the chi-square value interpreted?

Model answer: The calculated χ^2 is compared with the table value at the chosen probability (usually 5%) for the appropriate degrees of freedom. If the calculated value is smaller, the hypothesis is accepted; if larger, it is rejected.

Long-answer questions with model outlines

Q1. Explain the chi-square test and apply it to test a 3:1 ratio.

Outline: Define the test and formula; list the steps; take a numerical example with a table of O, E, $(O-E)^2/E$; find df and compare with the table; conclude accept or reject.

Q2. Discuss the rules of probability and their application in genetics.

Outline: Define probability; explain product and sum rules; apply them to monohybrid and dihybrid crosses; mention the binomial expansion for larger families; note the link to the chi-square test.

Lecture 13. Mendel's Experiments, Laws of Inheritance and Rediscovery of Mendel's Work

> **In one line:** Mendel crossed pure pea lines with contrasting characters and deduced the laws of dominance, segregation and independent assortment; his work, published in 1866, was rediscovered in 1900.

13.1 Mendel's material and method

Gregor Johann Mendel (1822–1884), an Austrian monk, worked in the monastery garden at Brünn with the garden pea, *Pisum sativum* (2n = 14). He chose seven pairs of clear contrasting characters and studied them one or two at a time, counting large numbers of offspring and applying simple arithmetic.

Table 13.1 — Mendel's seven pairs of contrasting characters in pea

Character	Dominant form	Recessive form
Seed shape	Round	Wrinkled
Seed colour (cotyledon)	Yellow	Green
Flower colour	Violet/purple	White
Pod shape	Inflated (smooth)	Constricted
Pod colour	Green	Yellow
Flower position	Axial	Terminal
Plant height	Tall	Dwarf

13.2 The monohybrid experiment and the law of dominance

Mendel crossed a pure tall plant with a pure dwarf plant. All the F_1 plants were tall. When the F_1 plants were selfed, the F_2 consisted of tall and dwarf plants in the ratio **3 : 1**. This showed that one character (tall) was **dominant** and the other (dwarf) **recessive**.

```
MONOHYBRID CROSS (TT × tt)

P:      TT (tall)   ×   tt (dwarf)
           |
F1:        Tt  (all tall)         ← law of dominance
           |  self
F2:  1 TT : 2 Tt : 1 tt
     (tall)   (tall)  (dwarf)
     Phenotypic ratio  = 3 tall : 1 dwarf
     Genotypic  ratio  = 1 TT : 2 Tt : 1 tt
```

Punnett square for the monohybrid F_2

	T	**t**
T	TT (tall)	Tt (tall)
t	Tt (tall)	tt (dwarf)

Law of dominance: when two unlike factors (alleles) for a character are present together, one (the dominant) expresses itself and the other (recessive) remains hidden.

13.3 The law of segregation

The law of segregation states that **the two alleles of a gene separate (segregate) during gamete formation, so that each gamete receives only one allele**. The F_1 hybrid (Tt) produces two kinds of gametes, T and t, in equal numbers (1 : 1). This is also called the law of purity of gametes. It is the most fundamental of Mendel's laws and is universally true.

13.4 The dihybrid experiment and the law of independent assortment

Mendel crossed pure round-yellow × pure wrinkled-green. The F_1 was all round-yellow. In the F_2 he obtained four phenotypes in the ratio **9 : 3 : 3 : 1**:

Table 13.2 — Mendel's dihybrid F_2 results

Phenotype	Ratio	Observed (Mendel)
Round-yellow	9	315
Round-green	3	108
Wrinkled-yellow	3	101
Wrinkled-green	1	32
Total	16	556

Punnett square for the dihybrid F_2 (RrYy × RrYy)

Gametes of each parent: RY, Ry, rY, ry (four kinds in equal numbers).

	RY	**Ry**	**rY**	**ry**
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

Phenotypes: round-yellow 9 (R_Y_), round-green 3 (R_yy), wrinkled-yellow 3 (rrY_), wrinkled-green 1 (rryy). So the **9:3:3:1** phenotypic ratio and the **genotypic ratio 1:2:2:4:1:2:1:2:1** result.

Law of independent assortment: when two or more pairs of characters are considered together, the alleles of one pair assort independently of the alleles of the other pair, because the genes lie on different chromosome pairs and orient randomly at meiosis.

13.5 Mendel's laws — summary

Table 13.3 — Mendel's laws

Law	Statement	Evidence	Exception
Law of dominance	In a heterozygote, the dominant allele masks the recessive	All F_1 tall in monohybrid	Incomplete dominance, codominance
Law of segregation	The two alleles separate during gamete formation; each gamete gets one	3:1 F_2; 1:1 test cross	None (universal)
Law of independent assortment	Genes on different chromosome pairs assort independently	9:3:3:1 F_2	Linkage (genes on the same chromosome)

13.6 Test cross — proof of segregation

Mendel crossed the F_1 tall (Tt) with the recessive parent (tt). The result was 1 tall : 1 dwarf, confirming that the F_1 produced two kinds of gametes in equal numbers.

	T	t
t	Tt (tall)	tt (dwarf)

Test cross ratio = **1 tall : 1 dwarf**. A dihybrid test cross (RrYy × rryy) gives **1 : 1 : 1 : 1**.

13.7 Why Mendel succeeded

- He chose clear-cut, qualitative (discontinuous) characters that gave unambiguous results.
- He studied one or two characters at a time and kept pure lines.
- He counted large numbers of offspring and used arithmetic (ratios), not just description.
- He kept careful records and repeated experiments across generations (P, F_1, F_2).
- His material, pea, is normally self-pollinated, so contamination was controlled.

13.8 Rediscovery of Mendel's work (1900)

Mendel's paper (1866) was ignored for 34 years. In 1900 three scientists, working independently on different plants, reached the same conclusions and then found Mendel's paper:

- **Hugo de Vries** (Holland) — working with evening primrose and other plants.
- **Carl Correns** (Germany) — working with pea and maize.
- **Erich von Tschermak** (Austria) — working with pea.

Their rediscovery marks the birth of the science of genetics. In 1902–03 Sutton and Boveri connected Mendel's laws with chromosome behaviour (chromosomal theory of inheritance), and Bateson coined the word "genetics" in 1905.

Worked example 13.1 — monohybrid with a test cross

Problem: A tall pea plant of unknown genotype is crossed with a dwarf plant. The progeny are 50 tall and 48 dwarf. What is the genotype of the tall parent?

Solution: A 1:1 ratio is the signature of a test cross in which the tall parent is **heterozygous (Tt)**. Tt × tt → 1 Tt (tall) : 1 tt (dwarf). A homozygous tall (TT) parent would give all tall progeny. Hence the tall parent is Tt.

Worked example 13.2 — probability in a dihybrid cross

Problem: In RrYy × RrYy, what is the probability of obtaining a plant that is round-seeded and green-cotyledon (R_yy)?

Solution: Round (R_) = 3/4; green (yy) = 1/4. By the product rule, P(R_yy) = 3/4 × 1/4 = **3/16**. This matches the round-green class of the 9:3:3:1 ratio.

Exam focus — Lecture 13

Key points

1. Mendel worked with garden pea (2n = 14) and seven contrasting characters.
2. Law of dominance: the dominant allele masks the recessive in the heterozygote.
3. Law of segregation: alleles separate in gamete formation; each gamete gets one; ratio 1:1.
4. Law of independent assortment: genes on different pairs assort independently; F_2 9:3:3:1.
5. Monohybrid F_2 = 3:1 phenotypic, 1:2:1 genotypic; test cross 1:1.
6. Dihybrid test cross = 1:1:1:1; dihybrid F_2 genotypic ratio 1:2:2:4:1:2:1:2:1.
7. Rediscovered in 1900 by de Vries, Correns and Tschermak.

Short-answer questions with model answers

Q1. State the law of segregation and give the gametic ratio it predicts.
Model answer: The law of segregation states that the two alleles of a gene separate during gamete formation so that each gamete receives only one allele. A heterozygote (Tt) therefore produces two kinds of gametes, T and t, in a 1:1 ratio.

Q2. Why did Mendel get a 9:3:3:1 ratio in the dihybrid F_2?
Model answer: Because the two gene pairs lie on different chromosome pairs and assort independently, so each character gives a 3:1 ratio; combining two independent 3:1 ratios by the product rule gives 9:3:3:1.

Q3. What is a test cross and what ratio does a monohybrid test cross give?
Model answer: A test cross is a cross of an individual of unknown genotype with the homozygous recessive parent. A monohybrid test cross (Tt × tt) gives a 1 tall : 1 dwarf ratio.

Long-answer questions with model outlines

Q1. Describe Mendel's monohybrid and dihybrid experiments and state the laws derived from them.
Outline: Describe the crosses with Punnett squares; give F_1 and F_2 results; derive dominance, segregation and independent assortment; mention the test cross as proof.

Q2. How was Mendel's work rediscovered and why was it ignored for 34 years?
Outline: Reasons for neglect (unusual method, poor circulation of the journal, no known chromosomes at the time, focus on blending inheritance); the 1900 rediscovery by de Vries, Correns and Tschermak; the consequences (birth of genetics, chromosomal theory).

Lecture 14. Genetic Terminologies

> **In one line:** These are the basic words of genetics — gene, allele, locus, genotype, phenotype and the terms for crosses — that must be used exactly.

14.1 Terms relating to the gene

Table 14.1 — Basic genetic terms and their meanings

Term	Meaning	Example
Gene	The unit of inheritance; a segment of DNA that codes for a polypeptide or a functional RNA	Gene for seed shape in pea
Allele (allelomorph)	Alternative forms of the same gene occupying the same locus on homologous chromosomes	T (tall) and t (dwarf)
Locus	The specific position of a gene on a chromosome	Locus of the T gene on pea chromosome
Homozygous	Having two identical alleles for a gene (pure breeding)	TT or tt
Heterozygous	Having two different alleles for a gene (hybrid)	Tt
Hemizygous	Having only one allele (no partner) for a gene, as in sex-linked genes in males	X^wY in white-eyed Drosophila male
Genotype	The genetic constitution of an individual (the alleles present)	TT, Tt, tt
Phenotype	The visible or measurable expression of the genotype	Tall or dwarf
Genome	The complete set of genes (or chromosomes) of an organism	Pea genome; human genome

14.2 Terms relating to alleles and expression

- **Dominant allele:** the allele that expresses itself in the heterozygote (written in capitals, T).
- **Recessive allele:** the allele whose expression is masked in the heterozygote and which shows only in the homozygous state (written in small letters, t).
- **Wild type:** the allele or phenotype most common in nature (often written with a + sign).
- **Mutant:** an allele or phenotype that differs from the wild type.

14.3 Terms relating to crosses

Table 14.2 — Terms describing crosses

Term	Meaning
P generation	The parental generation used in a cross
F_1 generation	The first filial generation — offspring of the P cross
F_2 generation	Offspring obtained by selfing or intercrossing the F_1
Test cross	Cross of an individual with the homozygous recessive parent, to reveal its genotype
Back cross	Cross of the F_1 (or any hybrid) back to either of its parents
Reciprocal cross	A cross in which the sexes (or the parents) of the two forms are reversed
Selfing	Crossing an individual with itself (natural in many crops)

Note the difference: every test cross is a back cross, but a back cross is not always a test cross. A back cross to the **dominant** parent is not a test cross; a back cross to the **recessive** parent is a test cross.

14.4 Terms for number of characters and gene pairs

Table 14.3 — Mono-, di-, tri- and polyhybrid crosses

Term	Number of gene pairs (heterozygous)	F_2 phenotypic ratio	Number of gamete types	Number of F_2 genotypes
Monohybrid	1 (Aa)	3 : 1	2	3
Dihybrid	2 (AaBb)	9 : 3 : 3 : 1	4	9
Trihybrid	3 (AaBbCc)	27 : 9 : 9 : 9 : 3 : 3 : 3 : 1	8	27
Polyhybrid	n	$(3:1)^n$	2^n	3^n

These numbers come from the rules: number of gamete types = 2^n, number of F_2 genotypes = 3^n, number of F_2 phenotypes = 2^n (for complete dominance), where n is the number of heterozygous gene pairs.

14.5 Terms for genes in the homozygous/heterozygous state and interactions

- **Pure line:** a group of individuals homozygous and identical for the characters studied, breeding true.
- **Hybrid:** an individual produced by crossing two genetically different parents.
- **Hemizygous:** having a single allele (sex-linked genes in the heterogametic sex).
- **Allele interaction:** interaction between the two alleles of one gene (dominance, incomplete dominance, codominance) — Lecture 15.
- **Non-allelic interaction (epistasis):** interaction between genes at different loci — Lectures 16–19.

Worked example 14.1 — writing genotypes and gametes

Problem: For the genotype AaBbCC, (a) state whether it is homozygous or heterozygous at each locus, (b) write all the gametes it can form, and (c) give the number of gamete types.

Solution: (a) Aa = heterozygous; Bb = heterozygous; CC = homozygous dominant. (b) Aa gives A, a; Bb gives B, b; CC gives only C. Combining: ABC, AbC, aBC, abC. (c) Number of gamete types = 2^2 = 4 (n = 2 heterozygous pairs).

Exam focus — Lecture 14

Key points

1. Gene = unit of inheritance; allele = alternative form at the same locus.
2. Homozygous = identical alleles; heterozygous = different alleles; hemizygous = single allele.
3. Genotype = genetic constitution; phenotype = its expression.
4. Test cross is always to the recessive parent; back cross is to either parent.
5. Number of gamete types = 2^n; F_2 genotypes = 3^n; F_2 phenotypes = 2^n.
6. Monohybrid 3:1, dihybrid 9:3:3:1, trihybrid 27:9:9:9:3:3:3:1.
7. Locus is the position of a gene on the chromosome.

Short-answer questions with model answers

Q1. Differentiate test cross from back cross.
Model answer: A back cross is a cross of the hybrid with either of its parents. A test cross is a cross of an individual of unknown genotype with the homozygous recessive parent. Every test cross is a back cross, but a back cross to the dominant parent is not a test cross.

Q2. What is a hemizygous condition? Give an example.
Model answer: Hemizygous means having only one allele for a gene instead of a pair. It occurs for sex-linked genes in the heterogametic sex, for example the white-eye gene (X^w) in a male *Drosophila* (X^wY).

Q3. How many gamete types and F_2 genotypes are expected in a trihybrid cross?
Model answer: 2^3 = 8 gamete types and 3^3 = 27 F_2 genotypes.

Long-answer questions with model outlines

Q1. Define and explain the important genetic terms with suitable examples.
Outline: Define gene, allele, locus, homozygous, heterozygous, hemizygous, genotype, phenotype; distinguish test cross and back cross; explain mono-, di-, tri- and polyhybrid crosses with their ratios; give examples for each.

Lecture 15. Chromosomal Theory of Inheritance; Allelic Interactions

> **In one line:** The chromosomal theory of inheritance states that genes are located on chromosomes and that their transmission follows chromosome behaviour; allelic interaction decides how the two alleles of one gene express themselves.

15.1 Chromosomal theory of inheritance

Proposed by **Walter Sutton** (1902) and **Theodor Boveri** (1903), the theory states that:

- Genes are located on chromosomes in a linear order.
- Homologous chromosomes pair and then separate during meiosis, so each gamete gets one chromosome of each pair — parallel to the segregation of alleles.
- Different chromosome pairs orient independently at meiosis — parallel to the independent assortment of genes.
- Chromosomes retain their individuality through cell division and are transmitted from generation to generation.

The theory was proved by: (i) the parallel behaviour of chromosomes and Mendel's factors; (ii) sex-linked inheritance in *Drosophila* (Morgan); (iii) Bridges' non-disjunction experiments linking the X chromosome with eye colour; and (iv) the cytological proof of crossing over (Creighton and McClintock in maize; Stern in *Drosophila*).

Table 15.1 — Parallel behaviour of chromosomes and genes

Behaviour of chromosomes	Behaviour of genes (Mendel's factors)
Occur in pairs (homologues) in diploids	Occur in pairs (alleles)
Separate during anaphase I of meiosis	Segregate during gamete formation
Different pairs orient independently	Different gene pairs assort independently
Carried in gametes, one of each pair	Gametes carry one allele of each pair
Maintain individuality through division	Genes are transmitted unchanged

15.2 Deviations from Mendelian inheritance

Mendel's ratios (3:1, 9:3:3:1) are modified by two broad kinds of deviation:

- **Allelic interaction** — interaction between the two alleles of the same gene (complete dominance, incomplete dominance, codominance).
- **Non-allelic interaction (epistasis)** — interaction between genes at different loci — Lectures 16–19.

15.3 Allelic interaction

(a) Complete dominance

One allele completely masks the expression of the other in the heterozygote. The heterozygote looks exactly like the homozygous dominant. Monohybrid F_2 = **3:1**; F_1 resembles one parent. *Example:* tall vs dwarf in pea; round vs wrinkled seed.

(b) Incomplete dominance (partial dominance)

Neither allele is completely dominant; the heterozygote shows an **intermediate** phenotype. The F_2 phenotypic ratio equals the genotypic ratio, **1 : 2 : 1**. *Examples:* pink flowers in *Antirrhinum majus* (snapdragon) and *Mirabilis jalapa* from a cross of red × white; and the four o'clock plant. This is also called **blending inheritance** but the alleles themselves remain separate.

```
INCOMPLETE DOMINANCE (red × white → pink)

P:   RR (red)   ×   rr (white)
F1:        Rr  (PINK, intermediate)
F2:   1 RR (red) : 2 Rr (pink) : 1 rr (white)
      Phenotypic ratio = 1 : 2 : 1  (= genotypic ratio)
```

(c) Codominance

Both alleles express themselves **fully and independently** in the heterozygote, so both parental phenotypes appear side by side. *Examples:* the AB blood group in humans (alleles I^A and I^B both expressed); roan coat colour in cattle (red and white hairs both present); and the speckled pattern in some flowers.

Table 15.2 — Complete dominance, incomplete dominance and codominance

Feature	Complete dominance	Incomplete dominance	Codominance
Heterozygote phenotype	Like dominant homozygote	Intermediate	Both parental phenotypes together
F_1 phenotype	One parent	Intermediate	Both parents
F_2 phenotypic ratio	3 : 1	1 : 2 : 1	1 : 2 : 1
Example	Pea tall/dwarf	Snapdragon flower colour; *Mirabilis*	AB blood group; roan cattle

(d) Threshold characters

A **threshold character** is one that is inherited in a discontinuous (Mendelian) way at the genetic level but is expressed only when a certain **threshold** of liability is crossed. Below the threshold the character does not appear; above it, the character appears fully. The liability itself is continuous (polygenic), but the expressed phenotype falls into two or a few distinct classes.

- *Example:* pyloric stenosis in humans — inherited as a threshold character, more common in males; the underlying liability is continuous, but only those crossing the threshold develop the condition.
- *In plants:* the development of a disease symptom, or bolting/flowering in response to accumulated temperature, can behave as a threshold character.
- Threshold characters bridge qualitative and quantitative genetics — they are "all-or-none" phenotypes underlain by continuous genetic and environmental variation.

Worked example 15.1 — incomplete dominance cross

Problem: In snapdragon, red (RR) × white (rr) gives pink F_1. What phenotypic and genotypic ratios are expected in the F_2, and how many of 400 F_2 plants would be pink?

Solution: F_1 = Rr (pink). $F_1 \times F_1$ (Rr × Rr):

	R	**r**
R	RR (red)	Rr (pink)
r	Rr (pink)	rr (white)

Phenotypic ratio = 1 red : 2 pink : 1 white = genotypic ratio. Pink plants = 2/4 = 1/2 of 400 = **200 plants**.

Worked example 15.2 — codominance in blood group

Problem: A man of blood group AB marries a woman of blood group O. What blood groups can their children have?

Solution: AB = I^AI^B; O = ii. Gametes: man gives I^A or I^B; woman gives only i.

	I^A	**I^B**
i	I^Ai (A)	I^Bi (B)

Children: 50% group A (I^Ai) and 50% group B (I^Bi). No child can be AB or O.

Exam focus — Lecture 15

Key points

1. Chromosomal theory (Sutton and Boveri): genes lie on chromosomes and follow chromosome behaviour.
2. Proved by sex linkage (Morgan), non-disjunction (Bridges) and cytological proof of crossing over.
3. Deviations are of two kinds: allelic interaction and non-allelic interaction (epistasis).
4. Complete dominance → F_2 3:1; heterozygote like the dominant homozygote.
5. Incomplete dominance → heterozygote intermediate; F_2 1:2:1.
6. Codominance → both alleles expressed; example AB blood group, roan cattle.
7. Threshold characters are all-or-none phenotypes underlain by continuous liability.

Short-answer questions with model answers

Q1. State the chromosomal theory of inheritance.
Model answer: Genes are located in a linear order on chromosomes; homologous chromosomes separate during meiosis as alleles segregate; different chromosome pairs assort independently as different gene pairs do; and chromosomes maintain their individuality through division and are transmitted from generation to generation.

Q2. Differentiate incomplete dominance from codominance.
Model answer: In incomplete dominance the heterozygote shows an intermediate phenotype (e.g. pink flowers from red × white). In codominance both alleles express fully in the heterozygote, so both parental phenotypes appear side by side (e.g. AB blood group, roan coat).

Q3. What is a threshold character?
Model answer: A threshold character is a discontinuous (all-or-none) phenotype that appears only when an underlying continuous liability crosses a certain threshold; for example pyloric stenosis in humans.

Long-answer questions with model outlines

Q1. Explain the chromosomal theory of inheritance and the evidence supporting it.
Outline: State the theory; give the parallel table between chromosomes and genes; describe the evidence from sex-linked inheritance, Bridges' non-disjunction and Creighton-McClintock/Stern.

Q2. Describe the types of allelic interaction with examples and F_2 ratios.
Outline: Define complete dominance, incomplete dominance, codominance and threshold characters; give F_1 and F_2 ratios; use a comparison table; give crop and animal examples.

Lecture 16. Non-Allelic Interaction Without Modification of Mendelian Ratio — Bateson and Punnett's Fowl Comb Experiment

> **In one line:** When two genes at different loci act together to produce a new phenotype but the F_2 still shows the familiar 9:3:3:1 numbers, the interaction is "simple" and does not modify the Mendelian ratio.

16.1 The idea of gene interaction

Mendel studied characters in which one gene controlled one character. But many characters are controlled by **two or more genes acting together**. When genes at two or more different loci influence the same character, they are said to show **non-allelic interaction** or **gene interaction**. The effect of one gene is influenced by the presence of another gene at a different locus.

Gene interaction is of two types:

- **Without modification of the Mendelian ratio:** the F_2 still gives the 9:3:3:1 numbers, but the four classes have new (combined) phenotypes. Example: fowl comb shape.
- **With modification of the Mendelian ratio:** the 9:3:3:1 numbers are regrouped into ratios such as 12:3:1, 9:3:4, 9:6:1, 15:1, 9:7 and 13:3. Example: epistasis (Lectures 17 and 19).

16.2 Bateson and Punnett's fowl comb experiment

Bateson and Punnett (1905–06) studied the shape of the comb in fowls. Four comb types occur:

- **Rose comb** — a broad comb with a ridge.
- **Pea comb** — three small ridges.
- **Walnut comb** — a broad, rounded comb (a new type).
- **Single comb** — a simple, upright comb.

Two genes are involved: **R** (rose) and **P** (pea). When both dominant alleles are present (R_P_), a new phenotype, the walnut comb, appears.

Table 16.1 — Genotypes and phenotypes of fowl comb shape

Genotype	Comb type	F_2 proportion
R_P_	Walnut (new type)	9
R_pp	Rose	3
rrP_	Pea	3
rrpp	Single	1

The crosses

```
Rose (RRpp)  ×  Pea (rrPP)
      |
F1:   RrPp  →  WALNUT comb (all)
      |  self
F2:   9 R_P_ (walnut) : 3 R_pp (rose) : 3 rrP_ (pea) : 1 rrpp (single)
     = 9 : 3 : 3 : 1  (new phenotypes in the old ratio)
```

Punnett square (RrPp × RrPp)

	RP	**Rp**	**rP**	**rp**
RP	RRPP (walnut)	RRPp (walnut)	RrPP (walnut)	RrPp (walnut)
Rp	RRPp (walnut)	RRpp (rose)	RrPp (walnut)	Rrpp (rose)
rP	RrPP (walnut)	RrPp (walnut)	rrPP (pea)	rrPp (pea)
rp	RrPp (walnut)	Rrpp (rose)	rrPp (pea)	rrpp (single)

16.3 Interpretation

- Each gene alone produces its own comb (R alone → rose; P alone → pea).
- When both dominant alleles occur together (R_P_), a **new phenotype (walnut)** is produced — this is the essence of gene interaction.
- When neither dominant allele is present (rrpp), a **single comb** results.
- Because the four classes still occur in 9:3:3:1 numbers, the Mendelian ratio is **not modified**; only the phenotypes are new. This is called simple (non-epistatic) gene interaction.

16.4 Why this is "without modification"

The test of "modification" is whether the four 9:3:3:1 classes remain distinguishable as four groups. In the fowl comb case they do (walnut, rose, pea, single), so the ratio is 9:3:3:1 and is unmodified. Only when two or more classes **merge** into a single phenotype (as in epistasis) does the ratio change to 12:3:1, 9:3:4, and so on.

Worked example 16.1 — predicting progeny of a walnut × single cross

Problem: A walnut-combed fowl (RrPp) is crossed with a single-combed fowl (rrpp). What comb types and ratio are expected among the progeny?

Solution: Gametes of RrPp: RP, Rp, rP, rp. Gametes of rrpp: rp only.

Gamete of walnut parent	**with rp**	**Genotype**	**Comb**
RP	RrPp	R_P_	Walnut
Rp	Rrpp	R_pp	Rose
rP	rrPp	rrP_	Pea
rp	rrpp	rrpp	Single

Expected ratio = **1 walnut : 1 rose : 1 pea : 1 single** (a 1:1:1:1 test-cross type ratio).

Exam focus — Lecture 16

Key points

1. Non-allelic interaction = effect of genes at different loci on one character.
2. In simple interaction the F_2 is 9:3:3:1 but with new phenotypes.
3. Fowl comb involves two genes, R (rose) and P (pea).
4. R_P_ = walnut (new); R_pp = rose; rrP_ = pea; rrpp = single.
5. Rose × pea → F_1 all walnut; F_2 9:3:3:1 (walnut:rose:pea:single).
6. The ratio is unmodified because the four classes remain distinct.

Short-answer questions with model answers

Q1. What is non-allelic gene interaction?
Model answer: Non-allelic (gene) interaction is the influence of genes at two or more different loci on the expression of the same character, so that the phenotype depends on the combined action of the genes rather than on one gene alone.

Q2. Give the genotypes for the four comb types in fowls.
Model answer: Walnut = R_P_; rose = R_pp; pea = rrP_; single = rrpp.

Q3. Why is the fowl comb ratio called unmodified?
Model answer: Because the F_2 still shows the four Mendelian classes in 9:3:3:1 numbers; the interaction only creates new phenotypes (walnut and single) without merging any of the classes, so the ratio itself is unchanged.

Long-answer questions with model outlines

Q1. Describe Bateson and Punnett's fowl comb experiment and explain the interaction.
Outline: Describe the four comb types; give the cross rose × pea; give F_1 walnut and F_2 9:3:3:1 with a Punnett square; explain the genotype-phenotype relationship; conclude that this is simple interaction without modification of the ratio.

Lecture 17. Non-Allelic Interaction With Modification of Mendelian Ratio (Epistasis)

> **In one line:** Epistasis is the masking of the expression of one gene by another gene at a different locus; it merges the 9:3:3:1 classes into ratios such as 12:3:1, 9:3:4, 9:6:1 and 15:1.

17.1 Epistasis — definition and types

Epistasis is the interaction in which one gene (the **epistatic** gene) suppresses or masks the expression of another gene at a different locus (the **hypostatic** gene). Epistasis is of two kinds:

- **Dominant epistasis:** a single dominant allele of one gene masks the effect of another gene. F_2 ratio **12:3:1**.
- **Recessive epistasis:** the homozygous recessive condition of one gene masks the effect of another gene. F_2 ratio **9:3:4**.

Related interactions are **duplicate interaction** (9:6:1), **duplicate dominant epistasis** (15:1), **duplicate recessive epistasis** (9:7) and **dominant and recessive epistasis** (13:3).

17.2 Dominant epistasis (12 : 3 : 1)

When the dominant allele of one gene masks the expression of another gene, the 9:3:3:1 classes become 12 (9 + 3) : 3 : 1.

Example — fruit colour in summer squash (*Cucurbita pepo*): gene W (white) is epistatic to gene Y (yellow). A plant with W_ is white whatever the Y locus; without W, yy gives green and Y_ gives yellow.

Table 17.1 — Dominant epistasis (12:3:1) in summer squash fruit colour

Genotype	Phenotype	Proportion
W_Y_ and W_yy	White (W masks both)	9 + 3 = 12
wwY_	Yellow	3
wwyy	Green	1

```
DOMINANT EPISTASIS (12 : 3 : 1)

White (WWyy) × Green (wwYY)
      |
F1:   WwYy (white, W is epistatic)
      | self
F2:   12 white (W_) : 3 yellow (wwY_) : 1 green (wwyy)
```

17.3 Recessive epistasis (9 : 3 : 4)

When the homozygous recessive condition of one gene masks the expression of another gene, the classes become 9 : 3 : 4 (3 + 1).

Example — coat colour in mice (or the Labrador retriever): gene C (colour) allows pigment deposition; cc (albino) blocks it. Gene B determines black (B_) or brown (bb).

Table 17.2 — Recessive epistasis (9:3:4) in mouse coat colour

Genotype	Phenotype	Proportion
B_C_ (e.g. black)	Agouti/black	9
bbC_ (brown)	Brown	3
_ _cc (albino)	Albino (cc is epistatic)	3 + 1 = 4

```
RECESSIVE EPISTASIS (9 : 3 : 4)

Black (BBCC) × Albino (bbcc)
      |
F1:  BbCc (black/agouti)
      | self
F2:  9 black (B_C_) : 3 brown (bbC_) : 4 albino (cc)
```

17.4 Duplicate (additive) interaction (9 : 6 : 1)

When two genes affect the same character in the same way and their effects are **additive**, the single-dose classes merge. The 9:3:3:1 becomes 9 : 6 (3 + 3) : 1.

Example — fruit shape in summer squash: disc shape requires at least one dominant allele at either locus; two dominant genes together give disc; a single dominant (in either gene) gives sphere; no dominant gives long.

Table 17.3 — Duplicate (additive) interaction (9:6:1) in summer squash fruit shape

Genotype	Phenotype	Proportion
A_B_	Disc	9
A_bb and aaB_	Sphere	3 + 3 = 6
aabb	Long	1

17.5 Duplicate dominant epistasis (15 : 1)

When a dominant allele at **either** of two loci produces the same phenotype, the 9:3:3:1 classes merge into 15 (9 + 3 + 3) : 1.

Example — capsule (pod) shape in shepherd's purse (*Capsella bursa-pastoris*): triangular capsule requires a dominant allele at either gene; only the double recessive gives the ovoid capsule.

Table 17.4 — Duplicate dominant epistasis (15:1) in *Capsella* capsule shape

Genotype	Phenotype	Proportion
A_B_, A_bb, aaB_	Triangular capsule	9 + 3 + 3 = 15
aabb	Ovoid capsule	1

17.6 Comparison of the ratios

Table 17.5 — Summary of the main epistatic ratios

Interaction	Ratio	Classes merged	Classic example
Dominant epistasis	12 : 3 : 1	9 + 3	Fruit colour in summer squash
Recessive epistasis	9 : 3 : 4	3 + 1	Coat colour in mice/Labrador
Duplicate (additive) interaction	9 : 6 : 1	3 + 3	Fruit shape in summer squash
Duplicate dominant epistasis	15 : 1	9 + 3 + 3	Capsule shape in *Capsella*
Duplicate recessive epistasis	9 : 7	3 + 3 + 1	Flower colour in sweet pea (Lecture 19)
Dominant and recessive epistasis	13 : 3	9 + 3 + 1	Plumage colour in fowl (Lecture 19)

Worked example 17.1 — dominant epistasis numerical

Problem: In summer squash, a cross of white (WwYy) × white (WwYy) gives 400 offspring. How many are expected to be white, yellow and green?

Solution: Ratio = 12 white : 3 yellow : 1 green (total 16).

- White = 12/16 × 400 = **300**
- Yellow = 3/16 × 400 = **75**
- Green = 1/16 × 400 = **25**

Worked example 17.2 — recessive epistasis numerical

Problem: In mice, BbCc × BbCc. Out of 64 offspring, how many are black, brown and albino?

Solution: Ratio = 9 black : 3 brown : 4 albino.

- Black = 9/16 × 64 = **36**
- Brown = 3/16 × 64 = **12**
- Albino = 4/16 × 64 = **16**

Exam focus — Lecture 17

Key points

1. Epistasis = masking of one gene's effect by another gene at a different locus.
2. Dominant epistasis → 12:3:1 (e.g. summer squash fruit colour).
3. Recessive epistasis → 9:3:4 (e.g. mouse/Labrador coat colour).
4. Duplicate (additive) interaction → 9:6:1 (e.g. summer squash fruit shape).
5. Duplicate dominant epistasis → 15:1 (e.g. *Capsella* capsule shape).
6. The epistatic gene masks; the hypostatic gene is masked.
7. All these ratios arise by merging classes of the 9:3:3:1 ratio.

Short-answer questions with model answers

Q1. Define epistasis and name its two chief types with their ratios.
Model answer: Epistasis is the masking of the expression of one gene by another gene at a different locus. Its two chief types are dominant epistasis (12:3:1) and recessive epistasis (9:3:4).

Q2. Explain recessive epistasis with an example.
Model answer: In recessive epistasis the homozygous recessive condition of one gene masks the effect of another gene. In mouse coat colour, the albino gene cc blocks pigment deposition whatever the B locus, so BbCc × BbCc gives 9 black : 3 brown : 4 albino.

Q3. What is duplicate dominant epistasis? Give its ratio and example.
Model answer: Duplicate dominant epistasis occurs when a dominant allele at either of two loci produces the same phenotype; the ratio is 15:1. Example: triangular vs ovoid capsule in *Capsella bursa-pastoris*, where aabb alone gives the ovoid capsule.

Long-answer questions with model outlines

Q1. Describe the different types of epistasis with ratios and examples.
Outline: Define epistasis; describe dominant (12:3:1), recessive (9:3:4), duplicate/additive (9:6:1) and duplicate dominant (15:1); for each give the genotypes, phenotypes and a crop/animal example; summarise in a table.

Lecture 18. Mid Semester Examination

In one line: This lecture slot in the syllabus is the Mid Semester Examination. It is used here as a revision chapter — the pattern of the paper, a formula sheet, and a checklist of everything covered so far (Lectures 1–17).

18.1 Purpose and pattern

The mid semester examination covers the first half of the syllabus (Lectures 1 to 17). It is a written test designed to check the understanding of basic concepts, definitions, comparisons, classical experiments and simple numericals. A typical paper carries a mix of very short answer, short answer and long answer questions, and one or two numerical problems.

Table 18.1 — Suggested revision checklist (Lectures 1–17)

Lecture	Must-know items
1	Definitions; milestones table; Sutton-Boveri; rediscovery year
2	Organelle functions; prokaryote vs eukaryote table; 70S/80S
3	Cell cycle stages; mitosis vs meiosis table; crossing over stage
4	Sporogenesis vs gametogenesis; double fertilisation; twins
5	Nucleosome; centromere; telomere; euchromatin vs heterochromatin; karyotype
6	Four shapes; polytene; lampbrush; B; ring; isochromosome
7	Deletion, duplication, inversion, translocation + pairing figures
8	Aneuploid types; non-disjunction; Klinefelter (47,XXY); Turner (45,X)
9	Haploid/DH; auto- vs allopolyploid; wheat; Triticale; cotton; tobacco; Brassica
10	Pre-Mendelian theories; germplasm theory; mutation theory
11	Model organisms and their contributions
12	Probability rules; chi-square formula and steps
13	Mendel's laws; mono- and dihybrid ratios; rediscovery
14	Terminologies; test cross vs back cross; 2^n, 3^n rules
15	Chromosomal theory; allelic interactions; threshold characters
16	Fowl comb; simple interaction
17	Epistasis ratios 12:3:1, 9:3:4, 9:6:1, 15:1

18.2 Formula and ratio sheet

Table 18.2 — Ratios and formulae to memorise

Item	Value / formula
Monohybrid F_2 (phenotypic)	3 : 1
Monohybrid F_2 (genotypic)	1 : 2 : 1
Dihybrid F_2	9 : 3 : 3 : 1
Trihybrid F_2	27 : 9 : 9 : 9 : 3 : 3 : 3 : 1
Monohybrid test cross	1 : 1
Dihybrid test cross	1 : 1 : 1 : 1
Gamete types	2^n
F_2 genotypes	3^n
F_2 phenotypes (complete dominance)	2^n
Chi-square	$\chi^2 = \Sigma(O-E)^2/E$
Degrees of freedom	Number of classes − 1
χ^2 table (P=0.05), df 1 / 2 / 3	3.84 / 5.99 / 7.82

18.3 Common mistakes to avoid

- Confusing the **phenotypic** ratio (3:1) with the **genotypic** ratio (1:2:1).
- Writing a test cross as a cross to the dominant parent — it must be to the recessive parent.
- Forgetting that in incomplete dominance the F_2 phenotypic ratio equals the genotypic ratio.
- Mixing up dominant epistasis (12:3:1) with recessive epistasis (9:3:4).
- Using the wrong degrees of freedom in the chi-square test.
- Writing the genome formula of wheat as AABB instead of AABBDD.

Exam focus — Lecture 18

Key points

1. The mid semester paper covers Lectures 1–17.
2. Revise definitions, comparison tables, classical experiments and ratios.
3. Practise chi-square and Punnett square numericals.
4. Learn the standard examples (wheat, cotton, tobacco, Brassica, fowl comb, squash, mice).
5. Keep answers neat: state the law, give the cross, show the ratio, give the example.

Short-answer questions with model answers

Q1. List any five ratios a first-half student must remember.

Model answer: Monohybrid F_2 3:1; monohybrid genotypic 1:2:1; dihybrid F_2 9:3:3:1; monohybrid test cross 1:1; dihybrid test cross 1:1:1:1.

Q2. State one classical experiment from Lectures 1–17 and its conclusion.

Model answer: Bateson and Punnett's fowl comb experiment — rose × pea gave all walnut F_1 and a 9:3:3:1 F_2 (walnut:rose:pea:single), showing that two genes can interact to produce a new phenotype without changing the Mendelian ratio.

Long-answer questions with model outlines

Q1. Prepare a revision plan for the mid semester examination covering Lectures 1–17.

Outline: Group the lectures into blocks (cell and chromosomes; Mendelism and probability; interactions); list the key tables and numericals for each block; allocate time; end with a self-test using the ratio sheet.

Lecture 19. Duplicate Recessive Epistasis (9:7); Dominant and Recessive Epistasis (13:3); Summary of Epistatic Ratios

In one line: Two more epistatic ratios complete the picture: duplicate recessive epistasis (9:7), where both genes must have a dominant allele for the character to appear, and dominant and recessive epistasis (13:3), where a dominant allele at one locus and a recessive at the other act together.

19.1 Duplicate recessive epistasis (9 : 7)

In this interaction the character appears only when a dominant allele is present at **both** loci (A_B_). If either locus is homozygous recessive (A_bb, aaB_, aabb), the character is not expressed. The classes merge into 9 : 7 (3 + 3 + 1).

Example — flower colour in sweet pea (*Lathyrus odoratus*), Bateson and Punnett: two genes, C and P, are both needed to make the purple pigment. If either is homozygous recessive, the flower is white.

Table 19.1 — Duplicate recessive epistasis (9:7) in sweet pea flower colour

Genotype	Phenotype	Proportion
C_P_	Purple (both dominant present)	9
C_pp, ccP_, ccpp	White	3 + 3 + 1 = 7

```
DUPLICATE RECESSIVE EPISTASIS (9 : 7)

Purple (CCpp) × White (ccPP)
      |
F1:  CcPp (purple)
      | self
F2:  9 purple (C_P_) : 7 white (C_pp, ccP_, ccpp)
```

19.2 Dominant and recessive epistasis (13 : 3)

In this interaction a **dominant allele at one locus** and the **homozygous recessive condition at the other locus** both produce the same effect (they are epistatic to the expression of the character), so the classes merge into 13 (9 + 3 + 1) : 3.

Example — plumage (feather) colour in the fowl (white Leghorn × white Wyandotte): the dominant allele I (inhibitor of colour) and the recessive allele cc both prevent colour, giving white; only the genotype C_ii gives coloured plumage.

Table 19.2 — Dominant and recessive epistasis (13:3) in fowl plumage

Genotype	Phenotype	Proportion
I_C_, I_cc, ii cc	White	9 + 3 + 1 = 13
iiC_	Coloured	3

19.3 Summary of all epistatic (modified) ratios

Table 19.3 — Complete summary of gene interaction ratios

Type of interaction	F_2 ratio	Class merged	Genotype giving the "special" class	Example
Simple interaction (no modification)	9 : 3 : 3 : 1	None	—	Fowl comb
Dominant epistasis	12 : 3 : 1	9 + 3	W_ masks Y	Summer squash fruit colour
Recessive epistasis	9 : 3 : 4	3 + 1	cc masks B	Mouse/Labrador coat
Duplicate (additive) interaction	9 : 6 : 1	3 + 3	A_B_ (additive)	Summer squash fruit shape
Duplicate dominant epistasis	15 : 1	9 + 3 + 3	aabb alone	*Capsella* capsule
Duplicate recessive epistasis	9 : 7	3 + 3 + 1	C_P_ required	Sweet pea flower colour
Dominant and recessive epistasis	13 : 3	9 + 3 + 1	iiC_ alone coloured	Fowl plumage

19.4 How to identify the interaction from a ratio

- If the total is 16 and the ratio is 9:3:3:1 — no interaction (simple) or duplicate dominant (if two classes merge).
- If the largest class is 12 — dominant epistasis (12:3:1).
- If there are two equal middle classes of 6 — duplicate/additive interaction (9:6:1).
- If the largest class is 15 — duplicate dominant epistasis (15:1).
- If the largest class is 9 and the rest total 7 — duplicate recessive epistasis (9:7).
- If the largest class is 13 — dominant and recessive epistasis (13:3).

Worked example 19.1 — identifying an interaction from data

Problem: In a cross between two purple-flowered sweet pea plants, the progeny were 118 purple and 89 white. What interaction is involved, and what is the expected ratio?

Solution: Total = 207. If the ratio is 9:7, expected purple = 9/16 × 207 = 116.4 and white = 7/16 × 207 = 90.6. The observed (118, 89) is very close. Test:

Class	O	E	$(O-E)^2/E$
Purple	118	116.4	0.022
White	89	90.6	0.028
$\chi^2 = 0.050$			df = 1; table 3.84

Since 0.050 < 3.84, the 9:7 ratio is accepted — the interaction is **duplicate recessive epistasis**.

Exam focus — Lecture 19

Key points

1. Duplicate recessive epistasis → 9:7; character needs a dominant allele at both loci (sweet pea).
2. Dominant and recessive epistasis → 13:3; both a dominant and a recessive condition can mask (fowl plumage).
3. All modified ratios are regroupings of 9:3:3:1 (total 16).
4. Recognise the interaction from the largest class (12, 13, 15, 9).
5. Always confirm with a chi-square test.

Short-answer questions with model answers

Q1. Explain duplicate recessive epistasis with an example and ratio.
Model answer: In duplicate recessive epistasis the character is expressed only when a dominant allele is present at both loci (A_B_); if either locus is homozygous recessive the character is absent, giving a 9:7 ratio. Example: purple flower colour in sweet pea requires both C and P; otherwise the flower is white.

Q2. What is the 13:3 ratio and in which organism was it demonstrated?
Model answer: The 13:3 ratio arises from dominant and recessive epistasis, in which a dominant allele at one locus and the homozygous recessive condition at another locus both prevent the expression of the character. It was demonstrated in plumage colour of the fowl.

Q3. If a cross gives 15 : 1, what interaction is it?
Model answer: Duplicate dominant epistasis, in which a dominant allele at either of two loci produces the same phenotype and only the double recessive gives the other phenotype.

Long-answer questions with model outlines

Q1. Give a complete summary of gene interaction ratios with examples.
Outline: Reproduce Table 19.3; explain each ratio briefly; give the genotype behind each special class; state the example; mention how the ratios arise from regrouping 9:3:3:1.

Lecture 20. Qualitative and Quantitative Traits; Polygenes; Multiple Factor Hypothesis

In one line: Qualitative traits fall into distinct classes and are controlled by one or a few genes; quantitative traits show continuous variation and are controlled by many genes (polygenes) each with a small effect.

20.1 Qualitative (discontinuous) traits

- Show **discontinuous variation** — individuals fall into clear-cut classes (tall or dwarf, red or white).
- Controlled by **one or a few genes** with large, easily recognised effects.
- Little influenced by the environment; the phenotype clearly reveals the genotype.
- Studied by counting and using Mendelian ratios (qualitative genetics).
- *Examples:* flower colour, seed shape, disease resistance (as a class), blood groups, comb shape.

20.2 Quantitative (continuous) traits

- Show **continuous variation** — individuals grade from one extreme to the other without distinct classes (plant height, grain yield, days to flowering).
- Controlled by **many genes (polygenes)**, each with a small effect, plus a large environmental effect.
- Measured, not counted; studied by statistical methods (means, variance, heritability).
- Also called **metric traits** or **polygenic traits**.

Table 20.1 — Qualitative versus quantitative traits

Feature	Qualitative (discontinuous)	Quantitative (continuous)
Variation	Discontinuous, distinct classes	Continuous, graded
Number of genes	One or a few	Many (polygenes)
Effect of each gene	Large	Small
Environmental effect	Small	Large
Method of study	Counting, Mendelian ratios	Measurement, statistics
Examples	Flower colour, seed shape, blood group	Yield, height, maturity, protein content

20.3 Polygenes and continuous variation

Polygenes (multiple genes, minor genes) are genes with small and similar effects that together control a quantitative character. Each polygene follows Mendelian inheritance, but because many of them act together (and the environment adds its own variation), the phenotype is distributed continuously. When the number of polygenes increases, the number of phenotypic classes increases and the distribution approaches a **normal (bell-shaped) curve**.

20.4 Transgressive segregation

Transgressive segregation is the appearance in the F_2 (or later generations) of individuals that are **more extreme than either parent** for a quantitative character. It happens because the favourable polygenes for a character are distributed between the two parents, and recombination in the F_2 can bring a greater number of favourable alleles together in one individual than either parent possessed. It is important in plant breeding, because it allows selection of lines superior to both parents.

```
TRANSGRESSIVE SEGREGATION
Parent A (has + + - - -)      Parent B (has - - - + +)
     │                                  │
     └──────────── F1 ──────────────────┘   (intermediate)
                     │ self
F2:  some individuals have + + - + +  ← MORE favourable alleles
     than either parent → transgressive (superior) segregants
```

20.5 Multiple factor hypothesis

Proposed by **Herman Nilsson-Ehle** (1908–09) to explain the inheritance of quantitative characters. Its main points are:

- A quantitative character is controlled by **two or more pairs of genes** (multiple factors).
- Each gene has a **small, cumulative (additive) effect** on the character.
- The genes are inherited in a Mendelian manner and **assort independently**.
- The effect of each dominant (or plus) allele adds to the character; the more such alleles present, the more extreme the phenotype.
- Because many genes and the environment act together, the variation is **continuous** and the F_2 shows a graded series of classes.

20.6 Nilsson-Ehle's experiment on wheat kernel colour

Nilsson-Ehle crossed red-kernel and white-kernel wheat. He found that the F_2 ratio depended on the number of gene pairs involved:

- With **one** gene pair, the F_2 gave 3 red : 1 white.
- With **two** gene pairs, the F_2 gave a ratio of **15 red : 1 white** (with several shades of red among the red class).
- With **three** gene pairs, the F_2 gave **63 red : 1 white**.

The number of shades and the ratios depend on the number of "plus" (contributing) alleles: with n gene pairs the F_2 shows (2n + 1) phenotypic classes, and the extreme classes are $(1/4)^n$ of the total. This showed that quantitative characters are inherited in a Mendelian way but through many genes.

Table 20.2 — Wheat kernel colour: number of genes and F_2 ratios

Number of gene pairs (n)	Extreme (white) class	F_2 ratio (coloured : white)	Phenotypic classes (2n+1)
1	1/4	3 : 1	3
2	1/16	15 : 1	5
3	1/64	63 : 1	7

20.7 Corolla length in *Nicotiana longiflora*

East and Jones (and earlier work on *Nicotiana longiflora*) studied the length of the corolla (flower tube) in tobacco. The character showed a **continuous, normal distribution** in the F_2, with the F_1 intermediate between the two parents. The variation was polygenic — many genes each with a small effect controlled corolla length, and the environment added further variation. This study, along with Nilsson-Ehle's wheat work, established the multiple factor (polygenic) basis of quantitative characters.

20.8 Significance in plant breeding

- Yield and quality traits are quantitative; their improvement depends on selection, which requires genetic variation and heritability.
- Transgressive segregation helps to obtain lines superior to both parents.
- Understanding polygenes helps in the choice of breeding methods (pure line selection, mass selection, hybridisation and selection).
- Polygenic disease resistance (slow rusting, field resistance) is more durable than single-gene resistance.

Worked example 20.1 — number of phenotypic classes

Problem: A quantitative character is controlled by four independent gene pairs (n = 4). How many phenotypic classes are expected in the F_2, and what fraction would be like the extreme parent?

Solution: Number of phenotypic classes = $2n + 1 = 2(4) + 1 =$ **9 classes**. Fraction like the extreme parent = $(1/4)^n = (1/4)^4 =$ **1/256**.

Worked example 20.2 — transgressive segregation logic

Problem: Two wheat varieties each give a moderate yield. Variety A carries the plus alleles at genes 1 and 2 (+ + − −), variety B at genes 3 and 4 (− − + +). What is the most favourable genotype obtainable in the F_2?

Solution: The most favourable genotype would combine all four plus alleles (+ + + +), which neither parent has. Such a combination can be recovered in the F_2 by recombination — an example of **transgressive segregation**. Its probability is $(1/4)^4 = 1/256$ for the fully homozygous plus genotype at all four loci.

Exam focus — Lecture 20

Key points

1. Qualitative traits: few genes, discontinuous variation, little environmental effect.
2. Quantitative traits: many polygenes, continuous variation, large environmental effect.
3. Polygenes have small, cumulative (additive) effects.
4. Transgressive segregation gives individuals more extreme than either parent.
5. Multiple factor hypothesis (Nilsson-Ehle): many genes, each additive, inherited in Mendelian fashion.
6. Wheat kernel colour: 1 gene → 3:1; 2 genes → 15:1; 3 genes → 63:1; classes = 2n + 1; extreme = $(1/4)^n$.
7. Corolla length in *Nicotiana longiflora* shows a continuous, polygenic distribution.

Short-answer questions with model answers

Q1. Differentiate qualitative from quantitative characters.
Model answer: Qualitative characters show discontinuous variation, are controlled by one or a few genes with large effects and are little influenced by the environment (e.g. flower colour). Quantitative characters show continuous variation, are controlled by many polygenes with small effects and are strongly influenced by the environment (e.g. grain yield).

Q2. What are polygenes?
Model answer: Polygenes are genes with small, similar and cumulative (additive) effects that together control a quantitative character; they are inherited in a Mendelian way but act in large numbers, producing continuous variation.

Q3. What is transgressive segregation and why is it useful?
Model answer: Transgressive segregation is the appearance of F_2 individuals more extreme than either parent for a quantitative character, caused by recombination bringing together more favourable polygenes than either parent had. It is useful because it allows selection of lines superior to both parents.

Long-answer questions with model outlines

Q1. Explain the multiple factor hypothesis with Nilsson-Ehle's wheat experiment.
Outline: State the hypothesis; describe the wheat kernel colour crosses for one, two and three gene pairs; give the ratios 3:1, 15:1, 63:1; explain (2n + 1) classes and $(1/4)^n$ extremes; conclude that quantitative inheritance is Mendelian but polygenic.

Q2. Describe the inheritance of quantitative characters with the *Nicotiana* corolla example.
Outline: Define quantitative characters and polygenes; describe continuous variation and the normal curve; present the *Nicotiana longiflora* corolla-length study; relate to plant breeding applications.

Lecture 21. Lethal Genes, Pleiotropism, Penetrance and Expressivity, Multiple Alleles, Blood Groups, Coat Colour in Rabbits, Self-Incompatibility, Pseudoalleles and Isoalleles

> **In one line:** This lecture covers gene effects that go beyond simple dominance — genes that kill, genes with many effects, genes whose expression varies, and genes with more than two alleles.

21.1 Lethal genes

A **lethal gene** is a gene that causes the death of the organism carrying it, usually when it is homozygous. Such genes are usually recessive, and their effect may act at any stage from the zygote to maturity. A **sub-lethal** gene kills some but not all individuals; a **semi-lethal** gene kills less than half; a **vital** gene has no lethal effect.

- **Recessive lethal:** kills only in the homozygous recessive condition; heterozygotes are normal. Example: the yellow coat colour gene (Y) in mice — YY is lethal, Yy is yellow, yy is normal agouti. A cross of two yellow mice (Yy × Yy) gives **2 yellow : 1 normal** (the 1/4 YY dies), a modified 2:1 ratio.
- **Dominant lethal:** kills even in the heterozygous condition; therefore it cannot be transmitted (because the carrier dies before reproducing). Example: Huntington's disease in humans acts late, so it is transmitted before death.
- **In plants:** chlorophyll-deficient (albino) seedlings, in maize and many crops, are recessive lethals; the seedling dies after the food reserve in the seed is exhausted. This is why a monohybrid cross for such a gene gives a 2:1 ratio among the surviving plants.

Table 21.1 — Effect of lethal genes on ratios

Situation	Cross	Normal ratio	Ratio with lethal gene
Recessive lethal (yellow mice)	Yy × Yy	3 : 1	2 yellow : 1 agouti
Recessive lethal (albino seedling)	Gg × Gg	3 : 1	2 green : 1 (albino dies)
Dominant lethal	not transmitted	—	carrier dies before reproduction

21.2 Pleiotropism

Pleiotropism is the ability of a **single gene to affect several apparently unrelated characters**. The gene is called a pleiotropic gene.

- Example: in *Drosophila*, the white-eye gene affects eye colour, and also fertility and the shape of the testes.
- In pea, the gene for purple flower colour also affects the colour of the seed coat and the leaf axil.
- In sickle-cell anaemia in humans, a single gene change affects the red blood cells and many organs.
- In maize, the gene for a plant colour also affects resistance to disease.

Pleiotropism arises because the gene product takes part in many pathways, so one change is expressed in several parts of the plant.

21.3 Penetrance and expressivity

- **Penetrance** is the **proportion of individuals with a given genotype that actually show the expected phenotype**. If all show it, penetrance is 100%; if only 80% show it, penetrance is 80% (the gene is incompletely penetrant).
- **Expressivity** is the **degree or intensity with which the phenotype is expressed** among those individuals that do show it. A gene may have variable expressivity, so the same genotype produces a mild or a severe form.

Both are caused by differences in the genetic background and in the environment. Example: the polydactyly gene in humans has incomplete penetrance (not every carrier has extra fingers) and variable expressivity (the extra digit may be a small tag or a full digit).

21.4 Multiple alleles

Multiple alleles are more than two alternative forms of a gene present in a population, though any one diploid individual can carry only two of them. Multiple alleles arise by repeated mutation of a gene and occupy the same locus. They show a definite **order of dominance** (an allelic series).

- Example: coat colour in rabbits (C series), blood groups in humans (I^A, I^B, i), and self-incompatibility in plants (S series).

Coat colour in rabbits (multiple allelic series)

The gene C has four alleles with the dominance order **C > c^{ch} > c^h > c**:

Table 21.2 — Multiple alleles for coat colour in rabbits

Allele	Phenotype	Phenotype in homozygous form
C (agouti)	Full colour (agouti/grey)	Agouti
c^{ch} (chinchilla)	Chinchilla (light grey)	Chinchilla
c^h (Himalayan)	Himalayan (white body, black extremities)	Himalayan
c (albino)	Albino (no pigment)	Albino

The dominance order is C > c^{ch} > c^h > c. A Himalayan (c^hc) crossed with an albino (cc) gives all Himalayan; the temperature-sensitive expression of the Himalayan allele is a classic example of the effect of environment on a gene.

21.5 Blood group genetics in humans

The ABO blood groups are controlled by a single gene with **three alleles**: I^A, I^B and i (I^O). I^A and I^B are codominant with each other, and both are dominant over i.

Table 21.3 — ABO blood groups, genotypes and antigens

Blood group	Possible genotypes	Antigen on red cells	Antibody in serum
A	I^AI^A, I^Ai	A	Anti-B
B	I^BI^B, I^Bi	B	Anti-A
AB	I^AI^B	A and B	None
O	ii	None	Anti-A and anti-B

The ABO locus shows **multiple alleles** and **codominance** at the same time. Blood groups are used in blood transfusion, in paternity testing and in population genetics. (The Rh factor, + or −, is controlled by a different gene.)

> **Worked example 21.1 — blood group inheritance**
>
> **Problem:** A mother of blood group O has a child of blood group A. Which blood groups are possible for the father?
>
> **Solution:** Mother = ii, so she gives only i. For the child to be A, it must be I^Ai, and it must receive I^A from the father. So the father must carry I^A — he can be group A (I^AI^A or I^Ai) or group AB (I^AI^B). He cannot be group O or group B (unless he is I^Bi, which has no I^A). So the father is **group A or group AB**.

21.6 Self-incompatibility in plants

Self-incompatibility is the failure of a fertile plant to set seed when self-pollinated. It is controlled by **S (self-incompatibility) genes** with multiple alleles (S_1, S_2, S_3...). It is a mechanism that prevents inbreeding and promotes cross-pollination.

- **Gametophytic self-incompatibility:** the pollen tube's growth is arrested if the S allele of the pollen (haploid) matches either S allele of the pistil. Example: *Nicotiana*, *Petunia*, and many Solanaceae; also the classic *Nicotiana* work.
- **Sporophytic self-incompatibility:** the incompatibility reaction depends on the S genotype of the pollen parent (sporophyte); pollen is rejected if it carries an S allele present in the pistil. Example: *Brassica*, *Antirrhinum*.

Self-incompatibility is used in hybrid seed production (for example, in *Brassica* vegetables) to prevent selfing and force crossing.

21.7 Pseudoalleles and isoalleles

- **Pseudoalleles:** closely linked genes that are so near each other that they behave almost like alleles of the same gene. They recombine only rarely (with a very low crossing-over frequency) and give a "compound" phenotype. Example: the *lozenge* (lz) locus in *Drosophila*; the *white* eye locus. Pseudoalleles show that what appears to be one gene can be a small cluster of related genes (a complex locus).
- **Isoalleles:** different alleles that produce the **same phenotype** but are distinguishable by their slightly different quantitative effect (for example, different intensities of the same colour). They occupy the same locus and cannot be told apart by ordinary dominance tests, only by their effect or by their product. Example: different isoalleles at the *white* locus in *Drosophila* give different amounts of eye pigment.

Exam focus — Lecture 21

Key points

1. Lethal genes kill the homozygote; recessive lethal in yellow mice gives a 2:1 ratio.
2. Pleiotropism = one gene affects many characters.
3. Penetrance = proportion showing the phenotype; expressivity = degree of expression.
4. Multiple alleles = more than two forms of a gene in a population; e.g. ABO blood groups, rabbit coat colour.
5. ABO blood groups: I^A and I^B codominant, both dominant over i.
6. Rabbit coat colour dominance order: C > c^{ch} > c^h > c.
7. Self-incompatibility (S alleles) may be gametophytic or sporophytic.
8. Pseudoalleles are closely linked genes behaving like alleles; isoalleles give the same phenotype with different intensity.

Short-answer questions with model answers

Q1. What is a lethal gene? Give an example and the modified ratio it produces.
Model answer: A lethal gene causes the death of the organism carrying it, usually in the homozygous condition. In mice the yellow gene Y is a recessive lethal: Yy × Yy gives 2 yellow : 1 agouti, because YY dies before birth.

Q2. Differentiate penetrance from expressivity.
Model answer: Penetrance is the proportion of individuals with a given genotype that show the expected phenotype; expressivity is the degree or intensity of the phenotype among those that show it.

Q3. Explain multiple alleles with the ABO blood group.
Model answer: The ABO blood group is controlled by a gene with three alleles — I^A, I^B and i. I^A and I^B are codominant and both are dominant over i, giving four phenotypes: A (I^AI^A, I^Ai), B (I^BI^B, I^Bi), AB (I^AI^B) and O (ii).

Long-answer questions with model outlines

Q1. Describe lethal genes, pleiotropism, penetrance and expressivity with examples.
Outline: Define each term; give recessive and dominant lethal examples and modified ratios; give pleiotropism examples; explain penetrance and expressivity with polydactyly; conclude with the importance of these concepts in breeding and in human genetics.

Q2. Explain multiple alleles with reference to blood groups and coat colour in rabbits.
Outline: Define multiple alleles and allelic series; describe the ABO system (genotypes, antigens, antibodies, codominance); describe the rabbit C series (four alleles, dominance order, phenotypes); mention self-incompatibility S alleles; give the significance of multiple alleles.

Lecture 22. Crossing Over — Mechanism, Significance, Cytological Proof (Stern's Experiment) and Factors

> **In one line:** Crossing over is the exchange of segments between non-sister chromatids of homologous chromosomes during pachytene of meiosis, producing new gene combinations.

22.1 Definition and occurrence

Crossing over is the interchange of corresponding (homologous) segments between two **non-sister chromatids** of a pair of homologous chromosomes. It occurs during **pachytene of prophase I** of meiosis, when the homologues are paired as bivalents. The visible result is the **chiasma** seen at diplotene.

```
CROSSING OVER (between non-sister chromatids)

Before:   ═══A═══B═══          (chromatid 1, from parent A)
          ═══a═══b═══          (chromatid 2, from parent A)
          ———A———B———          (chromatid 3, from parent B)
          ———a———b———          (chromatid 4, from parent B)

After (exchange between chromatid 2 and chromatid 3):
          ═══A═══B═══
          ═══a═══B═══   ← recombinant
          ———A———b———   ← recombinant
          ———a———b———
→ 2 parental and 2 recombinant chromatids
```

22.2 Mechanism of crossing over

The generally accepted mechanism is the **breakage and reunion** model:

1. **Synapsis:** homologous chromosomes pair during zygotene, forming a bivalent with a synaptonemal complex.
2. **Breakage:** at pachytene, corresponding chromatids break at identical points (or the DNA strands nick).
3. **Exchange (reunion):** the broken ends of non-sister chromatids rejoin in a crossed manner.
4. **Chiasma:** the point of exchange remains visible as a chiasma at diplotene, and terminalises at diakinesis.

At the molecular level, crossing over begins with a **double-strand break** in one chromatid, followed by strand invasion and resolution, giving a **Holliday junction**. The result is two parental and two recombinant chromatids.

22.3 Significance of crossing over

- **Produces new gene combinations (recombination):** this is a major source of genetic variation, on which selection, evolution and plant breeding depend.
- **Provides proof of linkage and gene order:** the frequency of crossing over is used to measure the distance between genes and to construct genetic maps.
- **Balances linkage:** it breaks unfavourable linkages and brings favourable alleles together.

- **Explains the formation of new varieties** by recombination of useful characters from different parents.
- Its absence (as in male *Drosophila* and in the heterogametic sex of some animals) is used as a tool in genetic analysis.

22.4 Cytological proof of crossing over — Stern's experiment (1931)

Curt Stern, working with *Drosophila melanogaster*, gave the first cytological proof that crossing over involves an actual exchange of chromosome segments. He used a female fly that was **heterozygous for two X-linked genes** — *carnation* (car, eye colour) and *Bar* (B, eye shape) — and whose two X chromosomes were **structurally different and could be recognised under the microscope** (one X carried a translocated piece of the Y chromosome at one end and was shorter at the other; the other X was normal).

By examining both the **phenotype** (eye colour and shape) and the **cytology** (the shape of the X chromosomes) of the progeny, Stern showed that whenever a recombinant phenotype appeared, the corresponding X chromosome was also a recombinant (structurally exchanged) chromosome. This proved that crossing over is the exchange of chromosomal segments.

```
STERN'S EXPERIMENT
Parent X chromosomes:
  X(car)  =  ———[car]——— with a translocated tip and a short end
  X(B)    =  ————————————— normal
Progeny:
  Non-recombinant phenotype → non-recombinant chromosome shape
  Recombinant phenotype     → recombinant chromosome shape
Conclusion: phenotype and chromosome structure change together = crossing over is real
```

22.5 Factors controlling crossing over

Table 22.1 — Factors affecting the frequency of crossing over

Factor	Effect
Distance between genes	Crossing over increases with distance; frequency is a measure of distance
Temperature	Extremes (very high or low) generally reduce crossing over; a suitable rise may increase it
Age of the organism	Older individuals may show altered (usually reduced) crossing over
Sex	Absent in male *Drosophila* and in some species; sex may change the frequency
Structural changes	Inversions and translocations suppress crossing over (crossover suppressors)
Radiation and chemicals	X-rays and some chemicals increase crossing over
Nutrition and water	Deficiency of certain nutrients (e.g. calcium, magnesium) may alter it
Centromere and heterochromatin	Crossing over is reduced near the centromere and in heterochromatic regions
Genotype	Specific genes control the frequency of crossing over

Worked example 22.1 — identifying recombinants

Problem: In a test cross of AB/ab × ab/ab, the progeny are: AB 420, ab 430, Ab 78, aB 72. Which are the parental and recombinant types, and what is the recombination frequency?

Solution: The most frequent classes (AB 420 and ab 430) are the **parental (non-recombinant)** types. The less frequent classes (Ab 78 and aB 72) are the **recombinant** types.

Total = 420 + 430 + 78 + 72 = 1000. Recombinants = 78 + 72 = 150.

Recombination frequency = 150/1000 × 100 = **15%** (15 map units).

Exam focus — Lecture 22

Key points

1. Crossing over is the exchange of segments between non-sister chromatids of homologous chromosomes.
2. It occurs at pachytene of prophase I; the visible result is the chiasma.
3. Mechanism: breakage and reunion (molecularly, double-strand break and Holliday junction).
4. It produces two parental and two recombinant chromatids.
5. Significance: variation, proof of linkage, gene mapping, breaking unfavourable linkages.
6. Stern's experiment (1931) gave cytological proof in *Drosophila*.
7. Frequency is affected by distance, temperature, sex, structure, radiation and genotype.

Short-answer questions with model answers

Q1. What is crossing over and when does it occur?
Model answer: Crossing over is the exchange of corresponding segments between non-sister chromatids of a pair of homologous chromosomes. It occurs during pachytene of prophase I of meiosis and becomes visible as a chiasma at diplotene.

Q2. How did Stern prove crossing over cytologically?
Model answer: Stern used a *Drosophila* female heterozygous for two X-linked genes and with two structurally distinguishable X chromosomes. He showed that every recombinant phenotype was accompanied by a recombinant chromosome structure, proving that crossing over involves the actual exchange of chromosome segments.

Q3. State any four factors that affect crossing over.
Model answer: Distance between genes, temperature, sex (absent in male *Drosophila*) and structural changes such as inversions.

Long-answer questions with model outlines

Q1. Describe the mechanism, significance and cytological proof of crossing over.
Outline: Define crossing over; describe synapsis, breakage, reunion and chiasma; explain its significance (variation, mapping, linkage breaking); describe Stern's experiment and its conclusion; mention the factors affecting it.

Lecture 23. Linkage and Its Estimation — Coupling and Repulsion, Bateson and Punnett's Experiment, Morgan's Chromosomal Theory

> **In one line:** Linkage is the tendency of genes on the same chromosome to be inherited together; it reduces recombination and is the reason many characters do not assort independently.

23.1 Linkage — definition

Linkage is the tendency of two or more genes located on the **same chromosome** to be transmitted together to the offspring, so that they do not assort independently. The genes involved are called **linked genes**, and the phenomenon is called linkage. It was discovered by **Bateson and Punnett** (1906) in the sweet pea and explained by **T. H. Morgan** (1911) on the basis of chromosomes.

23.2 Bateson and Punnett's experiment (coupling and repulsion)

Bateson and Punnett crossed sweet pea plants differing in two characters: **flower colour** (purple, P, dominant; red, p, recessive) and **pollen shape** (long, L, dominant; round, l, recessive).

Coupling phase

When the dominant alleles P and L were present in the **same parent** (PL/PL) and the recessives in the other (pl/pl), the F_1 (PpLl) was selfed. Instead of the 9:3:3:1 ratio, Bateson and Punnett obtained a large excess of the parental types (purple-long and red-round) and very few recombinants (purple-round and red-long). This is called the **coupling phase** (cis arrangement), because the dominants are coupled together in one parent.

Repulsion phase

When each parent carried one dominant and one recessive (Pl/Pl × pL/pL), the parental types were purple-round and red-long, again in excess, with few recombinants. This is the **repulsion phase** (trans arrangement), because the dominants are "repelled" into different parents.

Table 23.1 — Coupling and repulsion in sweet pea (Bateson and Punnett)

Phase	Parental combination	Excess (parental) types	Deficient (recombinant) types
Coupling (cis)	PL/PL × pl/pl	Purple-long, red-round	Purple-round, red-long
Repulsion (trans)	Pl/Pl × pL/pL	Purple-round, red-long	Purple-long, red-round

```
LINKAGE IN SWEET PEA (deviates from 9:3:3:1)

Coupling:  F1 = PL/pl  (P and L on one chromosome, p and l on the other)
           Gametes: mostly PL and pl (parental)
                    rarely Pl and pL (recombinant)
Result:    F2 shows an excess of purple-long and red-round
           → not 9:3:3:1, but a distorted ratio showing linkage
```

23.3 Chromosomal theory of linkage — Morgan

T. H. Morgan (1911), working with *Drosophila*, explained linkage on the basis of chromosomes. His theory states:

- Genes that are located on the **same chromosome** tend to be inherited together and are said to be linked.
- The **strength of linkage** depends on the distance between the genes: genes closer together are linked more strongly; genes farther apart are linked more weakly.
- Linked genes are arranged **in a linear order** on the chromosome.
- Linkage is broken by **crossing over**, which produces recombinants.
- Genes on different chromosomes show independent assortment; genes on the same chromosome show linkage.

Morgan and Sturtevant used the frequency of recombination to map genes (see Lecture 25).

23.4 Complete and incomplete linkage

Table 23.2 — Complete versus incomplete linkage

Feature	Complete linkage	Incomplete linkage
Definition	Linked genes always stay together; no recombinants are produced	Linked genes mostly stay together but some recombinants are produced
Crossing over	Does not occur between the genes	Occurs between the genes
Progeny	Only parental types	Mostly parental, few recombinant types
Distance	Genes very close together	Genes some distance apart
Example	Genes in male *Drosophila* (no crossing over)	Most linked genes in practice

Complete linkage gives only the parental types, so the test-cross ratio is 1:1 (two classes only). **Incomplete linkage** gives two large parental classes and two small recombinant classes.

23.5 Linkage group

A **linkage group** is the set of all genes located on one chromosome. The number of linkage groups equals the **haploid chromosome number (n)** of the species.

- Garden pea: n = 7 → 7 linkage groups.
- Maize: n = 10 → 10 linkage groups.
- *Drosophila*: n = 4 → 4 linkage groups.
- Human: n = 23 → 23 linkage groups (22 autosomes + 1 sex chromosome).

Linkage groups are identified by the symbol of the chromosome; the genes in a linkage group are arranged in a linear order and mapped by recombination frequencies.

23.6 Estimation of linkage

Linkage is estimated by the **test cross**: the F_1 heterozygote is crossed with the double recessive. The proportion of recombinant progeny gives the recombination frequency, which measures the distance between the genes.

Recombination frequency (%) = (Number of recombinants / Total progeny) × 100

A recombination frequency of 50% (or the 1:1:1:1 ratio) means the genes are unlinked (on different chromosomes or far apart); a frequency below 50% indicates linkage.

Worked example 23.1 — estimating linkage from a test cross

Problem: In maize, a plant heterozygous for coloured (C) and full (Sh) kernels was test crossed to a colourless-shrunken parent. The progeny were: coloured-full 4032, colourless-shrunken 4035, coloured-shrunken 149, colourless-full 152. Test whether the genes are linked and estimate the recombination frequency.

Solution: Total = 4032 + 4035 + 149 + 152 = 8368. The two large classes are parental (C-Sh and c-sh); the two small classes are recombinants (C-sh and c-Sh).

Recombinants = 149 + 152 = 301. Recombination frequency = 301/8368 × 100 = **3.6%**.

Since the frequency is far below 50% (and the ratio is not 1:1:1:1), the genes are **linked**, and the distance between them is about **3.6 map units**. This is close to the classic maize C-Sh distance of about 3.5 map units.

Exam focus — Lecture 23

Key points

1. Linkage = genes on the same chromosome transmitted together.
2. Discovered by Bateson and Punnett in sweet pea; explained by Morgan.
3. Coupling (cis) = dominants in the same parent; repulsion (trans) = dominants in different parents.
4. Morgan's theory: linked genes are linear; strength depends on distance; crossing over breaks linkage.
5. Complete linkage gives only parental types; incomplete linkage gives few recombinants.
6. Linkage group = genes on one chromosome; number equals the haploid number n.
7. Recombination frequency (%) = recombinants/total × 100; <50% indicates linkage.

Short-answer questions with model answers

Q1. Define linkage and linkage group.
Model answer: Linkage is the tendency of genes located on the same chromosome to be inherited together. A linkage group is the set of all genes on one chromosome; the number of linkage groups equals the haploid chromosome number of the species.

Q2. What are coupling and repulsion?
Model answer: Coupling (cis) is the arrangement in which two dominant alleles are present in the same parent (AB/AB). Repulsion (trans) is the arrangement in which each parent carries one dominant and one recessive allele (Ab/Ab × aB/aB). In both, parental types are in excess and recombinants are few.

Q3. How is linkage estimated?
Model answer: By a test cross of the F_1 heterozygote with the double recessive parent; the percentage of recombinant progeny gives the recombination frequency, which estimates the linkage (distance between the genes).

Long-answer questions with model outlines

Q1. Describe Bateson and Punnett's experiment and Morgan's chromosomal theory of linkage.
Outline: Describe the sweet pea cross and the coupling/repulsion phases; show the deviation from 9:3:3:1; state Morgan's theory (linear arrangement, distance, crossing over); explain complete and incomplete linkage; give the linkage group concept.

Lecture 24. Strength of Linkage and Recombination; Double Cross Over; Interference and Coincidence

In one line: The strength of linkage is measured by how close two genes are; recombination frequency measures their distance, and the occurrence of two crossovers between them is described by coincidence and interference.

24.1 Strength of linkage

The **strength of linkage** is the tendency of linked genes to remain together. It is **inversely related to the distance** between the genes:

- Genes very close together cross over rarely, so linkage is strong and recombination is low.
- Genes far apart cross over more often, so linkage is weak and recombination is high.
- When two genes are so far apart that crossing over occurs at least once between them in every meiosis, the recombination frequency reaches **50%** and the genes behave as if unlinked (they assort independently).

The strength of linkage is therefore the opposite of the recombination frequency: strong linkage = low recombination; weak linkage = high recombination.

24.2 Recombination frequency

The recombination frequency is the percentage of progeny that are recombinants. It is also called the **crossing-over value**. One unit of recombination frequency = one **map unit** (centimorgan, cM).

Recombination frequency (%) = (Total recombinants / Total progeny) × 100

Table 24.1 — Recombination frequency and linkage strength

Recombination frequency	Linkage	Interpretation
0%	Complete	Genes always together (no crossover between them)
Low (e.g. 5%)	Strong	Genes close together
Moderate (e.g. 25%)	Weak	Genes farther apart
50%	No detectable linkage	Genes far apart or on different chromosomes; independent assortment

24.3 Double cross over

A **double crossover** is the occurrence of **two crossovers** in the same chromosome pair between two genes of interest (in a three-point cross, one crossover between gene 1 and gene 2 and another between gene 2 and gene 3). Double crossovers produce the **least frequent** classes in a three-point test cross and are used to determine gene order.

```
DOUBLE CROSSOVER (between genes A and C, with B in the middle)

A ── B ── C
a ── b ── c
After two crossovers (A-B and B-C):
A ── b ── c   ← double crossover product (least frequent)
a ── B ── C   ← double crossover product
```

24.4 Interference

Interference is the phenomenon in which the occurrence of one crossover **reduces the chance** of another crossover nearby. It is measured as:

Interference (I) = 1 − Coefficient of coincidence (C)

- If the observed double crossovers equal the expected, then C = 1 and I = 0 (no interference).
- If fewer double crossovers occur than expected, then C < 1 and I is positive (interference is present, the usual case).
- If more double crossovers occur than expected, then C > 1 and I is negative (rare).

24.5 Coincidence

The **coefficient of coincidence (C)** is the ratio of the **observed** number of double crossovers to the **expected** number:

C = Observed double crossovers / Expected double crossovers

The expected number of double crossovers = (frequency of crossover in region 1) × (frequency of crossover in region 2) × total progeny.

Worked example 24.1 — coincidence and interference

Problem: In a three-point test cross with 1000 progeny, the recombination frequency between A and B is 20% and between B and C is 30%. The observed number of double crossovers is 42. Find the coefficient of coincidence and interference.

Solution:

- Expected double crossovers = 0.20 × 0.30 × 1000 = 60.
- Coefficient of coincidence (C) = observed / expected = 42 / 60 = **0.70**.
- Interference (I) = 1 − C = 1 − 0.70 = **0.30** (30% interference).

Interpretation: only 70% of the expected double crossovers occurred, so a crossover in one region interfered with a crossover in the other region.

Worked example 24.2 — strength of linkage

Problem: Two pairs of genes, A-B and X-Y, are on the same chromosome. A-B shows 6% recombination and X-Y shows 34% recombination. Which pair is more strongly linked?

Solution: A-B is more strongly linked, because its recombination frequency (6%) is much lower than that of X-Y (34%). Stronger linkage means the genes are closer together.

Exam focus — Lecture 24

Key points

1. Strength of linkage is inversely related to the distance between genes.
2. Recombination frequency (%) = recombinants/total × 100; one unit = one map unit (cM).
3. At 50% recombination the genes behave as unlinked.
4. Double crossover = two crossovers between the genes; gives the least frequent classes.
5. Coefficient of coincidence C = observed / expected double crossovers.
6. Interference I = 1 − C; positive interference is the usual case.

Short-answer questions with model answers

Q1. What is meant by the strength of linkage?
Model answer: The strength of linkage is the tendency of linked genes to stay together. It is inversely related to the distance between them: genes close together are strongly linked (low recombination), while genes far apart are weakly linked (high recombination).

Q2. Define coincidence and interference.
Model answer: The coefficient of coincidence is the ratio of observed to expected double crossovers. Interference is the reduction in the chance of a second crossover near the first, given by I = 1 − coefficient of coincidence.

Q3. Why do double crossovers give the least frequent classes?
Model answer: Because a double crossover requires two independent crossover events in the same region, whose joint probability is the product of two small probabilities; hence such progeny are rare and form the least frequent classes.

Long-answer questions with model outlines

Q1. Explain strength of linkage, recombination frequency, double crossover, coincidence and interference.
Outline: Define each term; show the relation between distance and recombination; explain double crossover and its use in gene order; give the formulae for coincidence and interference; work one numerical example.

Lecture 25. Two-Point and Three-Point Test Cross; Genetic Map and Physical Map

In one line: A two-point test cross measures the distance between two genes; a three-point test cross finds the order of three genes and their distances; the genetic map shows distances in map units, while the physical map shows actual DNA positions.

25.1 Two-point test cross

A **two-point test cross** involves **two** gene pairs. The F_1 heterozygote is crossed with the double recessive parent. The progeny fall into four classes: two large parental classes and two small recombinant classes (if the genes are linked).

The distance between the two genes is the recombination frequency:

Map distance (cM) = (recombinants / total progeny) × 100

Worked example 25.1 — two-point test cross

Problem: In a two-point test cross AB/ab × ab/ab, the progeny were: AB 210, ab 215, Ab 25, aB 30. Find the recombination frequency and the map distance.

Solution: Total = 210 + 215 + 25 + 30 = 480. Parental = AB, ab (large classes). Recombinants = Ab + aB = 25 + 30 = 55.

Recombination frequency = 55/480 × 100 = **11.46%** ≈ 11.5 map units (cM). The genes are linked, about 11.5 cM apart.

25.2 Three-point test cross

A **three-point test cross** involves **three** gene pairs. It is more efficient than a two-point cross because it gives the order of the three genes and the distances between them in a single experiment, and it also allows the detection of double crossovers.

The eight classes of progeny are arranged in four pairs:

- **Parental (most frequent):** two classes.
- **Single crossover in region I:** two classes.
- **Single crossover in region II:** two classes.
- **Double crossover (least frequent):** two classes.

Steps in a three-point test cross

1. Arrange the eight classes in descending order of number.
2. Identify the two most frequent classes as **parental** and the two least frequent as **double crossovers**.
3. Compare the double-crossover class with the parental class: the gene that differs is the **middle gene**, which fixes the gene order.
4. Calculate the distance between the first and middle gene (region I) and between the middle and third gene (region II), counting the double crossovers in both regions.
5. Add the two distances to get the distance between the two outer genes.

Worked example 25.2 — full three-point test cross

Problem: A three-point test cross (ABC/abc × abc/abc) gave the following progeny:

Phenotype	Number
A B C	320
a b c	305
A b c	130
a B C	125
A B c	35
a b C	30
A b C	10
a B c	5

Determine the gene order, the distances between the genes, and the coincidence and interference.

Solution:

Step 1 — Total and class sizes. Total = 320 + 305 + 130 + 125 + 35 + 30 + 10 + 5 = **960**.

Step 2 — Identify parental and double crossover classes. Most frequent: A B C (320) and a b c (305) → parental. Least frequent: A b C (10) and a B c (5) → double crossovers.

Step 3 — Find the middle gene. Compare double crossover (A b C) with parental (A B C): A matches A, C matches C, but B is replaced by b → the **middle gene is B**. So the order is **A — B — C**.

Step 4 — Distance A–B (region I). Crossovers in region I = A b c (130) + a B C (125) + double crossovers A b C (10) + a B c (5) = 270.

Distance A–B = 270/960 × 100 = **28.1 cM**.

Step 5 — Distance B–C (region II). Crossovers in region II = A B c (35) + a b C (30) + double crossovers A b C (10) + a B c (5) = 80.

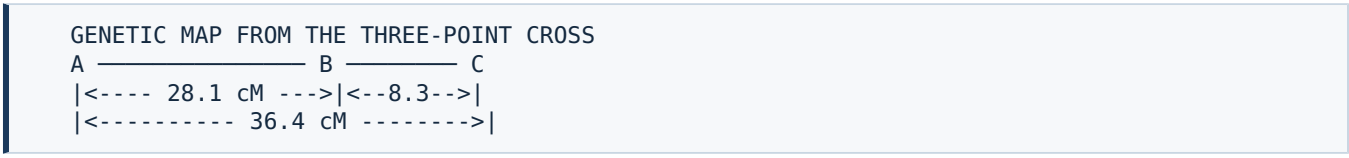
Distance B–C = 80/960 × 100 = **8.3 cM**.

Step 6 — Distance A-C. A–C = A–B + B–C = 28.1 + 8.3 = **36.4 cM**.

Step 7 — Coincidence and interference. Expected double crossovers = (270/960) × (80/960) × 960 = 22.5. Observed double crossovers = 10 + 5 = 15.

Coefficient of coincidence C = 15/22.5 = **0.67**. Interference I = 1 − 0.67 = **0.33** (33%).

Conclusion: Gene order A–B–C; distances A–B = 28.1 cM, B–C = 8.3 cM, A–C = 36.4 cM; coincidence 0.67 and interference 0.33.



25.3 Genetic map (linkage map)

A **genetic map** (linkage map) is a diagram that shows the **linear order and the relative distances** between genes on a chromosome, measured in **map units (centimorgans, cM)**. One map unit = 1% recombination frequency. The first genetic map was made by **Sturtevant** (1913) in *Drosophila*.

- **Construction:** from recombination frequencies obtained by two-point and three-point test crosses.
- **Uses:** to locate genes, to plan crosses, to study linkage and to compare species (comparative mapping).
- **Limitation:** the map distance is only an estimate; for genes far apart the frequency is underestimated because double crossovers restore the parental arrangement, so long distances must be built up from short segments.

25.4 Physical map

A **physical map** shows the **actual physical positions** of genes or DNA markers on a chromosome, measured in **base pairs (bp, kb, Mb)**. It is prepared by molecular methods such as:

- Restriction mapping and pulsed-field gel electrophoresis.
- DNA sequencing and genome assembly.
- Fluorescence in situ hybridisation (FISH) to locate a sequence on the chromosome.
- Contig maps from BAC/YAC libraries.

Table 25.1 — Genetic map versus physical map

Feature	Genetic map	Physical map
Basis	Recombination frequency	Actual DNA sequence position
Unit	Map unit (centimorgan, cM)	Base pairs (bp, kb, Mb)
Method	Test crosses, pedigree analysis	Sequencing, FISH, restriction mapping
Accuracy	Relative; affected by double crossovers	Absolute; gives exact position
Varies with species?	Yes (recombination differs)	No (physical distance is fixed)
Use	Gene order, linkage studies	Genome sequencing, map-based cloning

Relationship: physical distance and genetic (map) distance are not proportional — the same physical distance may correspond to different map distances in different regions, because recombination is not uniform along the chromosome (it is low near the centromere and in heterochromatin).

Exam focus — Lecture 25

Key points

1. Two-point test cross measures the distance between two genes.
2. Three-point test cross gives gene order and distances in one experiment.
3. Parental classes are most frequent; double crossovers are least frequent.
4. The middle gene is found by comparing the double-crossover class with the parental class.
5. Add the two region distances to get the outer-gene distance.
6. Genetic map is in map units (cM); physical map is in base pairs.
7. Sturtevant made the first genetic map (1913).

Short-answer questions with model answers

Q1. What is the advantage of a three-point over a two-point test cross?
Model answer: A three-point test cross gives the order of three genes and the distances between them in a single experiment, and it also detects double crossovers, which a two-point cross cannot. This makes it more efficient and more accurate.

Q2. How is the middle gene identified in a three-point cross?
Model answer: By comparing the least frequent (double-crossover) class with the parental class; the gene that is changed (reversed) between them is the middle gene.

Q3. Differentiate a genetic map from a physical map.
Model answer: A genetic map is based on recombination frequency and is measured in map units (cM); a physical map is based on the actual DNA sequence and is measured in base pairs. The two are not strictly proportional.

Long-answer questions with model outlines

Q1. Describe the method of a three-point test cross and its use in preparing a genetic map.
Outline: Explain the design; list the eight classes and their order; give the steps (parental, double-crossover, middle gene, region distances); work a numerical example; draw the resulting map; mention coincidence and interference.

Q2. Compare genetic and physical maps and explain why they are not proportional.
Outline: Define both maps; give basis, unit, method and accuracy; explain that recombination is non-uniform (low near centromere and heterochromatin), so the two maps differ; give their uses in breeding and genomics.

Lecture 26. Sex Determination

> **In one line:** Sex determination is the mechanism that decides whether an individual develops as male or female; it may be chromosomal, genic or environmental, and differs from organism to organism.

26.1 Autosomes and sex chromosomes

- **Autosomes:** chromosomes that are the same in both sexes and carry genes for ordinary (somatic) characters. In humans there are 22 pairs of autosomes.
- **Sex chromosomes (allosomes):** the chromosomes that differ between the sexes and determine sex. In humans these are the X and Y chromosomes (one pair).
- In a diploid individual, one sex usually has two identical sex chromosomes (homogametic) and the other has two different ones (heterogametic).

26.2 Chromosomal theory of sex determination

The sex of an individual is decided at fertilisation by the kind of sex chromosome carried by the gamete. There are four main systems.

Table 26.1 — Types of chromosomal sex determination

System	Male	Female	Example
XX – XY (male heterogametic)	XY	XX	Human, *Drosophila*, mammals
XX – XO (male heterogametic)	XO	XX	Grasshopper, some insects
ZZ – ZW (female heterogametic)	ZZ	ZW	Fowl, butterfly, moth, *Fumea*
Haplodiploidy	Haploid (from unfertilised egg)	Diploid (from fertilised egg)	Honey bee, ants, wasps

(a) Human (XX – XY)

Human females are XX (homogametic) and males are XY (heterogametic). The mother produces only X-bearing eggs; the father produces X- and Y-bearing sperms in equal numbers. Fertilisation by an X sperm gives a girl (XX) and by a Y sperm gives a boy (XY). The sex ratio is therefore 1:1. The **Y chromosome carries the gene (SRY)** that switches development towards maleness; hence an XXY individual (Klinefelter) is male.

(b) Fowl (ZZ – ZW)

In birds, the female is heterogametic (ZW) and the male is homogametic (ZZ). The egg decides the sex of the offspring: a Z egg fertilised by a Z sperm gives a male (ZZ), and a W egg fertilised by a Z sperm gives a female (ZW). The mother's gamete therefore determines sex.

(c) Butterfly and moth (ZZ – ZW)

In butterflies and moths, as in birds, the female is ZW (heterogametic) and the male is ZZ (homogametic). This is the reverse of the human system.

(d) Grasshopper (XX – XO)

In grasshoppers, the female is XX and the male has only one sex chromosome (XO). The male produces two kinds of sperm — one with an X and one without any sex chromosome. Fertilisation by an X sperm gives a female (XX); by a sperm with no sex chromosome gives a male (XO).

(e) Honey bee (haplodiploidy)

In the honey bee, sex is determined by fertilisation itself. A **fertilised egg (diploid)** develops into a **female** (queen or worker), and an **unfertilised egg (haploid)** develops into a **male (drone)** by parthenogenesis. Drones are haploid and produce sperm by mitosis; they have no father and cannot have sons, but they do have a grandfather. This is called haplodiploidy.

(f) *Fumea* (a moth, ZZ – ZW)

In *Fumea* (a bagworm moth) the female is the heterogametic sex (ZW) and the male is ZZ — the same system as in the fowl. It is cited in textbooks as a classic example of female heterogamety in the Lepidoptera.

26.3 Sex determination in plants

- ***Melandrium* (now *Silene*, a dioecious plant):** sex is determined by sex chromosomes. The male is heterogametic (XY) and the female is XX. The Y chromosome carries the genes for maleness; plants with Y are male, plants without Y are female.
- **Papaya (*Carica papaya*):** a trioecious plant (male, female and hermaphrodite plants). Sex is determined by a small pair of sex chromosomes; males are XY, females XX, and hermaphrodites are XY^h (a different Y). The sex is best recognised by the flower type.
- **Maize (*Zea mays*):** maize is monoecious (separate male and female inflorescences on the same plant) — tassel (male) and ear (female). Sex expression is **genic**, not chromosomal: genes such as *tasselseed* (ts) and *barren stalk* (ba) control the development of the two inflorescences. Mutations in these genes can convert tassels to ears or produce separate male and female plants.

26.4 Genic balance theory of Bridges

C. B. Bridges (1921), working with *Drosophila*, showed that sex is not decided simply by the presence of the Y chromosome but by the **balance (ratio) between the number of X chromosomes and the number of sets of autosomes**.

Sex index = (number of X chromosomes) / (number of autosome sets)

Table 26.2 — Genic balance theory (Bridges) in *Drosophila*

Chromosome constitution	X : A ratio	Sex
3A + 3X (3X, 3A)	1.0	Normal female (metafemale in some cases)
2A + 2X (normal female)	1.0	Female
3A + 2X (triploid intersex)	0.67	Intersex
2A + 1X (normal male)	0.5	Male
3A + 1X	0.33	Metamale

So a ratio of 1.0 (X : A) gives a female, 0.5 gives a male, values between give an intersex, and extreme values give metafemales or metamales. The Y chromosome in *Drosophila* does not determine sex; it is needed only for male fertility.

26.5 Gynandromorphs

A **gynandromorph** is an individual that has both male and female tissues in different parts of the body — for example, a *Drosophila* with one side male and the other female (a "bilateral gynandromorph"). It arises by the **loss of an X chromosome** in one cell during early cleavage (in an XX zygote), so that one group of cells becomes XO (male) and the rest remains XX (female). Gynandromorphs show that sex is determined **cell by cell** by the X : A balance, not by a hormone circulating through the whole body.

```
GYNANDROMORPH (bilateral)
XX zygote ──► during early division, one daughter cell loses an X
      │
      ├── XX cells  → female side  (left half)
      └── XO cells  → male side    (right half)
```

Worked example 26.1 — predicting sex ratio

Problem: In a mammal with the XX–XY system, a heterozygous carrier female (X^CX) is crossed with a normal male (XY). What proportion of the offspring will be male, and what proportion will inherit the X^C allele?

Solution: Gametes: female X^C, X; male X, Y.

	X	**Y**
X^C	X^CX (female, carrier)	X^CY (male, affected)
X	XX (female)	XY (male)

Sex ratio = 1 female : 1 male (50% each). Half of all offspring (both the X^CX female and the X^CY male) inherit the X^C allele, i.e. **50%**.

Exam focus — Lecture 26

Key points

1. Autosomes are the same in both sexes; sex chromosomes differ and determine sex.
2. Human: female XX, male XY (male heterogametic); Y carries the male-determining SRY gene.
3. Fowl, butterfly, moth and *Fumea*: female ZW, male ZZ (female heterogametic).
4. Grasshopper: female XX, male XO.
5. Honey bee: diploid = female, haploid = male (haplodiploidy, parthenogenesis).
6. Plants: *Melandrium* and papaya have sex chromosomes; maize sex is genic.
7. Bridges' genic balance: sex depends on the X : A ratio (1.0 female, 0.5 male, between = intersex).
8. Gynandromorphs arise by X-chromosome loss and show cell-by-cell sex determination.

Short-answer questions with model answers

Q1. Differentiate autosomes from sex chromosomes.
Model answer: Autosomes are the chromosomes that are identical in both sexes and carry genes for body (somatic) characters; sex chromosomes differ between the sexes and determine sex (e.g. X and Y in humans).

Q2. State Bridges' genic balance theory of sex determination.
Model answer: Bridges proposed that sex is determined by the ratio of the number of X chromosomes to the number of sets of autosomes. In *Drosophila* an X : A ratio of 1.0 gives a female, 0.5 gives a male, and intermediate values give an intersex.

Q3. How is sex determined in the honey bee?
Model answer: By haplodiploidy. A fertilised (diploid) egg develops into a female (queen or worker) and an unfertilised (haploid) egg develops parthenogenetically into a male drone.

Long-answer questions with model outlines

Q1. Describe the different chromosomal mechanisms of sex determination with examples.
Outline: Explain autosomes and sex chromosomes; describe the XX-XY (human), XX-XO (grasshopper), ZZ-ZW (fowl/butterfly/*Fumea*) and haplodiploid (honey bee) systems; give the sex ratio in each; end with sex determination in plants (*Melandrium*, papaya, maize).

Q2. Explain Bridges' genic balance theory and the origin of gynandromorphs.
Outline: State the X : A ratio; give the table of ratios and sexes; explain the role (or non-role) of the Y chromosome in *Drosophila*; describe how loss of an X chromosome in early cleavage produces a gynandromorph and what it proves.

Lecture 27. Sex-Linked Inheritance

> **In one line:** Sex-linked inheritance is the inheritance of genes located on the sex chromosomes; such genes show a criss-cross pattern and a difference between reciprocal crosses.

27.1 Types of sex-linked genes

Genes located on the sex chromosomes are called **sex-linked genes**. They are of three kinds:

- **X-linked genes:** genes on the X chromosome; these show the classic sex-linked (criss-cross) pattern. Most sex-linked characters studied are X-linked.
- **Y-linked (holandric) genes:** genes on the Y chromosome; they pass only from father to son.
- **Sex-influenced and sex-limited genes:** genes on autosomes whose expression depends on the sex of the individual (see section 27.5).

27.2 X-linked inheritance and criss-cross inheritance

In X-linked inheritance the **X chromosome of the father passes to his daughters** and the **X chromosome of the mother passes to both sons and daughters**. The result is a **criss-cross pattern**: a character is transmitted from father to grandson through the daughter, or from mother to son. The classic example is **white eye colour in *Drosophila***, discovered by Morgan (1910).

```
CRISS-CROSS INHERITANCE (white eye in Drosophila)

P:   White-eyed female (X^w X^w)  ×  Red-eyed male (X^+ Y)
     Eggs: X^w ; Sperms: X^+ and Y
F1:  X^+ X^w (red-eyed daughters)   and  X^w Y (white-eyed sons)
     → daughters get father's red eye; sons get mother's white eye

Reciprocal: Red-eyed female (X^+ X^+) × White-eyed male (X^w Y)
F1:  all red-eyed (daughters X^+X^w and sons X^+Y)
     → no white-eyed offspring (reciprocal difference)
```

27.3 Reciprocal difference

Reciprocal difference means that the results of a cross differ depending on which parent carries the trait. In autosomal inheritance, a cross AB × ab gives the same result as ab × AB. But in sex-linked inheritance, the two reciprocal crosses give **different results**, because the sexes differ in their sex chromosomes. This reciprocal difference is itself the proof that a gene is sex-linked.

Table 27.1 — Reciprocal crosses for white eye in *Drosophila*

Cross	Parents	F_1 daughters	F_1 sons
Cross 1	White female × red male	All red (carriers)	All white
Cross 2 (reciprocal)	Red female × white male	All red (carriers)	All red

The two crosses give different results — a clear reciprocal difference, showing that eye colour is X-linked.

27.4 Sex-linked inheritance in humans

- **Colour blindness (red-green):** a recessive X-linked character. A colour-blind man (X^cY) has all normal daughters (carriers) and no affected sons from a normal wife. A carrier woman (X^cX) has a 50% chance of an affected son.
- **Haemophilia:** a recessive X-linked bleeding disorder; the classic royal disease. Males are affected (X^hY); females are carriers (X^hX) or, rarely, affected (X^hX^h).
- **Duchenne muscular dystrophy:** recessive X-linked.

Because males have only one X, a recessive X-linked allele is expressed at once in males (they are hemizygous). This is why X-linked recessive disorders are much more common in males.

Worked example 27.1 — colour blindness pedigree

Problem: A normal-visioned man marries a woman who is a carrier of colour blindness. What is the probability that a son will be colour blind? What about a daughter?

Solution: Man = XY (normal); woman = X^cX (carrier).

	X	**Y**
X^c	X^cX (carrier daughter)	X^cY (colour-blind son)
X	XX (normal daughter)	XY (normal son)

Among sons: 1 of 2 (50%) will be colour blind. Among daughters: none will be colour blind (half will be carriers). Overall, 25% of all children are colour-blind sons and 25% are carrier daughters.

27.5 Holandric genes

Holandric genes are genes present on the **Y chromosome**. They are transmitted only from father to son ("holandric" = wholly male). A daughter never receives a Y chromosome, so she can neither show nor carry a holandric character. Examples in humans include the **SRY gene** (testis-determining factor) and some genes for hair on the ear (hypertrichosis of the pinna). Because only males carry them and they pass father to son, holandric characters never skip a generation among males.

27.6 Sex-influenced and sex-limited traits

Table 27.2 — Sex-linked, sex-influenced and sex-limited traits

Feature	Sex-linked	Sex-influenced	Sex-limited
Location of gene	On a sex chromosome	On an autosome	On an autosome
Expression	Depends on sex chromosome constitution	Expressed in both sexes but differently (dominance changes with sex)	Expressed in only one sex
Example	Colour blindness, haemophilia, white eye in *Drosophila*	Pattern baldness in humans; horns in sheep	Milk yield in cattle; egg production in fowl

Sex-influenced traits

These are autosomal genes whose **dominance changes with sex**. Example: **pattern baldness** in humans — the allele for baldness behaves as dominant in men and recessive in women, so a heterozygous man becomes bald while a heterozygous woman usually does not. Another example is the presence of horns in sheep (the horned allele is dominant in males and recessive in females).

Sex-limited traits

These are autosomal genes expressed in **only one sex**. Example: **milk yield** in cattle and **egg production** in fowl are expressed only in females; **semen characters** and some male secondary characters are expressed only in males. The genes are present in both sexes but are expressed in only one, usually because of hormones.

Exam focus — Lecture 27

Key points

1. Sex-linked genes lie on the sex chromosomes; most studied ones are X-linked.
2. Criss-cross inheritance: father's X goes to daughters, mother's X goes to sons.
3. Reciprocal crosses give different results in sex-linked inheritance (reciprocal difference).
4. X-linked recessive disorders (colour blindness, haemophilia) are commoner in males (hemizygous).
5. Holandric genes are on the Y chromosome; passed father to son (e.g. SRY).
6. Sex-influenced genes (autosomal) change dominance with sex (pattern baldness).
7. Sex-limited genes are expressed in only one sex (milk yield, egg production).

Short-answer questions with model answers

Q1. What is criss-cross inheritance?
Model answer: Criss-cross inheritance is the pattern in which an X-linked character passes from father to grandson through the daughter, or from mother to son. It occurs because the father gives his X only to his daughters and the mother gives her X to both sons and daughters.

Q2. What are holandric genes? Give an example.
Model answer: Holandric genes are genes located on the Y chromosome; they pass only from father to son. An example is the SRY gene (testis-determining factor) in humans.

Q3. Differentiate sex-influenced from sex-limited traits.
Model answer: Sex-influenced traits are autosomal genes expressed in both sexes but with different dominance according to sex (e.g. pattern baldness). Sex-limited traits are autosomal genes expressed in only one sex (e.g. milk yield in cattle).

Long-answer questions with model outlines

Q1. Explain sex-linked inheritance with the white-eye character in *Drosophila* and give examples in humans.
Outline: Define sex linkage; describe Morgan's white-eye crosses and the criss-cross pattern; show the reciprocal difference; give the human examples (colour blindness, haemophilia); explain why males are more often affected.

Q2. Describe holandric, sex-influenced and sex-limited inheritance with examples.
Outline: Define each; give the location of the gene and the pattern of expression; give examples; compare the three in a table.

Lecture 28. Mutation — Classification, Induction and Mutagenic Agents

> **In one line:** A mutation is a sudden, heritable change in the genetic material; it may occur spontaneously or be induced by physical and chemical mutagens, and it is the ultimate source of all new variation.

28.1 Mutation — definition

A **mutation** is a sudden, heritable change in the genetic material (DNA) that is not due to recombination or segregation. The term was coined by Hugo de Vries (1901). Mutations are the **ultimate source of genetic variation**; they produce new alleles on which selection and breeding act.

28.2 Classification of mutations

Table 28.1 — Classification of mutations

Basis	Types	Meaning / example
Origin	Spontaneous; Induced	Spontaneous = natural, low frequency; induced = by mutagens
Cell type	Somatic; Germinal	Somatic affects only the individual; germinal is transmitted
Size of change	Point (gene); Chromosomal	Point = change in one/few bases; chromosomal = structural/numerical
Direction	Forward; Reverse (back)	Forward = wild to mutant; reverse = mutant back to wild
Effect on phenotype	Dominant; Recessive	Recessive mutations are the commoner
Effect on function	Loss-of-function; Gain-of-function; Lethal	Loss = gene product absent; lethal = causes death
Base change	Transition; Transversion	Transition = purine↔purine or pyrimidine↔pyrimidine; transversion = purine↔pyrimidine

Point (gene) mutations

- **Substitution:** one base replaced by another. It may be a **silent** mutation (no amino-acid change), a **missense** mutation (one amino acid changed), or a **nonsense** mutation (a stop codon created, truncating the protein).
- **Frameshift:** insertion or deletion of one or a few bases, shifting the reading frame; usually very harmful.
- **Transition and transversion:** the two kinds of base substitution (see table).

Chromosomal mutations

These are the structural changes (deletion, duplication, inversion, translocation) of Lecture 7 and the numerical changes (aneuploidy, euploidy) of Lectures 8 and 9.

28.3 Methods of inducing mutations

Mutations are induced mainly by **physical** and **chemical** agents, collectively called **mutagens**. The deliberate use of mutagens to obtain useful mutants is called **mutation breeding**.

Table 28.2 — Mutagenic agents

Class	Agent	Mode of action
Physical — ionising radiation	X-rays, gamma rays, alpha and beta particles, neutrons	Break DNA strands; cause deletions, rearrangements and chromosome breaks
Physical — non-ionising radiation	Ultraviolet (UV) light	Forms pyrimidine dimers (thymine dimers); causes point mutations
Chemical — base analogues	5-bromouracil, 2-aminopurine	Mimic normal bases and pair wrongly, causing transitions
Chemical — deaminating agents	Nitrous acid	Deaminates bases (cytosine → uracil), causing transitions
Chemical — alkylating agents	EMS (ethyl methane sulphonate), MMS, mustard gas	Add alkyl groups to bases, causing mispairing; also cause breaks
Chemical — intercalating agents	Acridine orange, ethidium bromide, proflavin	Insert between bases; cause frameshift mutations
Biological	Some viruses, transposons (e.g. Ac-Ds in maize)	Insert into genes; cause insertion mutations

28.4 Induction of mutation — practical steps

1. Choose the plant material (seeds, pollen, buds, callus or cuttings) of a suitable variety.
2. Treat it with a suitable mutagen (physical or chemical) at a chosen dose and duration.
3. Raise the treated generation (M_1) and harvest seed from each plant separately (to avoid repeats).
4. Grow the M_2 generation and screen for visible (chlorophyll, morphological) mutants.
5. Select promising mutants, test them in the M_3 and later generations for stability, and evaluate for yield and quality.
6. Release the improved mutant as a variety, or use it as a parent in crossing.

28.5 Characteristics of mutations

- Mutations occur at **random** with respect to the gene and the organism's needs.
- They are usually **recessive** (so they are hidden in heterozygotes) and mostly **harmful**; only a few are useful.
- They are **recurrent** — the same mutation can occur again.
- They occur at a **low frequency** (spontaneous mutation rate is about 10^{-5} to 10^{-8} per gene per generation).
- They are **heritable** if they occur in the germ cells.

28.6 Mutation breeding in crops

- Mutation breeding has given improved varieties in rice, wheat, barley, groundnut, pulses, sugarcane and ornamentals. In India, the **IAEA/FAO and BARC** programmes have released mutants of rice (e.g. semi-dwarf and early types), groundnut and pulses.
- Gamma rays and EMS are the most commonly used mutagens.
- Mutation breeding is useful where the desired variation is not available in the germplasm, or to improve a single character (e.g. earliness, dwarfness, disease resistance) in an otherwise good variety.

Worked example 28.1 — classifying a mutation

Problem: A DNA base substitution changes the codon GAG (glutamic acid) to GTG (valine), producing sickle-cell haemoglobin. Classify the mutation and state its effect.

Solution: This is a **point mutation of the substitution type**, and specifically a **missense mutation**, because one amino acid (glutamic acid) is replaced by another (valine). It is also a **transition/transversion** at the base level (A → T here is a transversion). The change alters the protein (haemoglobin S), causing sickle-cell anaemia — a harmful but, in heterozygotes, partly protective mutation.

Exam focus — Lecture 28

Key points

1. Mutation = sudden heritable change in DNA; the term was coined by de Vries.
2. Mutations may be spontaneous or induced, somatic or germinal, point or chromosomal.
3. Point mutations: substitution (silent, missense, nonsense) and frameshift.
4. Mutagens are physical (X-rays, gamma, UV) or chemical (EMS, base analogues, acridine dyes).
5. UV causes thymine dimers; ionising radiation causes breaks; acridine dyes cause frameshifts.
6. Mutations are random, usually recessive and harmful, recurrent and rare.
7. Mutation breeding has produced improved crop varieties, especially in rice, groundnut and pulses.

Short-answer questions with model answers

Q1. Define mutation and state its importance.
Model answer: A mutation is a sudden, heritable change in the genetic material. It is important because it is the ultimate source of all new genetic variation, on which evolution and plant breeding depend.

Q2. Distinguish point mutation from chromosomal mutation.
Model answer: A point (gene) mutation is a change in one or a few bases of a gene (substitution or frameshift). A chromosomal mutation is a change in the structure or number of chromosomes (deletion, inversion, translocation, aneuploidy, polyploidy).

Q3. Name four chemical mutagens and their modes of action.
Model answer: Base analogues (5-bromouracil) mimic bases and cause transitions; nitrous acid deaminates bases; alkylating agents (EMS) add alkyl groups and cause mispairing; acridine dyes intercalate and cause frameshift mutations.

Long-answer questions with model outlines

Q1. Classify mutations and describe the methods of inducing them.
Outline: Define mutation; classify by origin, cell type, size, direction and effect; describe point mutations (substitution types and frameshift); list physical and chemical mutagens with their action; outline the steps of mutation breeding.

Q2. Describe mutation breeding and its achievements in crop improvement.
Outline: Define mutation breeding; describe the choice of material and mutagen; the M_1 to M_3 handling; give crop examples (rice, groundnut, pulses); note advantages and limitations.

Lecture 29. Maternal Effects and Cytoplasmic Inheritance

> **In one line:** Maternal effect is the influence of the mother's genotype on the offspring's phenotype before its own genes act; cytoplasmic inheritance is the transmission of characters through genes present in the cytoplasm (plastids, mitochondria, and other cytoplasmic particles).

29.1 Maternal effects

A **maternal effect** is a situation in which the **phenotype of the offspring is determined by the genotype of the mother**, not by its own genotype. The mother's gene products (mRNA or protein) are deposited in the egg and control early development before the embryo's own genes are switched on.

The classic example is **shell coiling in the snail *Limnaea peregra***:

- Snail shells coil either to the **right (dextral)** or to the **left (sinistral)**. Dextral (D) is dominant to sinistral (d), and the gene acts on the direction of coiling.
- The direction of coiling of an individual is decided not by its own genotype but by the **genotype of its mother**. So a snail with genotype Dd, whose mother was dd, coils to the left (sinistral), even though it carries D.
- The effect is therefore delayed by one generation: the maternal genotype controls the offspring's shell direction.

```
MATERNAL EFFECT (snail shell coiling, Limnaea peregra)

Mother's genotype decides offspring's coiling:
  Mother DD or Dd  → offspring dextral (right-coiled)
  Mother dd        → offspring sinistral (left-coiled)
The offspring's OWN genotype does not decide its coiling.
```

29.2 Cytoplasmic inheritance

Cytoplasmic inheritance (extranuclear inheritance, extrachromosomal inheritance) is the transmission of characters by genes located in the **cytoplasm** rather than in the nucleus. The genes involved are called **plasmagenes** or cytoplasmic genes, and they lie mainly in the **plastids and mitochondria**, and in some cytoplasmic particles.

Features of cytoplasmic inheritance

- **Maternal (usually) inheritance:** because the zygote receives almost all its cytoplasm from the egg, cytoplasmic characters are usually inherited from the **mother** only.
- **No Mendelian ratios:** the results do not fit Mendelian segregation, because the genes are not on nuclear chromosomes.
- **No reciprocal equality:** reciprocal crosses give different results (the character follows the female parent).
- **Independent of the nucleus:** the cytoplasmic genes replicate and are transmitted independently of nuclear genes.
- **Somatic segregation:** because plastids and mitochondria divide and are distributed randomly, cells can come to differ in the proportion of mutant and normal organelles (variegation).

- **Not affected by nuclear substitution** in most cases; the cytoplasmic element persists through many generations.

29.3 Plastid (chloroplast) inheritance — *Mirabilis jalapa*

The classic example is **plastid (leaf) colour in *Mirabilis jalapa*** (four o'clock plant), studied by Correns (1909). The plant may have green, white or variegated (green-and-white) leaves.

Table 29.1 — Plastid inheritance in *Mirabilis jalapa* (Correns)

Seed (female) parent	Pollen (male) parent	Progeny
Green	Any (green/white/variegated)	All green
White	Any	All white
Variegated	Any	Green, white and variegated (mixed)

Interpretation: the colour depends entirely on the **female parent**, not on the pollen. This is because the plastids are inherited through the egg cytoplasm. The variegated parent produces egg cells with different kinds of plastids (green, white or both), so the progeny are mixed. This is a clear case of cytoplasmic (plastid) inheritance and maternal transmission.

29.4 Mitochondrial inheritance and cytoplasmic male sterility (CMS) in maize

- **Mitochondrial inheritance:** mitochondria carry their own DNA (mtDNA), which is inherited from the mother. Mutations in mtDNA cause mitochondrial diseases in humans and male sterility in plants.
- **Cytoplasmic male sterility (CMS):** a condition in which the plant produces no functional pollen because of factors in the cytoplasm (mitochondria). It is inherited through the mother. CMS is widely used in hybrid seed production, because it removes the need for hand emasculation.
- In maize, CMS was discovered and used in hybrid seed production. Three main types were recognised: **Texas (T) cytoplasm**, **USDA (S) cytoplasm** and **Charrua (C) cytoplasm**. The T-type was widely used but had to be abandoned after the **Southern Corn Leaf Blight epidemic (1970–71)**, because the T cytoplasm also made the plants susceptible to the blight fungus *Bipolaris maydis*.
- CMS is often restored by nuclear **restorer (Rf) genes**, which bring back fertility; this CMS–restorer system is the basis of hybrid seed production in maize, sorghum, pearl millet, rice and sunflower.

29.5 Infective particles — kappa particles in *Paramecium*

In *Paramecium aurelia*, some strains (called "killer" strains) produce a toxic substance that kills sensitive strains. The killer character is due to **kappa particles** present in the cytoplasm.

- Kappa particles are **self-replicating cytoplasmic particles** (now known to be symbiotic bacteria). They are inherited through the cytoplasm, so the killer character shows **maternal (cytoplasmic) inheritance**.
- The particles require a nuclear gene (K) to be maintained; without the dominant K gene, the kappa particles are lost and the strain becomes sensitive.
- Because kappa is transmitted through the cytoplasm, a cross of killer × sensitive gives killer progeny only when the killer is the female (cytoplasmic donor).

- Kappa particles are an example of an **infective (symbiotic) particle** causing a heritable character — a bridge between cytoplasmic inheritance and infection.

Table 29.2 — Examples of cytoplasmic inheritance

Character	Organism	Cytoplasmic element
Plastid (leaf) colour	*Mirabilis jalapa*	Plastids (chloroplasts)
Male sterility (CMS)	Maize, sorghum, rice, sunflower	Mitochondria
Killer character	*Paramecium aurelia*	Kappa particles
Respiratory deficiency (petite)	Yeast	Mitochondria
Antibiotic resistance	Some fungi	Mitochondria

Worked example 29.1 — identifying cytoplasmic inheritance

Problem: A variegated female *Mirabilis* plant is crossed with a green male plant. The progeny are green, white and variegated. A green female crossed with a variegated male gives all green progeny. Explain.

Solution: The results show that the progeny follow the **female parent** and not the male parent. This is **cytoplasmic (plastid) inheritance**. A variegated female produces egg cells containing green, white or both kinds of plastids, so the progeny are mixed; a green female produces only green eggs, so all progeny are green, irrespective of the pollen.

Exam focus — Lecture 29

Key points

1. Maternal effect: the mother's genotype decides the offspring's phenotype (snail shell coiling).
2. Cytoplasmic inheritance is through plasmagenes in plastids and mitochondria.
3. Features: maternal transmission, no Mendelian ratios, reciprocal difference, somatic segregation.
4. Plastid colour in *Mirabilis jalapa* (Correns) follows the female parent.
5. CMS in maize is mitochondrial; T, S and C cytoplasms; T-type abandoned after the 1970 blight epidemic.
6. Kappa particles in *Paramecium* cause the killer character and are inherited through the cytoplasm.
7. Restorer (Rf) genes bring back fertility in CMS systems used for hybrid seed production.

Short-answer questions with model answers

Q1. What is a maternal effect? Give an example.
Model answer: A maternal effect is the control of the offspring's phenotype by the genotype of the mother rather than by its own genotype. Example: shell coiling in the snail *Limnaea peregra*, where the mother's genotype decides whether the offspring's shell coils to the right or the left.

Q2. Give any four features of cytoplasmic inheritance.
Model answer: (i) It is usually maternal; (ii) it does not follow Mendelian ratios; (iii) reciprocal crosses give different results; (iv) it is independent of nuclear genes and shows somatic segregation.

Q3. What is cytoplasmic male sterility and how is it used?
Model answer: CMS is male sterility caused by factors in the cytoplasm (mitochondria), inherited through the mother. It is used in hybrid seed production because it removes the need for hand emasculation; fertility is restored by nuclear restorer (Rf) genes.

Long-answer questions with model outlines

Q1. Explain cytoplasmic inheritance with the examples of *Mirabilis*, CMS in maize and kappa particles.
Outline: Define cytoplasmic inheritance and list its features; describe Correns' plastid colour work; describe CMS in maize (T, S, C types, blight epidemic, restorer genes); describe the kappa particle killer character in *Paramecium*; conclude with the significance of cytoplasmic inheritance in breeding.

Lecture 30. DNA as the Genetic Material; RNA as Genetic Material

> **In one line:** Griffith's transformation experiment, the Avery-MacLeod-McCarty work and the Hershey-Chase experiment proved that DNA is the genetic material; the tobacco mosaic virus work of Fraenkel-Conrat and Singer showed that RNA can also be the genetic material.

30.1 Early evidence — the search for the genetic material

By the early twentieth century it was known that chromosomes carry genes, and that chromosomes contain DNA and protein. For a long time protein was favoured as the genetic material because it is more complex (20 amino acids vs 4 bases). Three key experiments settled the question in favour of DNA.

30.2 Griffith's experiment (1928) — transformation

Frederick Griffith worked with *Streptococcus pneumoniae* (pneumococcus), which has two forms:

- **S (smooth) strain:** has a polysaccharide capsule, forms smooth colonies and is **virulent** (causes pneumonia).
- **R (rough) strain:** no capsule, rough colonies, **non-virulent**.

Table 30.1 — Griffith's transformation experiment

Injected into mice	Result	Interpretation
Live S strain	Mouse dies	S is virulent
Live R strain	Mouse lives	R is non-virulent
Heat-killed S strain	Mouse lives	Killed S is harmless
Heat-killed S + live R	Mouse dies; live S recovered	R was transformed into S

Conclusion: some substance (the "transforming principle") from the heat-killed S strain converted the harmless R strain into virulent S. Griffith had discovered **transformation**, but did not know what the transforming substance was.

30.3 Avery, MacLeod and McCarty (1944) — DNA is the transforming principle

Oswald Avery, Colin MacLeod and Maclyn McCarty purified the transforming substance and tested which component carried it. They treated the heat-killed S-strain extract with enzymes that destroy specific molecules:

- **Protease** (destroys protein) — transformation still occurred → protein was not the transforming principle.
- **RNase** (destroys RNA) — transformation still occurred → RNA was not the transforming principle.
- **DNase** (destroys DNA) — transformation **stopped** → DNA was the transforming principle.

Conclusion: DNA, not protein, is the substance that carries hereditary information. This was the first direct proof that DNA is the genetic material.

30.4 Hershey and Chase (1952) — confirmation with bacteriophage

Alfred Hershey and Martha Chase used the bacteriophage **T2**, which infects *E. coli*. A phage is made of DNA and protein coat only. They labelled the two components separately:

- Phages grown in medium with **radioactive phosphorus (^{32}P)** → the **DNA** was labelled (DNA contains phosphorus, protein does not).
- Phages grown in medium with **radioactive sulphur (^{35}S)** → the **protein coat** was labelled (protein contains sulphur, DNA does not).

After infection, they blended the mixture to strip off the phage coats and separated the bacteria by centrifugation:

Table 30.2 — Hershey-Chase results

Label	Where the radioactivity was found	Conclusion
^{32}P (DNA)	Inside the bacterial cells	DNA entered the bacterium
^{35}S (protein)	Outside, with the empty phage coats	Protein did not enter

Conclusion: only the DNA entered the bacterium and directed the production of new phages, so DNA is the genetic material.

```
HERSHEY-CHASE EXPERIMENT
T2 phage (DNA + protein coat)
  label DNA with 32P        label coat with 35S
        |                         |
  infect E. coli             infect E. coli
        |                         |
  blend + centrifuge         blend + centrifuge
        |                         |
32P in bacterial pellet     35S in supernatant (coats)
→ DNA entered the cell      → protein stayed outside
Conclusion: DNA is the genetic material
```

30.5 RNA as genetic material — Fraenkel-Conrat and Singer (1957)

Some viruses (most plant viruses and some animal viruses) have **RNA** as their genetic material. The classic proof came from work on the **tobacco mosaic virus (TMV)** by Heinz Fraenkel-Conrat and B. Singer.

TMV has a protein coat and an RNA core. Two strains were used — a common strain and a mutant strain with a different coat protein. The experiment separated the RNA and protein of each strain, then **reconstituted hybrid viruses** by mixing the RNA of one strain with the protein of the other:

- Hybrid virus with **RNA of strain A + protein of strain B** was used to infect tobacco plants.
- The progeny virus produced had the **protein of strain A** (matching the RNA), not strain B.

Conclusion: the **RNA determines the nature of the progeny virus** (including its protein coat), so RNA is the genetic material in TMV. This showed that RNA, like DNA, can carry genetic information.

30.6 Summary of the evidence

Table 30.3 — Experiments proving the nature of the genetic material

Year	Scientist(s)	Organism	Finding
1928	Griffith	*Streptococcus pneumoniae*	Transformation by a "transforming principle"
1944	Avery, MacLeod, McCarty	*S. pneumoniae*	DNA is the transforming principle
1952	Hershey and Chase	Phage T2 / *E. coli*	DNA enters the bacterium; DNA is the genetic material
1957	Fraenkel-Conrat and Singer	Tobacco mosaic virus	RNA is the genetic material in TMV

Worked example 30.1 — interpreting a labelling experiment

Problem: In a Hershey-Chase type experiment, a phage is labelled with ^{32}P. After infection and blending, 80% of the radioactivity is found in the bacterial pellet and 20% in the supernatant. What does this tell you, and why is some label in the supernatant?

Solution: ^{32}P labels DNA, and its presence mainly in the bacterial pellet shows that the **DNA entered the bacteria**, confirming DNA as the genetic material. The 20% in the supernatant is due to phages that did not attach or did not inject their DNA, so their labelled DNA remained outside.

Exam focus — Lecture 30

Key points

1. Griffith (1928) discovered transformation in pneumococcus.
2. Avery, MacLeod and McCarty (1944): DNase stopped transformation → DNA is the genetic material.
3. Hershey and Chase (1952): ^{32}P (DNA) entered the bacterium; ^{35}S (protein) did not.
4. Fraenkel-Conrat and Singer (1957): RNA of TMV determines the progeny → RNA can be the genetic material.
5. DNA contains phosphorus (not sulphur); protein contains sulphur (not phosphorus) — the basis of the labelling.
6. Most plant viruses have RNA as their genetic material.

Short-answer questions with model answers

Q1. How did Avery, MacLeod and McCarty prove that DNA is the genetic material?
Model answer: They treated the transforming extract of heat-killed S-strain pneumococci with protease, RNase and DNase separately. Protease and RNase did not stop transformation, but DNase stopped it, showing that DNA (destroyed by DNase) is the transforming principle.

Q2. Why were ^{32}P and ^{35}S chosen in the Hershey-Chase experiment?
Model answer: Because DNA contains phosphorus but no sulphur, and protein contains sulphur but no phosphorus. So ^{32}P labels DNA only and ^{35}S labels protein only, allowing the two components to be followed separately.

Q3. How did Fraenkel-Conrat and Singer show that RNA is the genetic material in TMV?
Model answer: They made hybrid viruses by combining the RNA of one TMV strain with the protein of another and infected tobacco plants. The progeny virus had the protein type of the strain that provided the RNA, showing that RNA carries the genetic information.

Long-answer questions with model outlines

Q1. Describe the experiments that proved DNA is the genetic material.
Outline: Griffith's transformation experiment (four treatments and the conclusion); the Avery-MacLeod-McCarty enzyme test; the Hershey-Chase labelling experiment (why P and S, the blending and centrifugation, the result); end with the conclusion that DNA is the genetic material.

Q2. Explain how RNA was shown to be the genetic material, and compare DNA and RNA as genetic materials.
Outline: Describe the TMV hybrid experiment of Fraenkel-Conrat and Singer; give the conclusion; then compare DNA (double-stranded, stable, deoxyribose, thymine) and RNA (usually single-stranded, uracil, ribose) as genetic materials; note that RNA is the genetic material in many viruses.

Lecture 31. Types of DNA and RNA; Structure of DNA (Watson-Crick Model); Proof for Semi-Conservative Replication

> **In one line:** DNA is a double helix of two antiparallel strands held by base pairing; RNA occurs as mRNA, tRNA and rRNA; the Meselson-Stahl experiment proved that DNA replicates semi-conservatively.

31.1 Types of DNA

- **Nuclear DNA (nDNA):** the main genetic material in the nucleus, arranged as linear chromosomes.
- **Extranuclear (organelle) DNA:** circular DNA in mitochondria (mtDNA) and plastids (cpDNA), inherited maternally.
- **A-DNA, B-DNA and Z-DNA:** forms of the double helix differing in shape. **B-DNA** (Watson-Crick) is the normal, right-handed form found in cells; **A-DNA** is a shorter, wider right-handed form seen in dehydrated samples; **Z-DNA** is a left-handed helix with a zig-zag backbone.
- **Satellite DNA:** highly repetitive, non-coding DNA, often in heterochromatin near the centromere.

31.2 Types of RNA

Table 31.1 — The three main types of RNA

Feature	mRNA (messenger RNA)	tRNA (transfer RNA)	rRNA (ribosomal RNA)
Function	Carries the genetic message from DNA to ribosome	Carries amino acids to the ribosome	Structural and functional part of the ribosome
Share of cell RNA	About 5%	About 15%	About 80%
Size	Variable (long)	Smallest	Variable (largest)
Structure	Single strand, linear	Clover-leaf shape (folded)	Folded, with protein
Stability	Least stable (short-lived)	Fairly stable	Most stable
Special features	Has codons; made by transcription	Has anticodon and amino-acid attachment site	Forms the ribosome; made in the nucleolus

Other RNAs include **hnRNA** (heterogeneous nuclear RNA, the precursor of mRNA), **snRNA** (small nuclear RNA, involved in splicing) and **miRNA/siRNA** (regulatory small RNAs).

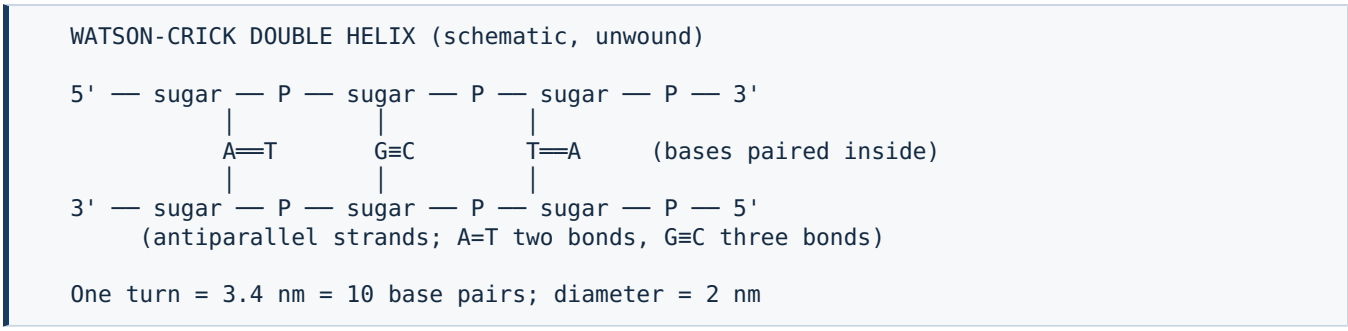
31.3 Structure of DNA — Watson and Crick model (1953)

James Watson and Francis Crick proposed the **double helix** model of DNA, using the X-ray diffraction data of Rosalind Franklin and Maurice Wilkins and the base-composition data of Erwin Chargaff.

Salient features

- DNA is a **double helix** of two polynucleotide chains wound around a common axis, like a twisted ladder.
- The two chains are **antiparallel** — one runs 5'→3' and the other 3'→5'.
- Each chain has a **sugar-phosphate backbone** on the outside and the **bases on the inside**.
- The bases pair by hydrogen bonds: **A pairs with T (two H-bonds)** and **G pairs with C (three H-bonds)**. This is **complementary base pairing**.
- The two chains are **complementary**: the sequence of one strand fixes the sequence of the other.
- One complete turn of the helix is **3.4 nm (34 Å)** long and contains **10 base pairs**; the diameter is **2 nm (20 Å)**. Each base pair is 0.34 nm apart.
- The two strands are held by hydrogen bonds between bases and by base-stacking (hydrophobic) interactions.

Chargaff's rules: in double-stranded DNA, the amount of adenine equals thymine (A = T) and guanine equals cytosine (G = C); hence purines (A + G) equal pyrimidines (T + C).



31.4 Semi-conservative replication — Meselson and Stahl (1958)

Three possible models of replication were proposed: **conservative** (the whole original molecule is conserved and a new one made), **semi-conservative** (each new molecule has one old and one new strand) and **dispersive** (old and new material scattered).

Meselson and Stahl's experiment:

1. They grew *E. coli* for many generations in a medium containing the heavy isotope ^{15}**N**, so that all the DNA was heavy.
2. They then transferred the bacteria to a medium with the normal light isotope ^{14}**N** and allowed them to divide.
3. They extracted DNA after each generation and separated it by **density-gradient centrifugation** (CsCl), which separates DNA by density (heavy, hybrid or light).

Table 31.2 — Meselson-Stahl results

Generation	DNA bands observed	Interpretation
Before transfer (all ^{15}N)	One heavy band	All DNA is heavy
After 1 generation in ^{14}N	One hybrid (intermediate) band	Each molecule has one heavy and one light strand
After 2 generations	One hybrid band + one light band (equal)	Hybrid molecules separate; light molecules appear

Conclusion: the results fit **semi-conservative replication** — each daughter DNA molecule consists of one parental (old) strand and one newly synthesised strand. This was confirmed when Taylor, Woods and Hughes (1957) obtained the same result in **Vicia faba** (broad bean), showing that plants also replicate DNA semi-conservatively.

```
SEMI-CONSERVATIVE REPLICATION (Meselson-Stahl)

Generation 0:   ═══HEAVY═══   (15N / 15N)
Generation 1:   ═HEAVY═
                ─LIGHT─      (one hybrid molecule per cell)
Generation 2:   ═HEAVY═    ─LIGHT─
                ─LIGHT─    ─LIGHT─
                (two hybrids + two light molecules)
```

Worked example 31.1 — Chargaff's rule

Problem: A double-stranded DNA molecule contains 30% adenine. Find the percentages of thymine, guanine and cytosine.

Solution: By Chargaff's rule, A = T, so T = 30%. Total A + T = 60%, leaving 40% for G + C. Since G = C, each is 40/2 = **20%**. So T = 30%, G = 20%, C = 20%.

Worked example 31.2 — length of a DNA molecule

Problem: A DNA molecule has 1000 base pairs. Find its length in nm and the number of turns of the helix.

Solution: Each base pair is 0.34 nm apart, so length = 1000 × 0.34 = **340 nm**. One turn contains 10 base pairs, so number of turns = 1000/10 = **100 turns**.

Exam focus — Lecture 31

Key points

1. DNA types: nuclear, organelle (mtDNA, cpDNA), A-, B-, Z-forms and satellite DNA.
2. mRNA carries the message (about 5%), tRNA carries amino acids (15%), rRNA forms ribosomes (80%).
3. Watson-Crick: double helix, antiparallel strands, sugar-phosphate backbone outside, bases inside.
4. A=T (2 H-bonds), G≡C (3 H-bonds); Chargaff's rules A=T and G=C.
5. One turn = 3.4 nm = 10 bp; diameter 2 nm; each bp 0.34 nm.
6. Meselson-Stahl (1958) proved semi-conservative replication using ^{15}N and CsCl centrifugation.
7. Taylor (1957) showed the same in *Vicia faba*.

Short-answer questions with model answers

Q1. State the main features of the Watson-Crick model of DNA.

Model answer: DNA is a double helix of two antiparallel polynucleotide strands with the sugar-phosphate backbone outside and paired bases inside; A pairs with T (two H-bonds) and G with C (three H-bonds); one turn is 3.4 nm with 10 base pairs, and the diameter is 2 nm.

Q2. Differentiate mRNA, tRNA and rRNA.

Model answer: mRNA carries the genetic message from DNA to the ribosome; tRNA carries amino acids to the ribosome and has an anticodon; rRNA forms the structural and functional part of the ribosome. They form about 5%, 15% and 80% of cellular RNA respectively.

Q3. How did Meselson and Stahl prove semi-conservative replication?

Model answer: They labelled *E. coli* DNA with heavy ^{15}N, transferred the cells to ^{14}N medium and followed the DNA density by CsCl centrifugation. After one generation a single hybrid band appeared and after two generations hybrid and light bands appeared in equal amounts, showing that each molecule has one old and one new strand.

Long-answer questions with model outlines

Q1. Describe the Watson-Crick model of DNA and the evidence that supports it.

Outline: Give the features of the double helix; explain complementary base pairing and Chargaff's rules; mention the dimensions; cite the X-ray data of Franklin and Wilkins; end with the biological significance of complementary strands for replication.

Q2. Describe the experiment that proved semi-conservative replication.

Outline: Explain the three models; describe the ^{15}N labelling and CsCl centrifugation; present the results after 0, 1 and 2 generations in a table; state the conclusion; mention the *Vicia faba* confirmation.

Lecture 32. Models of DNA Replication; Steps Involved in DNA Replication

In one line: DNA replication is semi-conservative, bidirectional and discontinuous on one strand; it needs many enzymes and proceeds in three stages — initiation, elongation and termination.

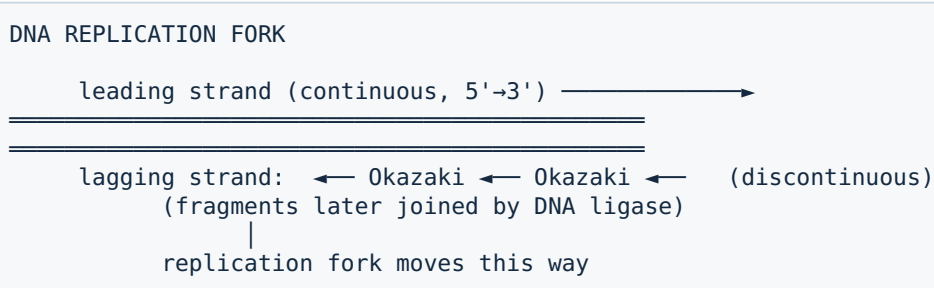
32.1 Models of DNA replication

Table 32.1 — Models of DNA replication

Model	Description	Status
Conservative	The original double helix is conserved intact; a completely new double helix is made	Rejected
Semi-conservative	Each new molecule has one old (parental) strand and one new strand	Accepted (Meselson-Stahl)
Dispersive	Old and new material is scattered in short segments in both strands	Rejected

Two further points describe **how** replication proceeds along the chromosome:

- **Unidirectional vs bidirectional:** replication is **bidirectional** — it starts at an origin and proceeds in both directions, forming a replication bubble (or "theta" structure in circular DNA).
- **Continuous vs discontinuous:** because DNA polymerase can add nucleotides only in the 5'→3' direction, one strand (the **leading strand**) is synthesised continuously and the other (the **lagging strand**) is synthesised discontinuously as short **Okazaki fragments**, later joined by ligase.



32.2 Enzymes and proteins of replication

Table 32.2 — Enzymes and proteins in DNA replication

Enzyme / protein	Function
DNA helicase	Unwinds the double helix at the replication fork
Single-strand binding (SSB) proteins	Keep the separated strands apart and prevent re-pairing
Topoisomerase / gyrase	Relieves supercoiling ahead of the fork
Primase	Synthesises a short RNA primer to start synthesis
DNA polymerase III (in *E. coli*)	Main enzyme that adds nucleotides 5'→3'
DNA polymerase I	Removes the RNA primer and fills the gap
DNA ligase	Joins the Okazaki fragments (seals nicks)

32.3 Steps involved in DNA replication

(a) Initiation

1. Replication begins at specific sites called **origins of replication** (ori).
2. **Helicase** unwinds the double helix, and **SSB proteins** keep the strands apart, forming a replication fork (or a bubble).
3. **Topoisomerase/gyrase** relieves the strain of supercoiling.
4. **Primase** lays down a short RNA primer to provide a free 3'-OH end.

(b) Elongation

1. **DNA polymerase** adds deoxyribonucleotides complementary to the template, always in the 5'→3' direction.
2. The **leading strand** is synthesised continuously toward the fork.
3. The **lagging strand** is synthesised discontinuously as **Okazaki fragments**, each needing its own primer.

(c) Termination

1. The RNA primers are removed (by DNA polymerase I) and the gaps filled with DNA.
2. **DNA ligase** joins the fragments into a continuous strand.
3. The two daughter double helices separate, each with one parental and one new strand (semi-conservative).

32.4 Replication in prokaryotes and eukaryotes

Table 32.3 — DNA replication in prokaryotes versus eukaryotes

Feature	Prokaryotes (*E. coli*)	Eukaryotes
DNA form	Single circular chromosome	Multiple linear chromosomes
Origin of replication	Single origin (oriC)	Many origins per chromosome
Direction	Bidirectional (theta structure)	Bidirectional from each origin
Speed	Faster (about 1000 nucleotides/second)	Slower (about 50 nucleotides/second)
Telomere problem	None (circular DNA)	Ends need telomerase to complete replication

Worked example 32.1 — number of DNA molecules after replication

Problem: A bacterial cell with one circular chromosome replicates by binary fission three times. How many chromosomes are present at the end, and how many are "new" molecules?

Solution: Each division doubles the number of chromosomes: after 3 divisions = 2^3 = **8 chromosomes**. Because replication is semi-conservative, all 8 molecules contain some new material, but 2 of them contain a strand from the original molecule (the others have only strands made after the first round).

Exam focus — Lecture 32

Key points

1. Three models: conservative (rejected), semi-conservative (accepted), dispersive (rejected).
2. Replication is bidirectional from an origin and forms a replication bubble/fork.
3. Leading strand is continuous; lagging strand is discontinuous (Okazaki fragments).
4. DNA polymerase adds nucleotides only in the 5'→3' direction.
5. Key enzymes: helicase, SSB proteins, topoisomerase, primase, DNA polymerase, ligase.
6. Stages: initiation, elongation and termination.
7. Prokaryotes have one origin; eukaryotes have many; telomerase solves the end-replication problem.

Short-answer questions with model answers

Q1. What are Okazaki fragments and why are they formed?
Model answer: Okazaki fragments are the short pieces of DNA formed on the lagging strand during replication. They are formed because DNA polymerase can add nucleotides only in the 5'→3' direction, so the lagging strand (which runs the other way) must be synthesised in short, discontinuous pieces that are later joined by DNA ligase.

Q2. State the functions of helicase, primase and ligase.
Model answer: Helicase unwinds the double helix at the fork; primase lays down a short RNA primer to start synthesis; ligase joins the Okazaki fragments (seals the nicks).

Q3. Why is replication called semi-conservative?
Model answer: Because each daughter DNA molecule keeps one strand of the parent molecule (conserved) and has one newly synthesised strand, so half of each molecule is conserved.

Long-answer questions with model outlines

Q1. Describe the models of DNA replication and the steps involved in replication.
Outline: Explain the three models; describe the origin, fork, leading and lagging strands; list the enzymes with functions; describe initiation, elongation and termination; draw the replication fork.

Q2. Explain DNA replication with the roles of the main enzymes.
Outline: Give the semi-conservative principle; describe unwinding (helicase, SSB, topoisomerase), priming (primase), elongation (polymerase, leading and lagging strands, Okazaki fragments) and termination (polymerase I, ligase); end with prokaryote vs eukaryote differences.

Lecture 33. Genetic Code; Cistron, Muton and Recon; The Gene Concept

> **In one line:** The genetic code is a triplet code in which three bases (a codon) specify one amino acid; a gene is a segment of DNA that codes for a polypeptide or functional RNA, and its finest units are the cistron, muton and recon.

33.1 The genetic code

The **genetic code** is the set of rules by which the sequence of bases in DNA (or mRNA) is translated into the sequence of amino acids in a protein. The code was worked out by Nirenberg, Khorana, Ochoa and colleagues in the 1960s.

Features of the genetic code

- **Triplet:** three bases (a **codon**) code for one amino acid; there are $4^3 = 64$ codons.
- **Degenerate:** most amino acids are coded by more than one codon (e.g. leucine by six codons), but each codon specifies only one amino acid.
- **Unambiguous and specific:** a given codon always codes for the same amino acid.
- **Non-overlapping and comma-less:** the codons are read one after another without overlap and without punctuation between them.
- **Universal:** the code is essentially the same in all organisms (with minor exceptions in mitochondria and a few organisms).
- **Start and stop codons: AUG** is the start codon (also codes methionine); **UAA, UAG and UGA** are the stop (nonsense) codons that do not code for any amino acid.
- **Colinearity:** the order of bases in DNA corresponds to the order of amino acids in the protein.

Table 33.1 — A few codons worth remembering

Amino acid	Codons
Methionine (start)	AUG
Tryptophan	UGG
Phenylalanine	UUU, UUC
Leucine	UUA, UUG, CUU, CUC, CUA, CUG
Glycine	GGU, GGC, GGA, GGG
Stop (nonsense)	UAA, UAG, UGA

33.2 Cistron, muton and recon

These are the three units of the gene, defined by the type of test used to detect them (Benzer's fine-structure work on the rII locus of phage T4):

Table 33.2 — Cistron, muton and recon

Unit	Definition	Detected by
Cistron	The unit of function — the segment of DNA that codes for one polypeptide (a gene)	Complementation test
Muton	The unit of mutation — the smallest segment that can mutate (one base pair)	Mutation test
Recon	The unit of recombination — the smallest segment that can recombine (one base pair)	Recombination test

The **cistron** is the largest of the three (a whole gene); the **muton** and **recon** are the smallest (one base pair). This shows that the gene is not an indivisible unit but has a fine internal structure.

33.3 The gene concept — development over time

Table 33.3 — Evolution of the gene concept

Concept	Period	Idea
Mendelian factor	1866	An indivisible unit of inheritance controlling one character
Classical gene	1910–40	A gene is a point on the chromosome; unit of function, mutation and recombination
One gene–one enzyme	1941	Beadle and Tatum: each gene controls one enzyme
Molecular (cistron) concept	1950s	A gene is a segment of DNA coding for a polypeptide; it has fine structure
One gene–one polypeptide	1960s	Refined from one gene–one enzyme
Modern concept	1970s–present	A gene is a DNA sequence (with exons and introns) that codes for a polypeptide or functional RNA; genes may overlap, be split or be mobile

33.4 Molecular concept and definition of a gene

- **Molecular concept:** a gene is a segment of DNA that carries the information for a functional product (a polypeptide or an RNA molecule). It has regulatory regions (promoter, terminator) and, in eukaryotes, coding regions (exons) separated by non-coding regions (introns).
- **Molecular definition:** a gene is the entire DNA sequence necessary for the synthesis of a functional polypeptide or RNA, including its regulatory sequences.

Structure of a gene (eukaryotic)

```
EUKARYOTIC GENE STRUCTURE

5' ──[ PROMOTER ]──[ EXON 1 ]──[ INTRON 1 ]──[ EXON 2 ]──[ INTRON 2 ]──[ EXON 3 ]──[ TERMINATOR ]── 3
      (start site)   coding     non-coding    coding      non-coding     coding

After transcription and splicing, the introns are removed and the exons are joined to form mature mRNA
Prokaryotic genes have no introns (continuous coding sequence).
```

33.5 Number of genes on a single chromosome

The number of genes on a chromosome varies with the size of the chromosome and the organism. Some standard values:

Table 33.4 — Approximate gene numbers

Organism	Number of genes (approximate)	Chromosomes (2n)
E. coli	About 4,400	1 (circular)
Yeast (*S. cerevisiae*)	About 6,000	32
Arabidopsis thaliana	About 25,000	10
Rice	About 35,000–40,000	24
Drosophila	About 14,000	8
Human	About 20,000–25,000	46

The average gene density (genes per unit length) differs greatly: *E. coli* is very gene-dense, whereas the human genome has much non-coding DNA, so its genes are more spread out.

33.6 Size of a gene

- An average gene contains about **1,000 to 1,500 base pairs** of coding sequence (enough for a protein of about 300–500 amino acids).
- Eukaryotic genes are larger because of introns; some genes (e.g. the human dystrophin gene) are enormous (over 2 million base pairs).
- The number of amino acids in a polypeptide × 3 = the number of coding bases (codons) needed, plus one stop codon.

33.7 Fine structure of a gene

The fine structure of a gene was worked out by **Seymour Benzer** using the rII region of bacteriophage T4. He showed that:

- A gene is not indivisible; it consists of many **mutable and recombinable sites** (mutons and recons).
- The smallest unit of mutation and of recombination is a **single base pair**.
- Different mutations within the same gene can be separated by recombination, and can be arranged in a linear order (fine-structure mapping).
- Two mutations in the same gene usually fail to complement each other, while mutations in different genes complement (this is the basis of the complementation test for defining a cistron).

33.8 Types of genes

Table 33.5 — Types of genes

Type	Description / example
Structural genes	Code for proteins (enzymes, structural proteins)
Regulatory genes	Code for products that control the expression of other genes (repressors, activators)
Housekeeping genes	Expressed in all cells for basic functions (e.g. ribosomal genes)
Tissue-specific genes	Expressed only in particular tissues (e.g. seed-storage protein genes)
Constitutive genes	Always active (not regulated)
Inducible / repressible genes	Switched on or off according to need (lac and trp operons)
Overlapping genes	A DNA sequence that codes for more than one protein (common in viruses)
Split genes	Genes with exons and introns (most eukaryotic genes)
Pseudo genes	Non-functional gene copies (dead genes)
Jumping (mobile) genes	Transposons that can move from place to place (Ac-Ds in maize; discovered by Barbara McClintock)
Lethal and semilethal genes	Genes whose expression causes death (see Lecture 21)

Worked example 33.1 — from DNA to protein length

Problem: A gene has a coding sequence of 900 bases. How many amino acids will the polypeptide have, ignoring the stop codon?

Solution: Each amino acid is coded by 3 bases (one codon). Number of codons = 900/3 = 300. So the polypeptide will have **300 amino acids** (the stop codon, if included, would be one of these 300 codons and would not add an amino acid).

Exam focus — Lecture 33

Key points

1. The genetic code is a triplet, degenerate, non-overlapping and universal code.
2. 64 codons; AUG is the start codon; UAA, UAG, UGA are stop codons.
3. Cistron = unit of function; muton = unit of mutation; recon = unit of recombination.
4. A gene is a DNA segment coding for a polypeptide or functional RNA.
5. Eukaryotic genes have exons (coding) and introns (non-coding); prokaryotic genes do not.
6. An average gene is about 1,000–1,500 base pairs; human genome has about 20,000–25,000 genes.
7. Benzer's fine-structure work showed the gene has internal structure (mutons and recons).
8. Types of genes: structural, regulatory, housekeeping, split, overlapping, pseudo and jumping genes.

Short-answer questions with model answers

Q1. State any four features of the genetic code.
Model answer: (i) It is a triplet code (three bases = one codon); (ii) it is degenerate (more than one codon per amino acid); (iii) it is non-overlapping and comma-less; (iv) it is universal and unambiguous, with AUG as the start codon and UAA, UAG, UGA as stop codons.

Q2. Differentiate cistron, muton and recon.
Model answer: A cistron is the unit of function (codes for one polypeptide); a muton is the unit of mutation (smallest mutable unit, one base pair); a recon is the unit of recombination (smallest recombinable unit, one base pair).

Q3. Give the molecular definition of a gene.
Model answer: A gene is the entire DNA sequence necessary for the synthesis of a functional polypeptide or RNA molecule, including its regulatory sequences; in eukaryotes it consists of exons separated by introns.

Long-answer questions with model outlines

Q1. Describe the genetic code and its features with examples.
Outline: Define the code; explain triplet nature, degeneracy, universality, non-overlap, start and stop codons; give a small codon table; explain how the code was deciphered (Nirenberg, Khorana).

Q2. Trace the evolution of the gene concept and describe the structure and types of genes.
Outline: Give the Mendelian, classical, one gene–one enzyme and molecular concepts; define a gene molecularly; draw the eukaryotic gene structure (promoter, exons, introns); explain cistron, muton and recon; list the types of genes with examples.

Lecture 34. Central Dogma; Transcription and Protein Synthesis; Regulation of Gene Expression; Operon Model (Lac and Trp Operons)

> **In one line:** The central dogma states that information flows from DNA to RNA to protein; the lac and trp operons show how genes are switched on and off according to the cell's needs.

34.1 The central dogma of life

Proposed by **Francis Crick (1958)**, the **central dogma** states that genetic information flows in one direction:

DNA → RNA → Protein

- **Replication:** DNA → DNA (copying the genetic material before cell division).
- **Transcription:** DNA → RNA (copying a gene into mRNA).
- **Translation:** RNA → Protein (reading the mRNA to build a polypeptide).
- **Reverse transcription** (RNA → DNA), found in retroviruses such as HIV, is an exception (now called "central dogma reversed").

```
CENTRAL DOGMA

DNA ──(replication)──► DNA
 │
 │ transcription
 ▼
RNA (mRNA)
 │
 │ translation (at the ribosome)
 ▼
PROTEIN

Exception: RNA ──(reverse transcriptase)──► DNA   (retroviruses)
```

34.2 Transcription

Transcription is the synthesis of an RNA molecule from a DNA template. It occurs in the nucleus (in eukaryotes). Only one strand of the DNA (the **template strand**) is read, and the RNA is built with bases complementary to it (A→U, T→A, G→C, C→G).

Steps of transcription

1. **Initiation:** RNA polymerase binds to the **promoter** region of the gene and the DNA strands separate.
2. **Elongation:** RNA polymerase moves along the template strand, adding ribonucleotides in the 5'→3' direction to form the RNA.
3. **Termination:** the polymerase reaches the **terminator** sequence, stops and releases the RNA.

Processing of mRNA (eukaryotes)

- **Capping:** a methylated guanosine cap is added at the 5' end.

- **Tailing:** a poly-A tail (many adenine residues) is added at the 3' end.
- **Splicing:** introns are removed and exons are joined.

34.3 Protein synthesis (translation)

Translation is the synthesis of a polypeptide on the ribosome, using the sequence of codons in mRNA. It needs mRNA, ribosomes, tRNA, amino acids, enzymes and energy (ATP/GTP).

Steps of translation

1. **Activation of amino acids:** each amino acid is attached to its specific tRNA (charging), forming aminoacyl-tRNA, using ATP.
2. **Initiation:** the ribosome binds the mRNA at the start codon (AUG), and the first tRNA (with methionine) pairs with it.
3. **Elongation:** the ribosome moves along the mRNA; each new tRNA brings its amino acid, and peptide bonds form between successive amino acids. The ribosome has a P site and an A site.
4. **Termination:** when a stop codon (UAA, UAG or UGA) is reached, the polypeptide is released and the ribosome dissociates.

The **codon–anticodon pairing** ensures that the amino acids are placed in the correct order, as specified by the mRNA.

34.4 Regulation of gene expression

Not all genes are active in every cell at all times. **Regulation of gene expression** means controlling which genes are switched on, when and how strongly. In prokaryotes this happens mainly at the level of **transcription** and is explained by the operon model. In eukaryotes regulation is more complex (transcriptional, processing, transport and translational control).

34.5 The operon model of Jacob and Monod

François Jacob and Jacques Monod (1961) proposed the **operon model** to explain the regulation of gene expression in bacteria, based on their work on the **lac operon** of *E. coli*.

Components of an operon

- **Structural genes:** genes that code for the enzymes of a metabolic pathway (in the lac operon: z, y and a).
- **Promoter (P):** the site where RNA polymerase binds to start transcription.
- **Operator (O):** the site to which the repressor protein binds, controlling transcription.
- **Regulator (i) gene:** a separate gene that codes for the repressor protein.

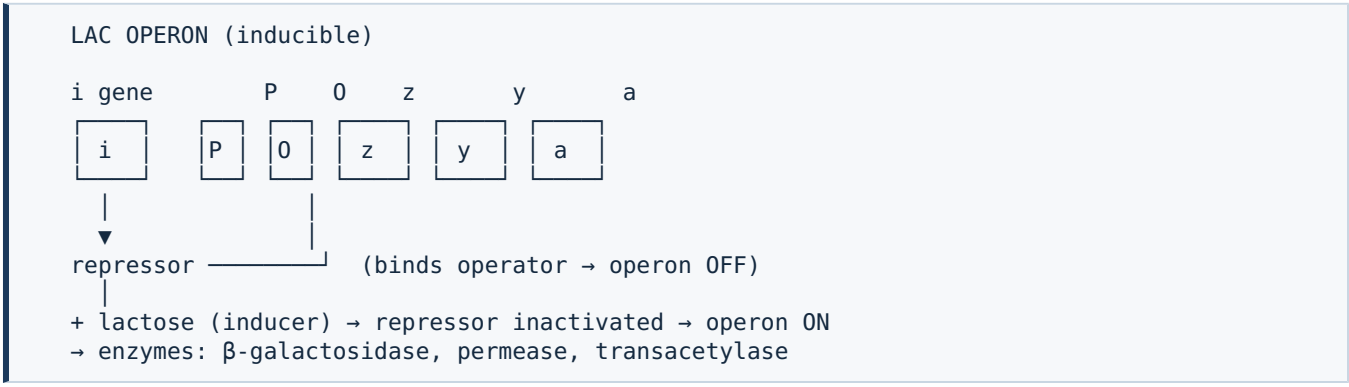
An operon is a group of structural genes with a common promoter and operator, transcribed together as a single **polycistronic mRNA**.

(a) The lac operon (inducible system)

The lac operon controls the metabolism of lactose in *E. coli*. It is an **inducible** system — it is switched on only when lactose is present.

- In the **absence of lactose**, the regulator gene produces a repressor protein that binds to the operator, blocking RNA polymerase, so the structural genes are **not transcribed** (the operon is off).
- In the **presence of lactose**, lactose (the **inducer**) binds to the repressor, changing its shape so that it cannot bind the operator. RNA polymerase can now transcribe the

structural genes, and the enzymes β-galactosidase (z), permease (y) and transacetylase (a) are produced. The operon is **on**.



(b) The trp operon (repressible system)

The trp operon controls the synthesis of the amino acid **tryptophan** in *E. coli*. It is a **repressible** system — it is normally on but is switched off when tryptophan is plentiful.

- In the **absence of tryptophan**, the repressor is inactive and cannot bind the operator, so the operon is **on** and the enzymes for tryptophan synthesis are made.
- When **tryptophan is abundant**, it acts as a **corepressor**: it binds the repressor, activating it, and the active repressor binds the operator, switching the operon **off**. The cell stops making tryptophan because it is already available.

Table 34.1 — Lac operon versus trp operon

Feature	Lac operon (inducible)	Trp operon (repressible)
Function	Metabolism (breakdown) of lactose	Synthesis of tryptophan
Normal state	Off (repressed)	On
Switched on by	Lactose (inducer)	Absence of tryptophan
Switched off by	Absence of lactose	Tryptophan (corepressor)
Regulatory molecule	Inducer (lactose)	Corepressor (tryptophan)
Type	Negative inducible control	Negative repressible control

34.6 Significance of gene regulation

- It allows the cell to use resources efficiently — enzymes are made only when needed.
- It is the basis of **differentiation**: different genes are switched on in different tissues, so a single genome gives rise to many cell types.
- Understanding operons is fundamental to molecular genetics and to genetic engineering, where promoters are used to control the expression of introduced genes in crops (for example, in Bt cotton and other GM crops).

Worked example 34.1 — predicting operon activity

Problem: In an *E. coli* cell growing in a medium with glucose and lactose, will the lac operon be strongly expressed? Explain.

Solution: Even though lactose is present (which would inactivate the repressor), the presence of **glucose** causes catabolite repression — the level of cyclic AMP (cAMP) is low, so the catabolite activator protein (CAP) cannot help RNA polymerase bind the promoter strongly. Hence the lac operon is only **weakly expressed** (low-level

transcription) when both glucose and lactose are present. The operon is fully expressed only when lactose is present and glucose is absent.

Exam focus — Lecture 34

Key points

1. Central dogma: DNA → RNA → Protein (Crick, 1958); reverse transcription is the exception.
2. Transcription: initiation, elongation and termination; mRNA is processed (capping, tailing, splicing).
3. Translation: amino-acid activation, initiation, elongation and termination at the ribosome.
4. An operon has structural genes, a promoter, an operator and a regulator gene; it gives polycistronic mRNA.
5. Lac operon is inducible: off normally; lactose (inducer) switches it on.
6. Trp operon is repressible: on normally; tryptophan (corepressor) switches it off.
7. Jacob and Monod proposed the operon model (1961).

Short-answer questions with model answers

Q1. State the central dogma of molecular biology.
Model answer: The central dogma, proposed by Crick, states that genetic information flows from DNA to RNA to protein — DNA is replicated, transcribed into RNA and translated into protein. Reverse transcription (RNA → DNA) in retroviruses is an exception.

Q2. What is an operon? Name its components.
Model answer: An operon is a group of structural genes with a common promoter and operator, transcribed together as a single polycistronic mRNA. Its components are the structural genes, the promoter, the operator and the regulator gene.

Q3. Differentiate the lac and trp operons.
Model answer: The lac operon is inducible and is normally off; it is switched on by lactose (the inducer). The trp operon is repressible and is normally on; it is switched off by tryptophan (the corepressor). The lac operon controls lactose breakdown, the trp operon controls tryptophan synthesis.

Long-answer questions with model outlines

Q1. Explain the central dogma, transcription and protein synthesis.
Outline: State the central dogma; describe transcription (initiation, elongation, termination and mRNA processing); describe translation (amino-acid activation, initiation, elongation, termination); mention the role of codons and anticodons.

Q2. Describe the operon model of Jacob and Monod with the lac and trp operons.
Outline: Define the operon and its components; describe the lac operon (inducible, lactose as inducer, the enzymes produced); describe the trp operon (repressible, tryptophan as corepressor); compare them in a table; conclude with the significance of gene regulation.

Glossary of Genetic Terms

This glossary collects the terms used in the lectures in one place, in alphabetical order. Each entry gives a short, exam-ready definition.

Term	Meaning
Allele	One of the alternative forms of a gene, occupying the same locus on homologous chromosomes.
Allopolyploid	A polyploid with chromosome sets derived from two or more different species (e.g. bread wheat, AABBDD).
Aneuploidy	Condition in which the chromosome number is not an exact multiple of the haploid number (e.g. 2n+1, 2n−1).
Anticodon	The three bases of tRNA that pair with the codon of mRNA during translation.
Autosome	A chromosome other than a sex chromosome; the same in both sexes.
Autopolyploid	A polyploid with multiple sets of chromosomes from the same species.
Back cross	A cross of a hybrid with either of its parents.
Barr body	The condensed, inactive X chromosome seen in the cells of female mammals.
Bivalent	A pair of homologous chromosomes joined during meiosis (prophase I).
C value	The amount of DNA in a haploid set of chromosomes.
Centromere	The primary constriction of a chromosome; the site of spindle attachment through the kinetochore.
Chiasma	The visible point of crossing over between non-sister chromatids of homologous chromosomes.
Chromatid	One of the two identical strands of a replicated chromosome, joined at the centromere.
Chromomere	A bead-like granule on the chromonema, seen at pachytene.
Chromonema	The fine, gene-bearing thread within a chromatid.
Chromosome	A thread-like structure of DNA and protein carrying the genes in a linear order.
Cistron	The unit of function — the segment of DNA coding for one polypeptide.
Codon	A triplet of bases in mRNA that specifies one amino acid.
Codominance	A condition in which both alleles of a heterozygote are fully expressed (e.g. AB blood group).
Coincidence	The ratio of observed to expected double crossovers.
Complementary base pairing	Pairing of A with T (or U) and G with C by hydrogen bonds.
Coupling (cis)	Arrangement in which two dominant alleles are present on the same chromosome.
Crossing over	The exchange of segments between non-sister chromatids of homologous chromosomes.
Cytogenetics	The branch combining cytology and genetics to study chromosomes and heredity.
Cytokinesis	The division of the cytoplasm during cell division.
Deletion	The loss of a segment of a chromosome.

Dihybrid	A hybrid heterozygous for two gene pairs.
Diploid	Having two sets of chromosomes (2n).
Dominance	The masking of the expression of one allele by another in a heterozygote.
Double crossover	Two crossovers between the genes of interest in a chromosome pair.
Duplication	The presence of a chromosomal segment more than once.
Epistasis	The masking of the expression of one gene by another gene at a different locus.
Euploidy	Condition in which the chromosome number is an exact multiple of the haploid number.
Euchromatin	Loosely packed, lightly stained, genetically active chromatin.
Exon	A coding region of a eukaryotic gene that is retained in mature mRNA.
Expressivity	The degree or intensity with which a genotype is expressed in the phenotype.
Gamete	A haploid reproductive cell (egg or sperm).
Gene	A segment of DNA that codes for a functional product (polypeptide or RNA).
Genetic code	The rules by which base triplets (codons) specify amino acids.
Genetic map	A diagram showing the linear order and distances of genes, measured in map units (cM).
Genome	The complete set of genes (or chromosomes) of an organism.
Genotype	The genetic constitution of an individual.
Germplasm	The hereditary material of the germ cells, which is continuous from generation to generation.
Gynandromorph	An individual with both male and female tissues in different parts of the body.
Haploid	Having a single set of chromosomes (n).
Hemizygous	Having only one allele for a gene, as in sex-linked genes in males.
Heterochromatin	Tightly packed, deeply stained, genetically inactive chromatin.
Heterozygous	Having two different alleles for a gene.
Holandric gene	A gene located on the Y chromosome; passed from father to son only.
Homozygous	Having two identical alleles for a gene.
Ideogram	A diagrammatic representation of a karyotype.
Incomplete dominance	A condition in which the heterozygote shows an intermediate phenotype.
Interference	The reduction in the chance of a second crossover near the first; I = 1 − coincidence.
Intron	A non-coding region of a eukaryotic gene that is removed during mRNA processing.
Inversion	A rearrangement in which a chromosomal segment is reversed.
Isoallele	

	An allele that gives the same phenotype as another but differs in its quantitative effect.
Isochromosome	An abnormal chromosome with two identical arms, formed by transverse misdivision of the centromere.
Karyotype	The complete set of chromosomes of a cell, arranged in a standard order.
Kinetochore	The protein structure at the centromere to which spindle fibres attach.
Lampbrush chromosome	A large chromosome with lateral loops, found in oocytes at diplotene.
Lethal gene	A gene that causes the death of the organism carrying it.
Linkage	The tendency of genes on the same chromosome to be inherited together.
Linkage group	The set of all genes on one chromosome; its number equals the haploid number n.
Locus	The specific position of a gene on a chromosome.
Map unit (centimorgan)	A unit of genetic distance equal to 1% recombination frequency.
Meiosis	The reduction division producing four haploid cells from one diploid cell.
Mitosis	The equational division producing two genetically identical diploid cells.
Monohybrid	A hybrid heterozygous for one gene pair.
Monosomic	An aneuploid lacking one chromosome (2n−1).
Multiple alleles	More than two alternative forms of a gene in a population (e.g. ABO blood groups).
Mutagen	An agent (physical or chemical) that induces mutations.
Mutation	A sudden, heritable change in the genetic material.
Muton	The unit of mutation — the smallest segment that can mutate (one base pair).
Non-disjunction	The failure of homologous chromosomes or sister chromatids to separate during meiosis.
NOR	The nucleolar organiser region, carrying rRNA genes; forms the nucleolus.
Nucleosome	The basic packing unit of chromatin: 146 bp DNA around a histone octamer.
Okazaki fragments	Short DNA pieces formed on the lagging strand during replication.
Operon	A group of structural genes with a common promoter and operator, transcribed together.
Penetrance	The proportion of individuals with a genotype that show the expected phenotype.
Phenotype	The visible or measurable expression of the genotype.
Pleiotropism	The ability of one gene to affect several characters.
Polygene	A gene with a small, cumulative effect; many polygenes control quantitative characters.
Polyploidy	The presence of more than two complete sets of chromosomes.
Polytene chromosome	A giant chromosome formed by repeated replication without division; found in *Drosophila* salivary glands.

Pseudoallele	Closely linked genes that behave almost like alleles of one gene.
Recombination	The formation of new gene combinations, chiefly by crossing over.
Recon	The unit of recombination — the smallest segment that can recombine (one base pair).
Repulsion (trans)	Arrangement in which each chromosome carries one dominant and one recessive allele.
Replication	The copying of DNA before cell division; it is semi-conservative.
Satellite chromosome	A chromosome with a terminal segment (satellite) separated by a secondary constriction.
Self-incompatibility	The failure of a fertile plant to set seed on self-pollination; controlled by S alleles.
Semi-conservative replication	Replication in which each new DNA molecule has one old and one new strand.
Sex chromosome	A chromosome (X or Y, Z or W) that differs between the sexes and determines sex.
Sex-influenced trait	An autosomal trait whose dominance changes with the sex of the individual.
Sex-limited trait	An autosomal trait expressed in only one sex.
Sporogenesis	The formation of spores by meiosis.
Synapsis	The pairing of homologous chromosomes during zygotene of prophase I.
Telomere	The end region of a chromosome that seals the ends and prevents fusion.
Test cross	A cross of an individual with the homozygous recessive parent to reveal its genotype.
Transcription	The synthesis of RNA from a DNA template.
Transgressive segregation	The appearance of offspring more extreme than either parent for a quantitative character.
Translation	The synthesis of a polypeptide on the ribosome using the mRNA sequence.
Translocation	The transfer of a chromosomal segment to a non-homologous chromosome.
Triploid	Having three sets of chromosomes (3n).
Trisomic	An aneuploid with one extra chromosome (2n+1).
Wild type	The allele or phenotype most common in nature.
X-linked gene	A gene located on the X chromosome; shows criss-cross inheritance.

References and Further Reading

The material in this handbook follows standard textbooks and university course material. The books below are the usual references for a course in the principles of genetics; students may consult any available edition.

Standard textbooks

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Plant breeding and applied genetics

1. Singh, B. D. *Plant Breeding: Principles and Methods*. Kalyani Publishers, New Delhi.
2. Allard, R. W. *Principles of Plant Breeding*. John Wiley & Sons, New York.
3. Poehlman, J. M. and Sleper, D. A. *Breeding Field Crops*. Iowa State University Press.
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University and institutional course material

1. Tamil Nadu Agricultural University (TNAU), Coimbatore — course material and e-course on *Principles of Genetics* (GPB 201), Department of Plant Breeding and Genetics.
2. ICAR e-Course / e-Krishi Shiksha material on *Principles of Genetics* and *Plant Breeding*, prepared under the National Agricultural Innovation Project / ICAR, New Delhi.
3. AgriMoon / e-Courses on *Fundamentals of Genetics* and *Cytogenetics*, compiled from the ICAR e-course material for agricultural universities.
4. Central Board of Secondary Education and State Board higher secondary biology textbooks, for basic cell biology and classical genetics.

Classical and landmark papers (for reference)

- Mendel, G. J. (1866). Versuche über Pflanzen-Hybriden (Experiments on Plant Hybrids).
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- Meselson, M. and Stahl, F. W. (1958). The replication of DNA in *Escherichia coli*. *Proceedings of the National Academy of Sciences*.
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  F. and Monod, J. (1961). Genetic regulatory mechanisms in the synthesis of proteins. *Journal of Molecular Biology*.

End of GPB 201 — Principles of Genetics, Lecture-wise Theory Study Material.
Prepared for B.Sc. (Hons.) Agriculture, Second Year Third Semester, TNAU, 2025 syllabus.